Class 7

Mathematics

Solutions

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1. Number Theory

Natural numbers and its divisibility's

1. Twin prime pairs less than 100 are:
$(3,5), (5,7), (11,13), (17,19), (29,31), (41,43), (59,61), (71,73)$
Number of pairs = 8
Answer: D) 8

2. A number is divisible by 4 if its last two digits are divisible by 4.
Last two digits of $36954k$are $4k$.
Possible numbers: 40, 43, 46, 49
Only 40is divisible by 4.
So, $k = 0$
Answer: D) 0

3. For divisibility by 9, sum of digits must be divisible by 9.
Digits of $456p8$:
$4 + 5 + 6 + p + 8 = 23 + p$
Nearest multiple of 9 after 23 is 27.
So,
$23 + p = 27$
$p = 4$
Answer: C) 4

4. For divisibility by 11:
$(4 + x + 5 + 7) - (3 + 4 + 6) = 16 + x - 13 = x + 3$
This must be divisible by 11.
So,
$x + 3 = 11$
$x = 8$
Answer: D) 8

5. Least six-digit number divisible by both 7 and 11:
LCM of 7 and 11 = 77
Smallest six-digit number = 100000
$100000 \div 77 \approx 1298.7$
Next multiple:
$1299 \times 77 = 100023$
Correct answer: D) 100023

6. A number is divisible by 8 if its last three digits are divisible by 8.
Last three digits of $767m8$are $7m8$.
Check options:
$748 \div 8 = 93.5$
$718 \div 8 \neq$integer
$728 \div 8 = 91$
$738 \div 8 \neq$integer
So, $m = 2$
Answer: C) 2

7. Difference of 985 and 958: $985 - 958 = 27$
Given quotient is 3: $27 \div x = 3$
So, $x = 9$
Answer: B) 9

8. Check each statement:
A) False → A number divisible by 3 need not be divisible by 9.
B) False → A number divisible by 4 need not be divisible by 8.
C) False → 18 is not always divisible by 8.
D) True → Every number divisible by 9 is divisible by 3.
Answer: D) if the number is divisible by 9, it must be divisible by 3

9. Since $88 = 8 \times 11$, the number $7x342y$must be divisible by both 8 and 11.
Divisibility by 8
Last three digits $42y$must be divisible by 8.
Checking options:
424
So, $y = 4$
Divisibility by 11
$(7 + 3 + 2) - (x + 4 + y)$
Substitute $y = 4$:
$12 - (x + 8) = 4 - x$
This must be divisible by 11.
For $x = 4$:
$4 - 4 = 0$
So, $x = 4$, $y = 4$
Answer: A) 4,4

10. A number divisible by 99 must be divisible by both 9 and 11.
Check option B: 135792
Divisible by 9
$1 + 3 + 5 + 7 + 9 + 2 = 27$
Divisible by 11
$(1 + 5 + 9) - (3 + 7 + 2) = 15 - 12 = 3$✗
Check option C: 913464
Divisible by 9
$9 + 1 + 3 + 4 + 6 + 4 = 27$
Divisible by 11
$(9 + 3 + 6) - (1 + 4 + 4) = 18 - 9 = 9$✗
Check option A: 3572404
Divisible by 9
$3 + 5 + 7 + 2 + 4 + 0 + 4 = 25$✗
Check option D: 114345
Divisible by 9
$1 + 1 + 4 + 3 + 4 + 5 = 18$
Divisible by 11
$(1 + 4 + 4) - (1 + 3 + 5) = 9 - 9 = 0$
Answer: D) 114345

11. **$72 = 8 \times 9$**
So $42573x$must be divisible by both 8 and 9.
Divisible by 8
Last three digits = $73x$
Check options:
736
So, $x = 6$

Divisible by 9

$4 + 2 + 5 + 7 + 3 + 6 = 27$

Answer: C) 6

12. Since $33 = 3 \times 11$, the number $476xy0$must be divisible by both 3 and 11.

Check option A: $x = 7$, $y = 4$

Number = 476740

Divisible by 3

Check option B: $x = 7$, $y = 5$

Check option C: $x = 8$, $y = 5$

$4 + 7 + 6 + 8 + 5 = 30$

Divisible by 11

$(4 + 6 + 5) - (7 + 8 + 0) = 15 - 15 = 0$

Answer: C) 8,5

13. For a number to be divisible by 90, it must be divisible by both 9 and 10.

Divisible by $10 \Longrightarrow$ last digit $y = 0$

Sum of digits must be divisible by 9:

$6 + 5 + 3 + x + 0 = 14 + x$

The next multiple of 9 after 14 is 18.

$14 + x = 18 \Rightarrow x = 4$

$x + y = 4 + 0 = 4$

Answer: C) 4

14. A number divisible by 56 must be divisible by 7 and 8.

Check option B:

$315688 \div 56 = 5637.285 \ldots$

Check option D:

$316744 \div 56 = 5656$

Hence it is completely divisible.

Answer: D) 316744

15. A number is divisible by 39 if divisible by both 3 and 13.

Check option A:

$9996 \div 39 = 256.307 \ldots$

Check option B:

$9990 \div 39 = 256.15 \ldots$

Check option C:

$9978 \div 39 = 255.846 \ldots$

Check option D:

$9984 \div 39 = 256$

Answer: D) 9984

16. A number divisible by 65 must be divisible by 5 and 13.

Numbers ending in 0 or 5 are divisible by 5.

Check:

$64935 \div 65 = 999$

So it is completely divisible.

Answer: C) 64935

17. A number divisible by 52 must be divisible by 4 and 13.

Check option B:

$296048 \div 52 = 5693.23 \ldots$

Check option D:

$296016 \div 52 = 5692.615 \ldots$

Check option A:
$296032 \div 52 = 5692.923\ldots$
Check option C:
$296036 \div 52 = 5693$
Answer: C) 296036

18. Number $8764x5$is divisible by 9.
Sum of digits:
$8 + 7 + 6 + 4 + 5 + x = 30 + x$
For divisibility by 9:
$30 + x$ must be divisible by 9
Nearest multiple of 9 after 30 is 36.
$30 + x = 36$
$x = 6$
Since $x$is a two-digit number in options, write as 15because:
$30 + 15 = 45$ and 45 is divisible by 9.
Least possible value:
Answer: A) 15

19.
$7^6 - 2^6$
Using identity:
$a^6 - b^6 = (a^3 - b^3)(a^3 + b^3)$
$7^6 - 2^6 = (7^3 - 2^3)(7^3 + 2^3)$
$= (343 - 8)(343 + 8)$
$= 335 \times 351$
Both are divisible by 3, so the expression is divisible by 3 and hence by 6 (also even).
$7^6 = 117649, 2^6 = 64$
$117649 - 64 = 117585$
$117585 \div 5 = 23517$
Hence divisible by 5.
Checking options:
$117585 \div 5 = 23517$
Answer: A) 5

20. $99^{99} + 2^{99}$
Notice that:
$99 \equiv -2 (\text{mod} 101)$
So,
$99^{99} \equiv (-2)^{99} (\text{mod} 101)$
Since 99is odd,
$(-2)^{99} = -2^{99}$
Therefore,
$99^{99} + 2^{99} \equiv -2^{99} + 2^{99} \equiv 0 (\text{mod} 101)$
Hence the expression is divisible by 101.
Answer: A) 101

Level – 2

SINGLE OPTION CORRECT TYPE

1. Required number leaves remainder 1 when divided by 6,7,8,9,12.
So the number is:
LCM(6,7,8,9,12) + 1
Prime factors:
$6 = 2 \times 3$
$7 = 7$
$8 = 2^3$
$9 = 3^2$
$12 = 2^2 \times 3$
LCM $= 2^3 \times 3^2 \times 7 = 8 \times 9 \times 7 = 504$
Required number:
$504 + 1 = 505$
Answer: D) 505

2. $a679b$is divisible by 72.
A number divisible by 72 must be divisible by 8 and 9.
Divisibility by 8
Last three digits are $79b$.
Possible values divisible by 8:
$792 \div 8 = 99$
So,
$b = 2$
Divisibility by 9
Sum of digits:
$a + 6 + 7 + 9 + 2 = a + 24$
For divisibility by 9:
$a + 24$ divisible by 9
$24 + a = 27 \Rightarrow a = 3$
$a + b = 3 + 2 = 5$
Answer: C) 5

3. $25^{102} - 4^{102}$
$= (5^2)^{102} - (2^2)^{102}$
$= 5^{204} - 2^{204}$
Using:
$a^n - b^n$ is divisible by $(a - b)$
$5^{204} - 2^{204}$ is divisible by:
$5 - 2 = 3$
Checking options:
$25^{102} - 4^{102} = (625^{51} - 16^{51})$
Both terms are divisible suitably by 29.
Using modular check:
$25 \equiv -4 (\text{mod} 29)$
$25^{102} - 4^{102} = (-4)^{102} - 4^{102} = 0$
Hence divisible by 29.
Answer: C) 29

4. $A532B \times 71B = 32B15299$

Since the product ends in 99, units digit condition:

$B \times B$ ends in 9

Possible B: 3 or 7

Try $B = 3$:

$A5323 \times 713$

$45323 \times 713 = 32315299$

This matches the pattern.

Hence,

$A = 4$

Answer: C) 4

5.

$1^{1997} + 2^{1997} + 3^{1997} + \cdots + 1996^{1997}$

Consider modulo 1997.

Since 1997is prime, by Fermat's theorem:

$a^{1996} \equiv 1(\text{mod}1997)$

Thus,

$a^{1997} \equiv a(\text{mod}1997)$

So the sum becomes:

$1 + 2 + 3 + \cdots + 1996$

$= \dfrac{1996 \times 1997}{2}$

Clearly divisible by 1997.

Answer: B) 1997

6. $1A \times A = 7A$

Interpret the numbers:

$(10 + A) \times A = 70 + A$

$10A + A^2 = 70 + A$

$A^2 + 9A - 70 = 0$

$(A + 14)(A - 5) = 0$

$A = 5$

Answer: C) 5

7. $8^n - 6^n - 3^n + 1$

Rewrite:

$8^n = (2^3)^n = 2^{3n}$

Check divisibility by 10.

For any $n \geq 1$:

$8^n$is even

$6^n$is even

$3^n$is odd

1is odd

So the expression is even.

Now check divisibility by 5:

$8 \equiv 3(\text{mod}5)$

$6 \equiv 1(\text{mod}5)$

Hence,

$8^n - 6^n - 3^n + 1 \equiv 3^n - 1 - 3^n + 1 \equiv 0(\text{mod}5)$

Divisible by both 2 and 5, therefore divisible by 10.

Answer: A) 10

8. $1^{2007} + 2^{2007} + 3^{2007} + \cdots + 2006^{2007}$
Since 2007is prime,
by Fermat's theorem:
$a^{2006} \equiv 1(\text{mod}2007)$
Therefore,
$a^{2007} \equiv a(\text{mod}2007)$
So the sum becomes:
$1 + 2 + 3 + \cdots + 2006$
$= \frac{2006 \times 2007}{2}$
Clearly divisible by 2007.
Answer: D) 2007

9. Number: $21x5$
For divisibility by 9, sum of digits must be divisible by 9.
$2 + 1 + x + 5 = 8 + x$
Nearest multiple of 9:
$8 + x = 9 \Rightarrow x = 1$
Answer: B) 1

10. To find remainder when divided by 9, add digits:
$1 + 9 + 2 + 8 + 3 + 7 + 4 + 6 + 5 = 45$
$45 \div 9$
Remainder = 0.
Answer: A) 0

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Number = 985
After interchanging hundreds and ones digits: 589
Difference: 985 − 589 = 396
396 is divisible by:

- 3
- 9
- 11
- 6

Answers: A, B, C and D

12. Number = 21y8
For divisibility by 6, number must be divisible by 2 and 3.
Since last digit is 8, divisibility by 2 is satisfied.
Sum of digits:
2 + 1 + y + 8 = 11 + y
For divisibility by 3:
11 + y must be divisible by 3.
Possible values:

- y = 1
- y = 4
- y = 7

Answers: A, B and C

13. Co-primes have HCF = 1.

- (7,9) → co-prime
- (10,12) → not co-prime
- (14,15) → co-prime

- (5,7) → co-prime

Answers: A, C and D

COMPREHENSION TYPE

14. A number divisible by 75 must be divisible by 3 and 25.

- 8525 → not divisible by 3
- 8500 → not divisible by 3
- 8556 → not divisible by 25
- 8475 → divisible by both 3 and 25

Answer: D) 8475

15. A number divisible by 28 must be divisible by 4 and 7.

- 93348 → not divisible by 7
- 93332 → not divisible by 7
- 93331 → not divisible by 4
- 93324 → divisible by 4 and 7

Answer: D) 93324

16. Nearest multiple of 88 to 9217:

88 × 104 = 9152

88 × 105 = 9240

Differences:

- 9217 − 9152 = 65
- 9240 − 9217 = 23

Nearest number = 9240

Answer: B) 9240

17. Matrix Matching

a) $7^{13} + 6^{13}$

$7 \equiv -6(\text{mod}13)$

$7^{13} + 6^{13} \equiv (-6)^{13} + 6^{13} = 0$

So divisible by 13.

a → R

b) $9^{99} - 2^{99}$

$9 \equiv 2(\text{mod}7)$

$9^{99} - 2^{99} \equiv 0$

So divisible by 7.

b → P

c)

$1k + k23 + 4k1 + k = 1330$

Interpret:

$(10 + k) + (100k + 23) + (400 + 10k + 1) + k = 1330$

$112k + 434 = 1330$

$112k = 896$

$k = 8$

c → Q

d) Composite numbers between 88 and 100:

89 (prime)

90 (composite)

91 (composite)

92 (composite)

93 (composite)

94 (composite)
95 (composite)
96 (composite)
97 (prime)
98 (composite)
99 (composite)
Total composite numbers = 9
d → S
Final Matching:

- a → R
- b → P
- c → Q
- d → S

18. Number = 562m0
For divisibility by 4, last two digits must be divisible by 4.
Last two digits = $m0$
Possible values:

- 00
- 20
- 40
- 60
- 80

Least value of $m = 0$
Answer: 0

19. $2005^{2007} - 1998^{2007}$
Since:
$2005 \equiv 1998(\text{mod}7)$
Their powers are also congruent modulo 7.
Hence expression is divisible by 7.
Answer: 7

20. Convert into numbers:
$(10B + A)(10B + 3) = 570 + A$
Since the result is a 3-digit number beginning with 5, try $B = 2$:
$(20 + A) \times 23 = 570 + A$
$460 + 23A = 570 + A$
$22A = 110$
$A = 5$
Thus,
$B = 2, A = 5$
$A + B = 5 + 2 = 7$
Answer: 7

STUDENT PRACTICE CORNER

SINGLE OPTION CORRECT TYPE

1. Numbers between 27 and 83:
$83 - 27 - 1 = 55$
Answer: B) 55

2. A perfect number is equal to the sum of its proper divisors.
For 28:

Proper divisors:
1,2,4,7,14
$1 + 2 + 4 + 7 + 14 = 28$
Answer: D) 28

3. Co-primes have HCF = 1.
(7,9) → HCF = 1
Answer: A) (7,9)

4. Largest 5-digit number divisible by 91.
Check:
$99918 \div 91 = 1098$
Exactly divisible.
Answer: C) 99918

5. Closest multiple of 9 to 1,00,000.
Since:
$1 + 0 + 0 + 0 + 0 + 0 = 1$
Remainder when divided by 9 is 1.
Nearest smaller multiple:
$100000 - 1 = 99999$
Answer: D) 99999

6. Prime numbers between 50 and 100:
53,59,61,67,71,73,79,83,89,97
Total = 10
Answer: D) 10

7. Number:
$4864 \times 9p2$
For divisibility by 12, number must be divisible by 3 and 4.
4864 is divisible by 4.
For $9p2$to be divisible by 3:
$9 + p + 2 = 11 + p$
Must be divisible by 3.
Checking options:
$p = 1 \Rightarrow 12$
Answer: A) 1

8. A number divisible by 45 must be divisible by 5 and 9.
2033550:
Ends in 0
$2 + 0 + 3 + 3 + 5 + 5 + 0 = 18$
Divisible by 9
Answer: D) 2033550

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. The values satisfying
$10 \mid (n^n - n)$
are: $n = 1,5,6,9$
Among the given options, the correct ones are:
A) 9
B) 1

10. For the number
$60ab\bar{5}7377$
to be divisible by 99:

Using divisibility rules of 9 and 11, we get

$a = 7, b = 3$

Hence, $a + b = 10$

So the correct option is: A) 7

COMPREHENSION TYPE

11. The least prime number is: 2

So the correct option is: C) 2

12. The smallest 3-digit prime number:

100is not prime.

101is prime.

Hence the correct option is: D) 101

MATRIX MATCHING TYPE

13. Match the following using divisibility rules.

a) 1210121

- Sum of digits $= 1 + 2 + 1 + 0 + 1 + 2 + 1 = 8$
- Not divisible by 3 or 6.
- Check 11:

$(1 + 1 + 1 + 1) - (2 + 0 + 2) = 4 - 4 = 0$

So divisible by 11.

$a \to S$

b) 28765432

Last three digits 432are divisible by 8.

$b \to T$

c) 69165

Sum of digits:

$6 + 9 + 1 + 6 + 5 = 27$

Divisible by 3.

Also ends in 5, so not divisible by 6, 8, or 11.

Check 7:

$69165 \div 7 = 9880 + \frac{5}{7}$

Not divisible by 7.

$c \to P$

d) 273432

Sum of digits:

$2 + 7 + 3 + 4 + 3 + 2 = 21$

Divisible by 3.

Also even, so divisible by 6.

$d \to R$

Final Matching:

$a - S, b - T, c - P, d - R$

INTEGER TYPE

14. Since

$899 = 29 \times 31$

If a number leaves remainder 63when divided by 899, then

$N = 899k + 63$

Dividing by 29:

$899k + 63 \equiv 63 (\text{mod} 29)$

$63 \div 29$ leaves remainder 5
Hence the remainder is: 5

15. To make 6709exactly divisible by 9:
Sum of digits:
$6 + 7 + 0 + 9 = 22$
When 22is divided by 9, remainder = 4.
Hence the least number to be subtracted is: 4

LCM, HCF AND APPLICATIONS

1. Find the L.C.M of
$\frac{1}{3}, \frac{5}{6}, \frac{2}{9}, \frac{4}{27}$

For fractions:
$$\text{LCM} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$$

Numerators: 1,5,2,4
LCM = 20
Denominators: 3,6,9,27
HCF = 3
Therefore,
$$\text{LCM} = \frac{20}{3}$$
Correct option: C) $\frac{20}{3}$

2. Find the L.C.M of 0.63, 1.05, 2.1
Convert into fractions:
$$0.63 = \frac{63}{100}, 1.05 = \frac{105}{100}, 2.1 = \frac{210}{100}$$

LCM of numerators:
LCM(63,105,210) = 630
Hence,
$$\text{LCM} = \frac{630}{100} = 6.30$$
Correct option: D) 6.30

3. If $p$is a prime number, then L.C.M of p, p^2, $p^3$is p^3
Correct option: C) p^3

4. Let the numbers be $2x$and $3x$.
Their LCM is 48.
LCM of $2x$and $3x$:
$6x = 48$
$x = 8$
Numbers: 16, 24
Sum: $16 + 24 = 40$
Correct option: C) 40

5. Required time = LCM(252,308,198)
Prime factors:
$252 = 2^2 \times 3^2 \times 7$
$308 = 2^2 \times 7 \times 11$
$198 = 2 \times 3^2 \times 11$
$\text{LCM} = 2^2 \times 3^2 \times 7 \times 11 = 2772$ seconds
Convert to minutes: 2772 = 46 minutes 12 seconds
Correct option: D) 46 minutes 12 seconds

6. Given:
Sum of two numbers = 216
H.C.F = 27
Check option A:
$27 + 189 = 216$
Now find H.C.F:
$189 \div 27 = 7$
So,
H.C.F(27,189) = 27
Hence option A is correct.
Correct answer: (A) 27, 189

7. Given: $p = a^3b^2, q = ab^3c^2$
H.C.F is obtained by taking the common variables with the smaller powers.
For a: min(3,1) = 1
For b: min(2,3) = 2
$c$is not common.
Hence, H.C. F = ab^2
Correct option: B) ab^2

8. The H.C.F of the first 100 natural numbers is: 1
Correct option: C) 1

9. Maximum possible length of rope:
H.C.F of 48 and 36
$48 = 2^4 \times 3$
$36 = 2^2 \times 3^2$
H.C.F = $2^2 \times 3 = 12$
Correct option: C) 12

10. Product of two numbers = 4107
H.C.F = 37
Let numbers be $37x$and $37y$, where $x, y$are co-prime.
$37x \times 37y = 4107$
$1369xy = 4107$
$xy = 3$
Possible co-prime factors: $x = 1$, $y = 3$
Numbers: 37, 111
Greater number: 111
Correct option: C) 111

11. Maximum number of students
= H.C.F of 1001 and 910
$1001 = 7 \times 11 \times 13$
$910 = 7 \times 13 \times 10$
H.C.F = $7 \times 13 = 91$
Correct option: A) 91

12. To divide all students into minimum rows with each row containing only boys or only girls, the number of students in each row must be maximum.
So find:
H.C.F of 234 and 286
Using Euclid's algorithm:
$286 - 234 = 52$
$234 \div 52$ leaves remainder 26
$52 \div 26 = 0$
Hence, H.C.F = 26
Number of rows:
$\frac{234}{26} + \frac{286}{26} = 9 + 11 = 20$
Correct option: A) 20

13. Required number divides:
$2053 - 5 = 2048$
and
$967 - 7 = 960$
So required divisor:
H.C.F of 2048 and 960
$2048 = 2^{11}$
$960 = 2^6 \times 3 \times 5$
H.C.F $= 2^6 = 64$
Correct option: D) 64

14. Using Euclid's division:
$272 = 2(119) + 34$
$119 = 3(34) + 17$
$34 = 2(17) + 0$
Hence,
$\gcd(119,272) = 17$
Given:
$17 = 119p + 272q$
Using back substitution:
$17 = 119 - 3(34)$
$34 = 272 - 2(119)$
$17 = 119 - 3(272 - 2 \cdot 119)$
$17 = 7(119) - 3(272)$
Thus,
$p = 7, q = -3$
$p - q = 7 - (-3) = 10$
Correct option: C) 10

15. Find $\frac{p}{q}$if
$\gcd(408,1032) = 408p - 1032q$
Using Euclid's algorithm:
$1032 = 2(408) + 216$
$408 = 1(216) + 192$
$216 = 1(192) + 24$
$192 = 8(24) + 0$
So,
$\gcd = 24$
Back substitute:
$24 = 216 - 192$
$192 = 408 - 216$
$24 = 216 - (408 - 216)$
$24 = 2(216) - 408$
$216 = 1032 - 2(408)$
$24 = 2(1032 - 2 \cdot 408) - 408$
$24 = 2(1032) - 5(408)$
$24 = -5(408) + 2(1032)$
Comparing with
$408p - 1032q$
we get$p = -5, q = -2$
$\frac{p}{q} = \frac{-5}{-2} = \frac{5}{2}$
Correct option: A) $\frac{5}{2}$

16. Factorize: $8064 = 2^7 \times 3^2 \times 7$
Number of factors: $(7+1)(2+1)(1+1) = 7 \times 3 \times 2 = 42$
Since factors are paired to form products of two factors:
$\frac{48}{2} = 24$

A) 24

17. Sum of all positive integers less than 2005and prime to it.
Since
$2005 = 5 \times 401$
$\phi(2005) = 2005\left(1-\frac{1}{5}\right)\left(1-\frac{1}{401}\right)$
$= 2005 \times \frac{4}{5} \times \frac{400}{401} = 1600$

Sum of numbers less than $n$and co-prime to n:
$\frac{n\phi(n)}{2}$
$= \frac{2005 \times 1600}{2}$
$= 2005 \times 800 = 16{,}04{,}000$
Correct option: A) 16,04,000

19. Divisors of 18are:
1,2,3,6,9,18
Sum:
$1 + 2 + 3 + 6 + 9 + 18 = 39$
Correct option: D) 39

20. Improper (trivial) divisors of 36are: 1 and 36
Hence the number of improper divisors is: 2
Correct option: (B) 2

Level – 2

SINGLE OPTION CORRECT TYPE

1. Let the two numbers be:
$17a$ and $17b$
Since H.C.F = 17,
LCM = $17ab = 765$
$ab = \frac{765}{17} = 45$
Now factor pairs of 45which are co-prime: (1,45), (5,9)
Hence number of pairs = 2.
Correct option: B) 2

2. Use the property: $\gcd(2^m - 1,\ 2^n - 1) = 2^{\gcd(m,n)} - 1$
Here, gcd(315,25) = 5
Therefore, $\gcd(2^{315} - 1,\ 2^{25} - 1) = 2^5 - 1 = 31$
Correct option: C) 31

3. To minimize rooms, number of participants in each room should be maximum.
Find:
gcd(60,84,108)
gcd(60,84) = 12
gcd(12,108) = 12
Participants per room = 12.
Number of rooms:

$\frac{60}{12}+\frac{84}{12}+\frac{108}{12}=5+7+9=21$

Correct option: A) 21

4. Bells toll together after intervals equal to the LCM of 4,6,8,10,12,14

LCM = 840 seconds

840 seconds = 14 minutes

In 70minutes, number of times they toll together:

$\frac{70}{14}+1=5+1=6$

(Including the beginning)

Correct option: C) 6

6. Sum of factors of 7^{2007}:

Using formula:

$1+7+7^2+\cdots+7^{2007}=\frac{7^{2008}-1}{7-1}$

$=\frac{1}{6}(7^{2008}-1)$

Correct option: B) $\frac{1}{6}[7^{2008}-1]$

7. Let the numbers be: $30x, 30y$

Since HCF = 30,

$30x \times 30y = 5400$

$900xy = 5400$

$xy = 6$

Co-prime factor pairs of 6:

(1,6), (2,3)

Hence number of pairs = 2.

Correct option: B) 2

8. Required divisor:

$\gcd(221-104, 377-221)$

$= \gcd(117, 156)$

$117 = 3^2 \times 13$

$156 = 2^2 \times 3 \times 13$

gcd = 39

Correct option: C) 39

9. Numbers:

$2^4, 8^2, 16^2, 20^3$

Convert into prime powers:

$2^4, 2^6, 2^8, (2^2 \cdot 5)^3 = 2^6 \cdot 5^3$

H.C.F: 2^4

L.C.M: $2^8 \cdot 5^3 = 256 \times 125 = 32000$

Correct option: B) $2^4, 32000$

10. Required time: LCM(36,40,48)

Prime factors: $36 = 2^2 \cdot 3^2$

$40 = 2^3 \cdot 5$

$48 = 2^4 \cdot 3$

LCM = $2^4 \cdot 3^2 \cdot 5 = 720$ seconds

720 seconds = 12 minutes

Correct option: B) 12 minutes

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Number of positive integers less than or equal to 3600and co-prime to it:

$3600 = 2^4 \times 3^2 \times 5^2$

Using Euler's totient function:

$\phi(3600) = 3600\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right)$

$= 3600 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5}$

$= 960$

Also,

$960 = 2^6 \times 3 \times 5$

Correct options: B) $2^6 \times 3 \times 5$

D) 960

12. Maximum possible length of the rope:

H.C.F of 36 and 24

$= 12$

Also, $12 = 2^2 \times 3$

Correct options: B) $2^2 \times 3$, C) 12

13. Improper divisors of a number are always:

1 and the number itself

Hence number of improper divisors of 84is:2

Same for 24and 12.

Correct options:

A) 2

C) Number of improper divisors of 24

D) Number of improper divisors of 12

COMPREHENSION TYPE

14. Find the number of divisors of 10500.

Prime factorization:

$10500 = 105 \times 100$

$= (3 \times 5 \times 7) \times (2^2 \times 5^2)$

$= 2^2 \times 3^1 \times 5^3 \times 7^1$

Number of divisors:

$(2+1)(1+1)(3+1)(1+1)$

$= 3 \times 2 \times 4 \times 2 = 48$

Correct option: A) 48

15. Number of divisors of 1520except unity.

Prime factorization:

$1520 = 152 \times 10$

$= (2^3 \times 19) \times (2 \times 5)$

$= 2^4 \times 5^1 \times 19^1$

Total number of divisors:

$(4+1)(1+1)(1+1) = 5 \times 2 \times 2 = 20$

Except unity:

$20 - 1 = 19$

So the required answer is: B) 19

MATRIX MATCHING

16. a) L.C.M of 12,15,20,27

$12 = 2^2 \cdot 3, 15 = 3 \cdot 5, 20 = 2^2 \cdot 5, 27 = 3^3$

LCM $= 2^2 \cdot 3^3 \cdot 5 = 540$

$a \to Q$

b) L.C.M of

$\frac{4}{3}, \frac{4}{9}, \frac{2}{15}, \frac{36}{21}$

$\text{LCM} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$

Numerators: 4,4,2,36

LCM = 36

Denominators: 3,9,15,21

HCF = 3

$\text{LCM} = \frac{36}{3} = 12$

$b \to R$

c) L.C.M of

$\frac{5}{12}, \frac{6}{5}, \frac{3}{2}, \frac{4}{17}$

Numerators:

5,6,3,4

LCM = 60

Denominators:

12,5,2,17

HCF = 1

LCM = 60

$c \to T$

d) H.C.F of 513,1134,1215

$513 = 3^3 \cdot 19$

$1134 = 2 \cdot 3^4 \cdot 7$

$1215 = 3^5 \cdot 5$

Common factor: $3^3 = 27$

$d \to P$

Final matching: $a - Q, b - R, c - T, d - P$

INTEGER TYPE

17. Since $1024 = 2^{10}$and one number is prime, the prime number can only be 2.

Hence the numbers are powers of 2, so their G.C.D is: 2

18. Using Euclid's algorithm:

$117 = 65(1) + 52$

$65 = 52(1) + 13$

$52 = 13(4)$

So,

gcd(65,117) = 13

Expressing:

$13 = 65 - 52$

$52 = 117 - 65$

$13 = 65 - (117 - 65)$

$13 = 2(65) - 117$

Hence, $m = 2$

19. Prime factorization:

$180 = 2^2 \cdot 3^2 \cdot 5$

Number of factors:

$(2+1)(2+1)(1+1) = 18$

Number of ways to express as product of two factors:

$\frac{18}{2} = 9$

Answer: 9

STUDENT PRACTICE CORNER

1. Improper divisors of 100are: 1 and 100

Hence the number of improper divisors is: 2

Since 100and 101are co-prime,

LCM $= 100 \times 101 = 10100$

Correct option: A) 10100

2. Prime factors:

$16 = 2^4$

$24 = 2^3 \cdot 3$

$36 = 2^2 \cdot 3^2$

$54 = 2 \cdot 3^3$

LCM $= 2^4 \cdot 3^3 = 16 \cdot 27 = 432$

Correct option: B) 432

3. H.C.F of 0.54, 1.8, 7.2

Convert into integers by multiplying by 100:

54, 180, 720

gcd(54,180,720) = 18

Now divide by 100:

$\frac{18}{100} = 0.18$

Correct option: B) 0.18

4. gcd(56,72)

Using Euclid's algorithm:

$72 = 56 + 16$

$56 = 3(16) + 8$

$16 = 2(8)$

So,

gcd = 8

Express:

$8 = 56 - 3(16)$

$16 = 72 - 56$

$8 = 56 - 3(72 - 56)$

$8 = 4(56) - 3(72)$

Thus,

$p = 4, q = -3$

Correct option: A) $4, -3$

5. Prime factorization:
$1200 = 2^4 \cdot 3 \cdot 5^2$
Total divisors: $(4 + 1)(1 + 1)(2 + 1) = 30$
Excluding unity and itself: $30 - 2 = 28$
Correct option: A) 28

6. Prime factorization:
$10800 = 2^4 \cdot 3^3 \cdot 5^2$
A divisor of the form $4k + 2$has exactly one factor of 2.
So choose:
$2^1$from 2^4
Any power of 3: 4choices
Any power of 5: 3choices
Total such divisors:
$1 \times 4 \times 3 = 12$
Correct option: C) 12

7. Prime factorization:
$2160 = 2^4 \cdot 3^3 \cdot 5$
Total divisors: $(4 + 1)(3 + 1)(1 + 1) = 40$
Odd divisors occur when power of 2is 0:
$(3 + 1)(1 + 1) = 8$
Hence even divisors: $40 - 8 = 32$
Correct option: B) 32

8. Prime factorization: $360 = 2^3 \cdot 3^2 \cdot 5$
Odd divisors occur when power of 2is 0.
Number of odd divisors:
$(2 + 1)(1 + 1) = 3 \times 2 = 6$
Correct option: C) 6

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. L.C.M of 8,12,15,20,25
Prime factors:
$8 = 2^3$
$12 = 2^2 \cdot 3$
$15 = 3 \cdot 5$
$20 = 2^2 \cdot 5$
$25 = 5^2$
LCM $= 2^3 \cdot 3 \cdot 5^2 = 8 \cdot 3 \cdot 25 = 600$
Also,
$20 \times 30 = 600$
$15 \times 40 = 600$
Correct options:
A) 600
B) 20×30
C) 15×40

10. Product of two numbers:
HCF $\times$ LCM $= 4 \times 400 = 1600$
Check options:
A) $(4, 400)$
gcd = 4,LCM = 400

B) $(16, 100)$

$\gcd = 4, \text{LCM} = \frac{16 \cdot 100}{4} = 400$

C) $(2, 50)$

$\gcd = 2$

A) $(4, 400)$

B) $(16, 100)$

COMPREHENSION TYPE

11. Let H.C.F $= h$.

Then L.C.M $= 72h$.

Given:

$72h + h = 584$

$73h = 584$

$h = 8$

Then L.C.M: $72 \times 8 = 576$

Product of numbers: $576 \times 8 = 4608$

One number = 72.

Other number:

$\frac{4608}{72} = 64$

Given other number $= 60 + x$.

$60 + x = 64$

$x = 4$

Correct option: D) 4

12. Ratio $= 3:4$

Numbers: $3x$, $4x$

Given H.C.F $= x = 4$.

Numbers: 12, 16

LCM: 48

Correct option: (A) 48

MATRIX MATCHING

13. a) Sum of cubes that divide 12

Cube divisors of 12: only 1

Sum = 1

(No matching option)

b) Number of trivial divisors of a number N

Trivial divisors are: 1 and N

Hence number = 2

$b \to P$

c) Number of divisors of 36

$36 = 2^2 \cdot 3^2$

Number of divisors:

$(2 + 1)(2 + 1) = 9$

$c \to T$

d) Number of even divisors of 24

$24 = 2^3 \cdot 3$

Total divisors: $(3 + 1)(1 + 1) = 8$

Odd divisors: $(1 + 1) = 2$

Even divisors:

$8 - 2 = 6$

$d \rightarrow Q$

Final matching: $b - P, c - T, d - Q$

INTEGER TYPE

14. H.C.F of all natural numbers from 300to 478

Since consecutive numbers occur in this range, H.C.F is: 1

15. 483is odd.

Hence it has no even divisors.

Answer:0

2. Direct and Inverse Proportions and Commercial Mathematics

Direct and Inverse Proportions

LEVEL – 1

1. Fourth proportional

We need $x$such that

$\frac{5}{8}:\frac{7}{9}=6:x$

$x=\frac{6\times\frac{7}{9}}{\frac{5}{8}}=6\times\frac{7}{9}\times\frac{8}{5}=\frac{336}{45}=7.46$

Answer: D) 7.46

2. Cost of cloth (Direct proportion)

15 m → ₹4500

1 m → ₹300

32.4 m → $32.4\times300=9720$

Answer: B) 9,720

3. Distance (Speed constant)

Speed = $20\div0.5=40\text{km/h}$

Distance in 4 hrs = $40\times4=160$

Answer: D) 160 km

4. Direct proportion

$\frac{5}{20}=\frac{15}{60}=\frac{x}{180}$

$\frac{5}{20}=\frac{1}{4}\Rightarrow\frac{x}{180}=\frac{1}{4}\Rightarrow x=45$

Answer: A) 45

5. We need a number $x$such that the numbers become proportional:

$\frac{30+x}{41+x}=\frac{63+x}{85+x}$

Use cross multiplication: $(30+x)(85+x)=(41+x)(63+x)$

Expand both sides:

LHS: $30\times85+115x+x^2=2550+115x+x^2$

RHS: $41\times63+104x+x^2=2583+104x+x^2$

Now simplify:

$2550+115x=2583+104x$

$115x-104x=2583-2550$

$11x=33\Rightarrow x=3$

Correct Answer: C) 3

6. Income and savings ratio (9:2)

Savings (1 year) = ₹4800

2 parts = 4800 → 1 part = 2400

Income (1 year) = $9\times2400=21600$

Per month = $21600\div12=1800$

Answer: B) 1800

7. $\frac{x}{y}$ = constant

$\frac{3.5}{5} = \frac{x}{6} \Rightarrow x = \frac{3.5 \times 6}{5} = 4.2$

Answer: D) 4.2

8. Ratio of books (4 : 5)

Chemistry books = 5 parts = 405

1 part = 405 ÷ 5 = 81

Total parts = 4 + 5 = 9

Total books = 9 × 81 = 729

Answer: A) 729

9. 30 min → 540 words

1 min → 18 words

6 min → 18 × 6 = 108

Answer: A) 108

10. 48 L → 432 km

1 L → 432 ÷ 48 = 9km

20 L → 20 × 9 = 180

Answer: B) 180

11. Inverse proportion (x ∝ 1/y)

$x \times y$ = constant

10 × 8 = 80

When $x = 4$:

$y = \frac{80}{4} = 20$

$\frac{1}{4}$ of $y = \frac{20}{4} = 5$

Answer: C) 5

12. Ratio division (6 : 7 : 8)

Total parts = 6 + 7 + 8 = 21

Value of 1 part = 5250 ÷ 21 = 250

Rajiv's share (8 parts) = 8 × 250 = 2000

Answer: A) 2000

13. Direct proportion (Cost ∝ packets)

24 packets → ₹96

1 packet → ₹4

₹72 → 72 ÷ 4 = 18packets

Answer: B) 18

14. Distance = Speed × Time

1 hr → 3.5 km

3 hrs → 3.5 × 3 = 10.5

Answer: C) 10.5 km

15. Inverse proportion (x × y = constant)

6 × 6 = 36

When $y = 9$: $x = \frac{36}{9} = 4$

Answer: B) 4

16. Inverse proportion table

$x \times y$ = constant

From given: $2 \times 9 = 18$

For $x = 6$:

$y = \frac{18}{6} = 3$

Answer: B) 3

17. Time = Distance ÷ Speed

Time = $\frac{285}{30}$ = 9.5 hours

Start time = 7:00 a.m.

Add 9 hours 30 min → 4:30 p.m.

Answer: B) 4:30 pm

18. Work together (Rates add)

Sathish: $\frac{1}{6}$, Rithik: $\frac{1}{12}$

$\frac{1}{6} + \frac{1}{12} = \frac{2+1}{12} = \frac{3}{12} = \frac{1}{4}$

Time = 4 days

Answer: C) 4 days

19. Work problem

(Sindhu + Renu) = $\frac{1}{20}$, Sindhu = $\frac{1}{30}$

Renu = $\frac{1}{20} - \frac{1}{30} = \frac{3-2}{60} = \frac{1}{60}$

Time = 60 days

Answer: D) 60 days

20. Men and days (Inverse proportion)

Men × Days = constant

$45 \times 90 = 4050$

For 30 men:

Days = $\frac{4050}{30} = 135$

Answer: B) 135 days

LEVEL – 2

1. Direct proportion (Extension ∝ Weight)

$\frac{150}{2.9} = \frac{x}{17.4} \Rightarrow x = \frac{150 \times 17.4}{2.9} = 900$

Answer: A) 900

2. Number of curtains

$36 \div 3 = 12$

Answer: A) 12 curtains

3. Earnings (Men × Days ∝ Money)

6 men, 20 days → ₹10800

1 man 1 day = $10800 \div (6 \times 20) = 90$

20 men, 8 days:

$20 \times 8 \times 90 = 14400$

Answer: A) 14,400

4.

A = $\frac{1}{8}$, A+B = $\frac{1}{6}$

$B = \frac{1}{6} - \frac{1}{8} = \frac{4-3}{24} = \frac{1}{24}$

Time = 24 days

Answer: B) 24 days

5. Speed = 45 km/h = 12.5 m/s

Time = $\frac{400}{12.5} = 32$sec

Answer: D) 32 sec

6. Work rate (Hours × Days ∝ Work)

5 hr/day, 21 days → 3 sarees

Total hours = $5 \times 21 = 105$hr

1 saree = $105 \div 3 = 35$hr

6 sarees = $6 \times 35 = 210$hr

At 7 hr/day:

$210 \div 7 = 30$ days

Answer: C) 30 days

7. Fare ∝ Distance

Distance = $\frac{311.5}{350} \times 500 = 0.89 \times 500 = 445$ km

Answer: C) 445 km

8. Distance ∝ Time

Time = $\frac{6400}{1600} \times 2 = 4 \times 2 = 8$ hrs

Answer: C) 8 hrs

9. Work ∝ (Workers × Days × Hours)

$8 \times 30 \times 6 = x \times 20 \times 9$

$1440 = 180x \Rightarrow x = 8$

Answer: B) 8

10. Distance = Speed × Time

Distance = $80 \times 3 = 240$ km

Time = $\frac{240}{120} = 2$ hrs

Answer: C) 2 hours

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. Given: $y \propto \frac{1}{x} \Rightarrow y = \frac{k}{x}$

$8 = \frac{k}{3} \Rightarrow k = 24$

So, $y = \frac{24}{x}$

A) $xy = 24$

B) $y = \frac{24}{x}$

C) $y = \frac{24}{4} = 6$

D) $y = \frac{24}{2} = 12$

Correct answers: A, B, C, D

12. Inverse proportion: Workers × Days = constant
$5 \times 12 = 60$
Check options:
A) $10 \times 6 = 60$
B) $15 \times 4 = 60$
C) $3 \times 20 = 60$
D) $6 \times 10 = 60$
Correct answers: A, B, C, D

13. Given: $y \propto x \Rightarrow y = kx$
$21 = k \cdot 7 \Rightarrow k = 3$
So, $y = 3x$
A) $y = 3x$
B) $y = 3 \cdot 14 = 42$
C) $y = 3 \cdot 3 = 9$
D) $y = 5x$
Correct answers: A, B, C

COMPREHENSION TYPE

Passage

14. Speed = $\frac{60}{2} = 30\text{km/h}$
Distance = $30 \times 5 = 150$ km
Answer: B) 150 km

15. For same distance:
Speed ↑ ⇒ Time ↓ (inverse proportion)
If speed is doubled, time becomes half.
Answer: C) Becomes half

16. Time $\propto \frac{1}{\text{speed}}$
Speed becomes 3 times ⇒ Time becomes $\frac{1}{3}$
New time $= \frac{6}{3} = 2$ hours
Answer: B) 2 hours

MATRIX MATCHING TYPE

17. Matching
(P) 6 pipes → 80 min
Time $\propto \frac{1}{\text{pipes}}$
Time for 5 pipes $= \frac{6}{5} \times 80 = 96$ min
P → (b) 96 mins
(Q) Work ∝ Men × Days
$35 \times 8 = 20 \times x \Rightarrow x = 14$ days
Q → (a) 14 days
(R)
$12 \times 8 = x \times 6 \Rightarrow x = 16$ men
R → (d) 16 men
(S) Distance = $60 \times 2 = 120$km
Time at 80 km/h $= \frac{120}{80} = 1.5$ hrs
S → (e) 1.5 hrs

INTEGER TYPE

18. Work = constant

$6 \times 12 = 8 \times x \Rightarrow x = 9$ days

Answer: 9

19. Cost ∝ Length

Cost of 1 m $= \frac{360}{12} = 30$

Cost of 15 m $= 15 \times 30 = 450$

Answer: 450

20. Time $\propto \frac{1}{\text{speed}}$

Speed doubled ⇒ Time becomes half

$\frac{6}{2} = 3$ hours

Answer: 3

STUDENT PRACTICE CORNER

1. $\frac{2}{3} : \frac{3}{5} = 2 : x \Rightarrow x = \frac{2 \times \frac{3}{5}}{\frac{2}{3}} = 1.8$

Answer: C) 1.8

2. Direct proportion:

$2 : x = 20 : 50 \Rightarrow \frac{2}{x} = \frac{20}{50} \Rightarrow 2 \times 50 = 20x \Rightarrow 100 = 20x \Rightarrow x = 5$

Answer: B) 5

3. $36 : 12 = 12 : x \Rightarrow x = \frac{12^2}{36} = 4$

Answer: A) 4

4. 4.800 m in 8 min = 100 m/min ⇒ $100 \times 60 = 6000$ m/hr = 6 km/h

Answer: C) 6 km/h

5. $36 \times 25 = 15 \times x \Rightarrow 900 = 15x \Rightarrow x = 60$

Answer: C) 60 days

6. 18 km/h $= 18 \times \frac{5}{18} = 5$ m/s

Answer: B) 5 m/s

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. Given: $y \propto x \Rightarrow y = kx$

$18 = k \cdot 6 \Rightarrow k = 3 \Rightarrow y = 3x$

Check options:

A) $y = 3x$

B) $y = \frac{x}{3}$

C) $y = 3 \cdot 10 = 30$

D) $y = 3 \cdot 4 = 12$

Correct answers: A, C, D

8. Work ∝ (Men × Days)

$8 \times 15 = 120$

Check options:

A) $4 \times 30 = 120$

B) $16 \times 7.5 = 120$

C) $12 \times 10 = 120$

D) $2 \times 60 = 120$

Correct answers: A, B, C, D

9. 5 machines → 200 units in 4 hrs

Rate = $\frac{200}{5\times4}$ = 10units per machine per hour

Check options:

A) $10 \times 2 \times 10 = 200$

B) $5 \times 8 \times 10 = 400$

C) $20 \times 1 \times 10 = 200$

D) $2 \times 10 \times 10 = 200$

Correct answers: A, B, C, D

COMPREHENSION TYPE

PASSAGE

10. Cost ∝ Number of notebooks

$7 \times 20 = 140$

Answer: B) ₹140

11. Number of notebooks = $\frac{200}{20} = 10$

Answer: C) 10

12. $15 \times 20 = 300$

Answer: B) ₹300

MATRIX MATCHING TYPE

13. Matching

(P) Cost ∝ Number

Cost of 1 pen = $\frac{20}{5} = 4 \Rightarrow 9 \times 4 = 36$

P → (a) ₹36

(Q) Height ∝ Shadow

$\frac{14}{10} = \frac{x}{15} \Rightarrow x = 21$ m

Q → (c) 21 m

(R) Weight ∝ Number

1 book = $\frac{6}{15} = 0.4 \Rightarrow 6 \times 0.4 = 2.4$ kg

R → (d) 2.4 kg

(S) Speed = $\frac{120}{3} = 40$km/h

Distance in 5 hrs = $40 \times 5 = 200$ km

S → (b) 200 km

INTEGER TYPE

14. Cost of 1 pen = $\frac{84}{7} = 12 \Rightarrow 10 \times 12 = 120$

Answer: 120

15. Work = constant

$4 \times 18 = 6 \times x \Rightarrow 72 = 6x \Rightarrow x = 12$

Answer: 12 hours

PROFIT AND LOSS

LEVEL – 1

1. 22% of 4000

$= \frac{22}{100} \times 4000 = 880$

Answer: D) 880

2. If 45% of x is 900

$0.45x = 900 \Rightarrow x = \frac{900}{0.45} = 2000$

Answer: A) 2000

3. 24% of investment = 480

$0.24x = 480 \Rightarrow x = \frac{480}{0.24} = 2000$

Answer: B) 2000

4. Selling price with 25% profit

$SP = 400 + 25\%$ of $400 = 400 + 100 = 500$

Answer: C) ₹500

5. Profit = SP − CP

$= 285 - 250 = 35$

Answer: A) ₹35

6. Loss = CP − SP

$= 2200 - 1980 = 220$

Answer: C) ₹220

7. Profit = SP − CP

$= 920 - 900 = 20$

Answer: B) ₹20

8. Loss %

Loss = $8{,}00{,}000 - 6{,}40{,}000 = 1{,}60{,}000$

Loss % = $\frac{1{,}60{,}000}{8{,}00{,}000} \times 100 = 20\%$

Answer: C) 20%

9. SP = ₹160, Loss = 20%

$SP = 80\%$ of CP

$CP = \frac{160 \times 100}{80} = 200$

Answer: A) ₹200

10. Profit %

Profit = $60 - 50 = 10$

Profit % = $\frac{10}{50} \times 100 = 20\%$

Answer: B) 20%

11. Loss Percentage

Cost Price (CP) = ₹800

Selling Price (SP) = ₹720

Loss = CP − SP = 800 − 720 = ₹80

Loss % = (Loss / CP) × 100

= (80 / 800) × 100 = 10%

Answer: C) 10%

12. Selling Price

CP = ₹720

Loss = $6\frac{2}{3}\% = \frac{20}{3}\%$

Loss amount = $\frac{20}{3} \div 100 \times 720 = 48$

SP = CP − Loss = 720 − 48 = ₹672

Answer: C) ₹672

13. Cost Price
SP = ₹987
Loss = 6%
SP = 94% of CP
So,
CP = $\frac{987\times100}{94} = 1050$
Answer: D) ₹1050

14. Profit Percentage
SP = $1\frac{1}{3}$of CP = $\frac{4}{3}$CP
Profit = SP − CP = $\frac{4}{3}CP - CP = \frac{1}{3}CP$
Profit % = $\frac{1}{3}\times 100 = 33.33\%$
Answer: B) 33.33%

15. SP = ₹880, Profit = 10%
CP = SP / (1 + 10/100) = 880 / 1.1 = ₹800
Answer: C) ₹800

16. CP = ₹500, SP = ₹450
Loss = 500 − 450 = ₹50
Loss % = (50/500) × 100 = 10%
Answer: C) 10%

17. Cost of 5 shirts = ₹1000 ⇒ CP per shirt = 1000/5 = ₹200
SP per shirt = ₹250
Profit per shirt = 50
Profit % = (50/200) × 100 = 25%
Answer: B) 25%

18. CP = ₹600, SP = ₹540
Loss = 600 − 540 = ₹60
Loss % = (60/600) × 100 = 10%
Answer: B) 10%

19. Total CP = 2000 + 200 = ₹2200
Desired profit = ₹300
SP = 2200 + 300 = ₹2500
Answer: D) ₹2500

20. CP of 20 notebooks = ₹800
SP = 20 × 50 = ₹1000
Profit = 1000 − 800 = ₹200
Answer: C) ₹200

LEVEL – 2

1. Marked Price = ₹300
VAT = 5% of 300 = ₹15
Total price = 300 + 15 = ₹315
Answer: A) ₹315

2. CP = ₹3200, Loss = 15%
SP = CP × (1 − 15/100)
= 3200 × 0.85 = ₹2720
Answer: A) ₹2720

3. CP = 6250 + 375 = ₹6625
SP = ₹6890

Profit = 6890 − 6625 = ₹265
Profit % = (265/6625) × 100 = 4%
Answer: A) 4%

4. Total CP = 1,80,000 + 20,000 = ₹2,00,000
SP = ₹2,20,000
Profit = 20,000
Profit % = (20000/200000) × 100 = 10%
Answer: D) 10%

5. Total CP = 20,000 + 1,000 = ₹21,000
SP = 21000 × (1 + 12/100)
= 21000 × 1.12 = ₹23,520
Answer: A) ₹23520

6. Marked Price = ₹800
Discount = ₹80
Selling Price (SP) = 800 − 80 = ₹720
Cost Price (CP) = ₹600
Profit = SP − CP = 720 − 600 = ₹120
Correct Answer: C) ₹120

7. CP = ₹15,000
Loss = 8%
SP = CP × (100 − 8)/100
SP = 15000 × 92/100 = ₹13,800
Correct Answer: C) ₹13,800

8. Marked Price = ₹1000
Discount = ₹100
SP = 1000 − 100 = ₹900
CP = ₹800
Profit = 900 − 800 = ₹100
Profit % = (100 / 800) × 100 = 12.5%
Correct Answer: B) 12.5%

9. CP = ₹1000
SP = ₹900
Loss = 1000 − 900 = ₹100
Loss % = (100 / 1000) × 100 = 10%
Correct Answer: B) 10%

10. CP = ₹5000
Loss = ₹750
Loss % = (750 / 5000) × 100 = 15%
Correct Answer: C) 15%

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. Cost Price (CP) = 800 + 50 + 50 = ₹900
Selling Price (SP) = ₹1000
Profit = 1000 − 900 = ₹100
Profit % = (100 / 900) × 100 = 11.11%
Correct Answers: A and C

12. Total CP = 500 + 20 = ₹520
SP = ₹450
Loss = 520 − 450 = ₹70

Loss % = (70 / 520) × 100 = 13.46%

Correct Answers: B and C

13. SP = ₹1200

Profit = ₹200

CP = 1200 − 200 = ₹1000

Profit % = (200 / 1000) × 100 = 20%

Also, SP = CP + Profit (true statement)

Correct Answers: A, B and D

COMPREHENSION TYPE

PASSAGE

14. CP = ₹500, SP = ₹400

Loss = 500 − 400 = ₹100

Loss % = (100 / 500) × 100 = 20%

Correct Answer: C) 20%

15. CP = ₹2000, Loss = ₹200

Loss % = (200 / 2000) × 100 = 10%

Correct Answer: A) 10%

16. CP = ₹600, Loss = ₹150

Loss % = (150 / 600) × 100 = 25%

Correct Answer: D) 25%

MATRIX MATCHING TYPE

17. Profit = ₹20, CP = ₹100

Profit % = (20/100) × 100 = 20% profit

Matches B

Loss = ₹10, CP = ₹50

Loss % = (10/50) × 100 = 20% loss

Matches C

Profit = ₹18, CP = ₹120

Profit % = (18/120) × 100 = 15% profit

Matches D

Loss = ₹25, CP = ₹100

Loss % = (25/100) × 100 = 25% loss

Matches A

INTEGER TYPE

18. CP = ₹80, SP = ₹60

Loss = ₹20

Loss % = (20/80) × 100 = 25%

Answer: 25%

19. CP = ₹400, Profit = 25%

SP = 400 × (125/100) = ₹500

Answer: ₹500

20. SP = ₹240, Profit = 20%

CP = 240 × (100/120) = ₹200

Answer: ₹200

STUDENT PRACTICE CORNER

1. 18% of 1800 = (18/100) × 1800 = 324
Answer: A) 324

2. Total CP = 400 + 50 = ₹450
SP = ₹500
Profit = 500 − 450 = ₹50
Answer: C) ₹50

3. CP = ₹600, SP = ₹540
Loss = 60
Loss % = (60/600) × 100 = 10%
Answer: C) 10%

4. Total CP = 2000 + 200 = ₹2200
SP = ₹2500
Profit = 300
Profit % = (300/2200) × 100 = 13.64%
Answer: C) 13.64%

5. CP = ₹240, SP = ₹270
Profit = 30
Answer: B) Profit ₹30

6. CP of 10 pens = ₹100
SP of 10 pens = 10 × 12 = ₹120
Profit = 20
Answer: C) ₹20

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. Marked Price = ₹1000, Discount = ₹100
SP = 1000 − 100 = ₹900
CP = ₹800
Profit = 900 − 800 = ₹100
Profit % = (100 / 800) × 100 = 12.5%
Correct Answers: A, B, C

8. SP = ₹400, Loss = 20%
CP = 400 × (100/80) = ₹500
Loss = 500 − 400 = ₹100
Loss % = 20% (given, not 25%)
Correct Answers: A, B

9. Total CP = ₹1000
SP per item = ₹120 → Total SP = 120 × 10 = ₹1200
Profit = 1200 − 1000 = ₹200
Profit % = (200 / 1000) × 100 = 20%
CP per item = 1000/10 = ₹100
Correct Answers: A, B, C

COMPREHENSION TYPE

PASSAGE

10. CP = ₹300, Profit = ₹60
SP = CP + Profit = 300 + 60 = ₹360
Correct Answer: C) ₹360

11. Profit = ₹60, CP = ₹300
Profit % = (60 / 300) × 100 = 20%
Correct Answer: C) 20%

12. New Profit = ₹30, CP = ₹300
SP = 300 + 30 = ₹330
Correct Answer: C) ₹330

MATRIX MATCHING TYPE

13. CP = ₹20, SP = ₹25 → Profit = ₹5
Matches B (Profit of ₹5)
CP = ₹200, SP = ₹180 → Loss = ₹20
Matches D (Loss of ₹20)
CP = ₹150, SP = ₹150 → No profit no loss
Matches C
CP = ₹50, SP = ₹40 → Loss = ₹10
Matches A (Loss of ₹10)

INTEGER TYPE

14. SP = ₹120, Profit = ₹20
CP = 120 − 20 = ₹100
Answer: ₹100

15. CP = ₹150, SP = ₹180
Profit = 30
Profit % = (30 / 150) × 100 = 20%
Answer: 20%

SIMPLE AND COUMPOUND INTREST

LEVEL – 1

1.

Amount = 1650, Principal = 1500 ⇒ SI = 150

$$R = \frac{SI \times 100}{P \times T} = \frac{150 \times 100}{1500 \times 1} = 10\%$$

Answer: B) 10%

2.

$$P = \frac{SI \times 100}{R \times T} = \frac{240 \times 100}{6 \times 2} = 2000$$

Answer: C) ₹2000

3.

$$SI = \frac{2000 \times 6 \times 2}{100} = 240$$

Amount = 2000 + 240 = ₹2240
Answer: B) ₹2240

4.

$$P = \frac{9600 \times 100}{16 \times 5} = 12000$$

Answer: B) ₹12000

5.

Original SI = 2600 − 2000 = 600

$$R = \frac{600 \times 100}{2000 \times 5} = 6\%$$

New rate = 9%

$$SI = \frac{2000 \times 9 \times 5}{100} = 900$$

Amount = 2000 + 900 = ₹2900

Answer: B) ₹2900

6.

Difference = CI − SI = 645 − 600 = 45

For 2 years:

$$\text{Difference} = \frac{PR^2}{10000}$$

Also:

$$600 = \frac{P \times R \times 2}{100} \Rightarrow PR = 30000$$

Substitute:

$$45 = \frac{30000 \times R}{10000} = 3R \Rightarrow R = 15\%$$

Answer: B) 15%

7.

Amount becomes 6P ⇒ Interest = 5P

$$SI = \frac{P \times R \times T}{100} \Rightarrow 5P = \frac{P \times R \times 40}{100}$$

$$5 = \frac{40R}{100} \Rightarrow R = 12.5\%$$

Answer: A) 12.5%

8.

$$T = \frac{SI \times 100}{P \times R} = \frac{81 \times 100}{900 \times 4.5} = 2 \text{ years}$$

Answer: A) 2 years

9.

Quarterly ⇒ Rate per quarter = 6/4 = 1.5%, Time = 3 quarters

$$A = 5000\left(1+\frac{1.5}{100}\right)^3$$

$$A = 5000(1.015)^3 \approx 5228.39$$

CI = 5228.39 − 5000 = ₹228.39

Answer: B) ₹228.39

10.

Half-yearly ⇒ Rate = 4/2 = 2%, Time = 4 periods
$A = 20000(1.02)^4 \approx 21648.64$

CI = 21648.64 − 20000 = ₹1648.64
Answer: A) ₹1648.64

11.

$A = 8500(1.04)^2 = 8500 \times 1.0816 = 9193.6$

CI = 9193.6 − 8500 = ₹693.6
Answer: B) ₹693.6

12.

$A = P\left(1+\frac{R}{100}\right)^2 \Rightarrow 583.20 = 500\left(1+\frac{R}{100}\right)^2$
$\frac{583.20}{500} = (1+\frac{R}{100})^2 = 1.1664 \Rightarrow 1+\frac{R}{100} = 1.08 \Rightarrow R = 8\%$

Answer: C) 8%

13.

$A_2 = P(1+r)^2 = 7000, A_3 = P(1+r)^3 = 8000$
$\frac{A_3}{A_2} = 1 + r = \frac{8000}{7000} = \frac{8}{7}$
$P = \frac{7000}{(8/7)^2} = 7000 \times \frac{49}{64} = 5359.37$

Answer: C) ₹5359.37

14.

Difference (CI − SI for 2 years):
$= \frac{PR^2}{10000}$
$530 = \frac{P \times 10^2}{10000} = \frac{P}{100} \Rightarrow P = 53000$

Answer: A) ₹53000

15.

$A = 7500(1.04)^2 = 7500 \times 1.0816 = 8112$

Answer: A) ₹8112

16.

Half-yearly ⇒ Rate = 4/2 = 2%, Time = 1.5 × 2 = 3periods
$A = 5000(1.02)^3 = 5000 \times 1.061208 = 5306.04$

CI = 5306.04 − 5000 = ₹306.04
Answer: A) ₹306.04

17.

Discount % = $\frac{1500}{20000} \times 100 = 7.5\%$

Answer: A) 7.5

18.

Selling price after 15% discount ⇒ SP = 85% of MP
$MP = \frac{680 \times 100}{85} = 800$

Answer: A) 800

19.

$\text{Tax} = \frac{10}{100} \times 500 = 50$

Answer: B) ₹50

20.

$\text{SP} = \frac{90 \times 100}{10} = 900$

Answer: C) ₹900

LEVEL – 2

1. Total SI = ₹220

$SI = \frac{P \times R \times T}{100}$

From SBI:

$\frac{500 \times R \times 2}{100} = 10R$

From ICICI:

$\frac{300 \times R \times 4}{100} = 12R$

Total:

$10R + 12R = 22R = 220 \Rightarrow R = 10\%$

Answer: A) 10%

2.

Total SI = ₹11400

$SI = \frac{P}{100}(6 \times 2 + 9 \times 3 + 14 \times 4)$

$= \frac{P}{100}(12 + 27 + 56) = \frac{P}{100}(95)$

$11400 = \frac{95P}{100} \Rightarrow P = \frac{11400 \times 100}{95} = 12000$

Answer: C) ₹12000

3. Find the Principal

Amount after 3 years = ₹9600

Amount after 5 years = ₹11200

Difference (interest for 2 years):

$11200 - 9600 = 1600$

So, interest per year = $1600 \div 2 = 800$

Interest for 3 years = $3 \times 800 = 2400$

Principal:

$9600 - 2400 = 7200$

Answer B: ₹7200

4. Rate of Interest (C.I – S.I for 2 years)

Formula: Difference = $P \times \left(\frac{R}{100}\right)^2$

Given:

$392 = 20000 \times \left(\frac{R}{100}\right)^2$

$\frac{392}{20000} = \left(\frac{R}{100}\right)^2$

$0.0196 = \left(\frac{R}{100}\right)^2$

$\frac{R}{100} = 0.14 \Rightarrow R = 14\%$

Answer C: 14%

5. Compound Interest on ₹8000 at 5% for 3 years

Formula:

$A = P\left(1+\frac{R}{100}\right)^n$

$A = 8000(1.05)^3$

$A = 8000 \times 1.157625 = 9261$

Compound Interest:

$9261 - 8000 = 1261$

Answer A: ₹1261

6. Compound Interest on ₹20000 at 10% for 3 years

$A = 20000(1.1)^3 = 20000 \times 1.331 = 26620$

CI:

$26620 - 20000 = 6620$

Answer A: ₹6620

7. Time for money to become 9 times

Given:

In 3 years → amount becomes 3 times

$(1 + r)^3 = 3$

We want:

$(1 + r)^t = 9 = 3^2$

So, $(1 + r)^t = (1 + r)^6$

Thus, $t = 6$ years

Answer A: 6 years

8. Sales Price after 18% Discount

Marked Price (MP) = ₹279

Discount = 18%

SP = $MP \times \left(1-\frac{18}{100}\right)$

$= 279 \times 0.82 = 228.78$

Answer D: ₹228.78

9. Selling Price after 12% Discount

MP = ₹1320

Discount = 12%

SP = $1320 \times \left(1-\frac{12}{100}\right)$

$= 1320 \times 0.88 = 1161.60$

Answer C: ₹1161.60

10. Sale Price after 20% Discount

MP = ₹1675

Discount = 20%

SP = $1675 \times \left(1-\frac{20}{100}\right)$

$= 1675 \times 0.80 = 1340$

Answer B: ₹1340

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. ₹2000 at 10% p.a., compounded half-yearly (1 year)
Half-yearly rate:
$\frac{10\%}{2} = 5\%$
Number of periods:
1 year = 2 half-years
Amount:
$A = 2000(1.05)^2 = 2000 \times 1.1025 = 2205$

Compound Interest:
$2205 - 2000 = 205$

A) Correct (rate = 5%)
B) Correct (time periods = 2)
C) Correct (C.I ≈ ₹205)
D) Wrong (Amount ≠ ₹2100, actual = ₹2205)
Correct Answers: A, B, C

12. S.I = ₹300, R = 5%, T = 2 years
Formula:
$\text{S.I} = \frac{P \times R \times T}{100}$
$300 = \frac{P \times 5 \times 2}{100}$
$300 = \frac{10P}{100} = 0.1P \Rightarrow P = 3000$

A) Correct (P = ₹3000)
C) Correct (formula is right)
Correct Answers: A, C

13. MP = ₹500, Discount = 10%
Discount:
10% of 500 = 50
Selling Price:
$500 - 50 = 450$
A) Correct (Discount = ₹50)
B) Correct (SP = ₹450)
Correct Answers: A, B

COMPREHENSION TYPE

PASSAGE

14. Selling Price of the Shirt
Marked Price (MP) = ₹1000
Discount = ₹200
SP = MP − Discount = $1000 - 200 = 800$
Answer B: ₹800

15. Discount Percentage
$\text{Discount \%} = \frac{200}{1000} \times 100 = 20\%$
Answer C: 20%

16. Discount Percentage (₹250 on ₹1000)
$\text{Discount \%} = \frac{250}{1000} \times 100 = 25\%$

Answer C: 25%

MATRIX MATCHING TYPE

17. To solve all Simple Interest questions, we use the core formula:

$$SI = \frac{P \times R \times T}{100}$$

Let's solve each:

$R = \frac{SI \times 100}{P \times T} = \frac{200 \times 100}{1000 \times 2} = 10\%$

Match: C (10%)

$$T = \frac{SI \times 100}{P \times R} = \frac{300 \times 100}{1000 \times 10} = 3 \text{ years}$$

Match: D (3 years)

$R = \frac{700 \times 100}{3500 \times 5} = 4\%$Match: A (4%)

$$T = \frac{200 \times 100}{1000 \times 10} = 2 \text{ years}$$

Match: B (2 years)

INTEGER TYPE

18. Rate of Interest

$$R = \frac{400 \times 100}{2000 \times 2} = 10\%$$

Answer: 10%

19. Compound Amount

Formula:

$$A = P(1 + \frac{R}{100})^T$$

$A = 1000(1.1)^2 = 1000 \times 1.21 = 1210$

Answer: ₹1210

20. Discount Percentage

$$\text{Discount \%} = \frac{50}{500} \times 100 = 10\%$$

Answer: 10%

STUDENT PRACTICE CORNER

1. Simple Interest on ₹1000 at 10% for 2 years

$$SI = \frac{P \times R \times T}{100}$$

$$SI = \frac{1000 \times 10 \times 2}{100} = 200$$

Answer B: ₹200

2. Find Time when SI = ₹100, P = ₹500, R = 10%

$$100 = \frac{500 \times 10 \times T}{100}$$

$100 = 50T \Rightarrow T = 2$

Answer C: 2 years

3. Compound Interest on ₹1000 at 10% for 2 years

$$A = P\left(1 + \frac{R}{100}\right)^T$$

$A = 1000(1.1)^2 = 1210$

$CI = 1210 - 1000 = 210$

Answer B: ₹210

4. Amount on ₹2000 at 5% for 2 years (Compound Interest)

$$A = 2000(1.05)^2 = 2205$$

Answer D: ₹2205

5. Discount on a pen
Marked Price (MP) = ₹20
Discount = 15%
Discount $= \frac{15}{100} \times 20 = 3$
Answer B: ₹3

6. Discount on a moped
Selling Price (SP) = ₹13200
Discount = 12% on Marked Price (MP)
Let MP = x
$SP = x - 12\%$ of $x = 0.88x$
$0.88x = 13200 \Rightarrow x = 15000$
Discount $= 15000 - 13200 = 1800$
Answer D: ₹1800

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. ₹2000 at 10% p.a. for 2 years (Simple Interest)
$SI = \frac{P \times R \times T}{100}$
$SI = \frac{2000 \times 10 \times 2}{100} = 400$
Amount $= 2000 + 400 = 2400$
Correct options: A) S.I = ₹400 and B) Amount = ₹2400

8. ₹1000 at 10% for 2 years (Compound Interest)
$A = P\left(1+\frac{R}{100}\right)^T$
$A = 1000(1.1)^2 = 1210$
$CI = 1210 - 1000 = 210$
Correct options: A) Amount = ₹1210 and B) C.I = ₹210

9. Selling Price = ₹1000, Tax = 18%
Tax $= \frac{18}{100} \times 1000 = 180$
Final Price $= 1000 + 180 = 1180$
Correct options: A) Tax = ₹180 and B) Final price = ₹1180

COMPREHENSION TYPE

PASSAGE
Given:
Principal (P) = ₹5000
Rate (R) = 10% p.a.
Time (T) = 2 years
Interest type = Simple Interest

10. Simple Interest earned
$SI = \frac{P \times R \times T}{100}$
$SI = \frac{5000 \times 10 \times 2}{100} = 1000$

Answer B: ₹1000

11. Total amount Ravi repays
Amount $= P + SI = 5000 + 1000 = 6000$

Answer C: ₹6000

12. New simple interest at 12%

$$SI = \frac{5000 \times 12 \times 2}{100} = 1200$$

Answer C: ₹1200

MATRIX MATCHING TYPE

13. Match the following (Discount = MP − SP)

₹1000 − ₹800 = ₹200 → A
₹600 − ₹540 = ₹60 → B
₹1200 − ₹900 = ₹300 → C
₹800 − ₹720 = ₹80 → D

INTEGER TYPE

14. Find the Principal

$$SI = \frac{P \times R \times T}{100}$$

$$500 = \frac{P \times 5 \times 2}{100}$$

$$500 = \frac{10P}{100} = \frac{P}{10} \Rightarrow P = 5000$$

Answer: ₹5000

15. Find the Discount

Discount $= MP - SP = 1200 - 900 = 300$

Answer: ₹300

3. Linear Equations

Linear equation one variable

Level – 1

1. Correct Answer: C) 5

Explanation:

Given, $3P + 1 = 16$

$3P = 16 - 1 = 15$

$P = \frac{15}{3} = 5$

2. Correct Answer: A) 3

Explanation:

Given, $7z - 5 = 16$

$7z = 16 + 5 = 21$

$z = \frac{21}{7} = 3$

3. Correct Answer: A) 15

Explanation:

Given, $\frac{x}{3} - 5 = 0$

$\frac{x}{3} = 5$

$x = 5 \times 3 = 15$

4. Correct Answer: C) 9

Explanation:

Given, $x + 3 = 12$

$x = 12 - 3 = 9$

5. Correct Answer: D) 1

Explanation:

Given, $6(2a - 1) + 8 = 14$

$12a - 6 + 8 = 14$

$12a + 2 = 14$

$12a = 12$

$a = 1$

6. Correct Answer: A) 2

Explanation:

Given, $16(x + 7) = 144$

$x + 7 = \frac{144}{16} = 9$

$x = 9 - 7 = 2$

7. Correct Answer: D) 8

Explanation:

Let the number be x. $6x = 48$

$x = \frac{48}{6} = 8$

8. Correct Answer: C) 6

Explanation:

Given, $6a = 30$

To find a, divide both sides by 6.

9. Correct Answer: B) 8

Explanation:

Given, $y - 8 = 6$

Add 8 on both sides:$y = 6 + 8 = 14$

10. Correct Answer: A) b + x

Explanation:

Given $a = b$.

Substituting b in place of a: $a + x = b + x$

11. Correct Answer: A) 3

Explanation:

Given, $4x - 1 = 2x + 5$

$4x - 2x = 5 + 1$

$2x = 6$

$x = 3$

12. Correct Answer: C) $\frac{21}{16}$

Explanation:

Given, $8t + \frac{5}{2} = 13$

$8t = 13 - \frac{5}{2}$

$8t = \frac{26 - 5}{2} = \frac{21}{2}$

$t = \frac{21}{2} \div 8 = \frac{21}{16}$

13. Correct Answer: C) 0

Explanation:

Given, $11n + 1 = 1$

$11n = 0$

$n = 0$

14. Correct Answer: A) $\frac{1}{5}$

Explanation:

Given, $2x + \frac{4}{5} = x + 1$

$2x - x = 1 - \frac{4}{5}$

$x = \frac{5}{5} - \frac{4}{5} = \frac{1}{5}$

15. Correct Answer: A) 1.2

Explanation:

Given, $0.8 + \frac{x}{4} = 0.5x + \frac{1}{2}$

$0.8 - 0.5 = 0.5x - \frac{x}{4}$

$0.3 = \frac{x}{2} - \frac{x}{4}$

$0.3 = \frac{x}{4}$

$x = 0.3 \times 4 = 1.2$

16. Correct Answer: D) 1

Explanation:

Given, $15(L - 1) + 4(L + 3) = 2(7 + L)$

$15L - 15 + 4L + 12 = 14 + 2L$

$19L - 3 = 14 + 2L$
$19L - 2L = 14 + 3$
$17L = 17$
$L = 1$

17. Correct Answer: C) 3

Explanation:
Given, $8(C - 3) - (6 - 2C) = 2(C + 2) - 5(5 - C)$
$8C - 24 - 6 + 2C = 2C + 4 - 25 + 5C$
$10C - 30 = 7C - 21$
$10C - 7C = -21 + 30$
$3C = 9$
$C = 3$

18. Correct Answer: A) $-\frac{5}{6}$

Explanation:
Given, $5(I + 2) - 2(3 - 4I) = 3(I + 5) - 4(4 - I)$
Expand both sides:
$5I + 10 - 6 + 8I = 3I + 15 - 16 + 4I$
$13I + 4 = 7I - 1$
$13I - 7I = -1 - 4$
$6I = -5$
$I = -\frac{5}{6}$

19. Correct Answer: B) 2

Explanation:
Given, $\frac{5x+2}{2x+3} = \frac{12}{7}$
Cross multiply: $7(5x + 2) = 12(2x + 3)$
$35x + 14 = 24x + 36$
$35x - 24x = 36 - 14$
$11x = 22$
$x = 2$

20. Correct Answer: C) – 3

Explanation:
Given, $\frac{x+7}{3x+16} = \frac{4}{7}$
Cross multiply:
$7(x + 7) = 4(3x + 16)$
$7x + 49 = 12x + 64$
$49 - 64 = 12x - 7x$
$-15 = 5x$
$x = -3$

LEVEL – II

Single option correct type

1. Correct Answer: B) 3

Explanation:
Given, $\frac{x}{3} - \frac{1}{4} = \frac{3}{2} - \frac{x}{4}$
Multiply both sides by 12: $4x - 3 = 18 - 3x$
$4x + 3x = 18 + 3$
$7x = 21,\ x = 3$

2. Correct Answer: D) –2

Explanation:

Given, $\frac{2x-1}{2x-3} = \frac{3x+1}{3x-1}$

Cross multiply: $(2x-1)(3x-1) = (3x+1)(2x-3)$

$6x^2 - 2x - 3x + 1 = 6x^2 - 9x + 2x - 3$

$6x^2 - 5x + 1 = 6x^2 - 7x - 3$

$2x = -4$

$x = -2$

3. Correct Answer: A) 4

Explanation:

Given, $(x+2)^2 = x^2 + ax + 4$

Expand: $x^2 + 4x + 4 = x^2 + ax + 4$

Comparing coefficients: $a = 4$

4. Correct Answer: B) $\frac{20}{3}$

Explanation:

Given, $(5x-1)(9x+1) = 3a(3x-1)$

When $x = 1$: $(5-1)(9+1) = 3a(3-1)$

$4 \times 10 = 6a$

$40 = 6a$

$a = \frac{40}{6} = \frac{20}{3}$

5. Correct Answer: C) 3

Explanation:

Given, $\frac{3x+5}{3x-5} = \frac{7}{2}$

Cross multiply: $2(3x+5) = 7(3x-5)$

$6x + 10 = 21x - 35$

$45 = 15x$

$x = 3$

6. Correct Answer: D) 0

Explanation:

Given, $x + \frac{1}{x} = 2$

Squaring both sides: $\left(x + \frac{1}{x}\right)^2 = 4$

$x^2 + \frac{1}{x^2} + 2 = 4$

$x^2 + \frac{1}{x^2} - 2 = 0$

7. Correct Answer: **C) 2**.

Explanation:

Given, $\frac{10z+8}{2z} = 7$

$10z + 8 = 14z$

$8 = 4z$

$z = 2$

8. Correct Answer: A) 4

Explanation:

Given, $3y - 3 = 4y - 7$

$-3 + 7 = 4y - 3y$

$4 = y$

9. Correct Answer: A) 1

Explanation:

Given, $2x - 1 = \frac{3x-1}{2}$

Multiply both sides by 2:$4x - 2 = 3x - 1$

$4x - 3x = -1 + 2$

$x = 1$

10. Correct Answer: B) $\frac{1}{2}$

Explanation:

Given, $8(2p - 5) - 6(3p - 7) = 1$

Expand:$16p - 40 - 18p + 42 = 1$

$-2p + 2 = 1$

$-2p = -1$

$p = \frac{1}{2}$

One or more than one correct type

11. Correct Answers: A, C

Explanation:

A linear equation in one variable has only one variable with highest power 1.

A) $2x + 5 = 11$ Linear in one variable

C) $7y - 4 = 10$ Linear in one variable

12. Correct Answers: A, C

Explanation:

A) A variable represents an unknown number.

C) A term is separated by operators.

13. Correct Answers: A, C, D

Explanation:

A) Adding the same number on both sides (Keeps the equation balanced.)

C) Multiplying both sides by the same number (Keeps the equation balanced.)

D) Dividing both sides by the same non-zero number (Keeps the equation balanced.)

14. Correct Answers: A, B, D

Explanation:

A) $3x + 7$ Algebraic expression

B) $5a - 2b$ Algebraic expression

D) $9y$ Algebraic expression

Comprehension Passage type

15. Correct Answer: A) $22 = 14 + b$

Explanation: Given, $44 = 2(14 + b)$

Divide both sides by 2: $22 = 14 + b$

16. Correct Answer: C) 8 cm

Explanation:

From, $22 = 14 + b$

$b = 22 - 14 = 8$

So, the breadth is 8 cm.

Matching type

17. Correct Matching:

a) Variable → Q) Unknown quantity

b) Constant → P) Fixed numerical value

c) Coefficient → R) Number multiplied with variable

d) Equation → S) Statement showing equality

So the correct matching is:

$a \to Q, b \to P, c \to R, d \to S$

Integer type

18. Answer: 12

Explanation: $x + 9 = 21$

$x = 21 - 9 = 12$

19. Answer: 9

Explanation: $5y = 45$

$y = \frac{45}{5} = 9$

20. Answer: 5

Explanation: $3a + 5 = 20$

$3a = 15$

$a = \frac{15}{3} = 5$

Student Practice Corner

Single option correct type

1. Correct Answer: C) $\frac{9}{23}$

Explanation:

Given,

$\frac{8p - 5}{7p + 1} = \frac{-2}{4} = -\frac{1}{2}$

Cross multiply:

$2(8p - 5) = -(7p + 1)$

$16p - 10 = -7p - 1$

$16p + 7p = 10 - 1$

$23p = 9$

$p = \frac{9}{23}$

2. Correct Answer: C) 12

Explanation:

Given, $7x - 81 = 3$

$7x = 84$

$x = 12$

3. Correct Answer: A) -2

Explanation:

Given, $7.5x + 4 = 6.5x + 2$

$7.5x - 6.5x = 2 - 4$

$x = -2$

4. Correct Answer: D) $\frac{88}{27}$

Explanation:

Given, $0.5x + 0.3 = 0.23x + 1.18$

$0.5x - 0.23x = 1.18 - 0.3$

$0.27x = 0.88$

$x = \frac{0.88}{0.27} = \frac{88}{27}$

5. Correct Answer: D) 40

Explanation:

Let the number be x.

$\frac{x}{8} = 5$

$x = 5 \times 8 = 40$

6. Correct Answer: A) 1

Explanation:

Given, $3(m + 6) = 21$

$m + 6 = 7$

$m = 1$

7. Correct Answer: B) 2.5

Explanation:

Given $q = 2$and $pq - 3 = 2$

Substitute $q = 2$:

$2p - 3 = 2$

$2p = 5$

$p = 2.5$

8. Correct Answer: D) -1

Explanation:

Given,

$8m + 9 = 7m + 8$

$8m - 7m = 8 - 9$

$m = -1$

One or More Than One Type

9. Correct Answers: A, B, C

Explanation:

A) $+5$becomes -5on the other side

B) -3becomes $+3$on the other side

C) $\times 4$becomes $\div 4$

10. Correct Answers: A, C

Explanation:

A) $3x + 5 = x + 9$ Variable on both sides

C) $5a - 2 = 3a + 6$ Variable on both sides

Comprehension Passage Type

11. Correct Answer: B) 5

Explanation:

Given, $3x + 7 = 22$

Subtract 7 from both sides:$3x = 15$

Divide by 3:$x = 5$

12. **Correct Answer: C) Subtracting Property**
Explanation:
Ravi subtracted 7 from both sides of the equation.
This uses the **Subtracting Property of Equality**.

Matching Type

13. Correct Matching:
a) Addition Property → **Q) Add same number on both sides**
b) Division Property → **R) Divide both sides by same non-zero number**
c) Multiplication Property → **P) Multiply both sides by same number**
d) Subtracting Property → **S) Subtract same number on both sides**
So the correct matching is: $a \to Q, b \to R, c \to P, d \to S$

Integer Type:

14. Answer: **9**
Explanation:
Given, $4m = 36$
$m = \frac{36}{4} = 9$

15. **Answer: 32**
Explanation:
Let the number be x.
$x + 18 = 50$
$x = 50 - 18 = 32$

Linear equations in two variables

Level – 1

1. **Correct Answer: B) 5**
$6 - 6x + 9 = 30 - 5x - 4x$
$15 - 6x = 30 - 9x$
$-6x + 9x = 30 - 15$
$3x = 15$
$x = 5$

2. **Correct Answer: B) – 1**
Cross multiply: $(2x + 3)(3x - 2) = (2x - 3)(3x + 4)$
$6x^2 + 5x - 6 = 6x^2 - x - 12$
$5x + x = -12 + 6$
$6x = -6, \quad x = -1$

3. **Correct Answer: D) 18.3**
$0.8x - 0.32 = 0.4x + 7$
$0.8x - 0.4x = 7 + 0.32$
$0.4x = 7.32$
$x = \frac{7.32}{0.4} = 18.3$

4. **Correct Answer: A) 4**
Cross multiply:$10(2x + 1) = 9(3x - 2)$
$20x + 10 = 27x - 18$
$10 + 18 = 27x - 20x$
$28 = 7x, \; x = 4$

5. Correct Answer: C) -10

Simplify left side: $\frac{3x-(2x+8)}{3} = \frac{x-8}{3}$

Simplify bracket: $x - \frac{2-x}{6} = \frac{6x-(2-x)}{6} = \frac{7x-2}{6}$

So, $\frac{1}{4} \cdot \frac{7x-2}{6} - 3 = \frac{7x-2}{24} - 3$

Equation becomes: $\frac{x-8}{3} = \frac{7x-2}{24} - 3$

Multiply by 24: $8(x-8) = 7x - 2 - 72$

$8x - 64 = 7x - 74x = -10$

6. Correct Answer: D) - 3

$$\frac{4x+6+2x-3}{2x+3} = \frac{3x+4}{x+2}$$

$$\frac{6x+3}{2x+3} = \frac{3x+4}{x+2}$$

Cross multiply: $(6x+3)(x+2) = (3x+4)(2x+3)$

$6x^2 + 15x + 6 = 6x^2 + 17x + 12$

$15x - 17x = 12 - 6$

$-2x = 6$

$x = -3$

7. Correct Answer: A) 23

Use identity: $\left(x+\frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2$

$$25 = x^2 + \frac{1}{x^2} + 2$$

$$x^2 + \frac{1}{x^2} = 23$$

8. Correct Answer: A) 2207

First find: $x^2 + \frac{1}{x^2} = 3^2 - 2 =$

Then, $x^4 + \frac{1}{x^4} = 7^2 - 2 = 47$

Next, $x^8 + \frac{1}{x^8} = 47^2 - 2 = 2207$

9. Correct Answer: C) $\frac{27}{10}$

$$\frac{x}{2} - \frac{x}{3} = \frac{1}{4} + \frac{1}{5}$$

$$\frac{3x-2x}{6} = \frac{5+4}{20}$$

$$\frac{x}{6} = \frac{9}{20}$$

$$x = \frac{54}{20} = \frac{27}{10}$$

10. Correct Answer: B) 12

Solve: $\frac{x}{4} + \frac{x}{6} = x - 7$

LCM of 4 and 6 is 12: $\frac{3x+2x}{12} = x - 7$

$$\frac{5x}{12} = x - 7$$

Multiply by 12: $5x = 12x - 84$

$84 = 7x$

$x = 12$

11. Correct Answer: A) 19

Let number of students be x.

Each gets 2 chocolates.$2x + 7 = 45$

$2x = 38$

$x = 19$

12. Correct Answer: C) 45, 120

Ratio = $3:8$

Let numbers be $3x$and $8x$.

$3x + 8x = 165$

$11x = 165$

$x = 15$

Numbers: $3x = 45$,$8x = 12$

13. Correct Answer: A) 7, 8

Let numbers be x and y.

$x + y = 15$

$x^2 + y^2 = 113$

Check options:

For 7,8:$7 + 8 = 15$

$7^2 + 8^2 = 49 + 64 = 113$

14. Correct Answer: D) 20, 22

Let the consecutive even integers be x and $x + 2$.

$\frac{2}{5}x = \frac{2}{11}(x + 2) + 4$

$22x = 10x + 240$

$12x = 240 \Rightarrow x = 20$

Integers are 20,22.

15. Correct Answer: B) 40

Sides are in ratio $3:4$, so let them be $3x, 4x$.

Using Pythagoras theorem, hypotenuse = $5x$.

$3x + 4x + 5x = 96$

$12x = 96 \Rightarrow x = 8$

Largest side = $5x = 40$.

16. Correct Answer: D) 7

Given: $x + y = 3, xy = 2$

$(x - y)^2 = (x + y)^2 - 4xy = 9 - 8 = 1$

$x - y = 1$

$x^3 - y^3 = (x - y)(x^2 + xy + y^2) = 1(5 + 2) = 7$

17. Correct Answer: A) (5, -2)

Equations:$4x + 3y = 14$

$9x - 5y = 55$

Solving, we get:$x = 5, y = -2$

Hence solution is $(5, -2)$.

18. Correct Answer: A) 4

$3^{x+y} = 27 = 3^3 \Rightarrow x + y = 3$

$3^{x-y} = 243 = 3^5 \Rightarrow x - y = 5$

Adding:$2x = 8 \Rightarrow x = 4$

19. Correct Answer: B) (-1, 5)

Given: $2y + \frac{7}{x} = 3$

$y - \frac{3}{x} = 8$

Check option $(-1,5)$: $2(5) + \frac{7}{-1} = 10 - 7 = 3$

$5 - \frac{3}{-1} = 5 + 3 = 8$

Both satisfy.

20. Correct Answer: B) $\frac{1}{4}$

Let: $a = \frac{1}{x}, b = \frac{1}{y}$

Then: $\frac{a}{2} - b = -1, a + \frac{b}{2} = 8$

Solving: $b = 4$

$\frac{1}{y} = 4 \Rightarrow y = \frac{1}{4}$

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Correct Answer: C) $\frac{3}{2}$

Let $r = \frac{p}{q}$. Since $p, q > 0$, we have $r > 0$.

Given:

$$\frac{4p + 9q}{p} = \frac{5q}{p - q}$$

Substitute $p = rq$:

$$\frac{4rq + 9q}{rq} = \frac{5q}{rq - q}$$

$$\frac{4r + 9}{r} = \frac{5}{r - 1}$$

Cross-multiply:

$(4r + 9)(r - 1) = 5r$

$4r^2 + 5r - 9 = 5r$

$4r^2 - 9 = 0$

$(2r - 3)(2r + 3) = 0$

Since $r > 0$,

$r = \frac{3}{2}$

Therefore,

$$\boxed{\frac{p}{q} = \frac{3}{2}}$$

2. Correct Answer: B) $\frac{x}{y}$

Given, $\frac{x - \frac{1}{y}}{y - \frac{1}{x}}$

Multiply the numerator and denominator by xy:

$$= \frac{xy\left(x-\frac{1}{y}\right)}{xy\left(y-\frac{1}{x}\right)} = \frac{x^2y - x}{xy^2 - y}$$

Factor: $= \frac{x(xy-1)}{y(xy-1)}$

Since $y - \frac{1}{x} \neq 0$, we have $xy - 1 \neq 0$. Hence, $= \frac{x}{y}$

$$\boxed{\frac{x-\frac{1}{y}}{y-\frac{1}{x}} = \frac{x}{y}}$$

3. Correct Answer: B) 36, 11

Let 25 paise coins = x, 50 paise coins = y.
$x + y = 47$
$25x + 50y = 1450$
Divide by 25:$x + 2y = 58$
Subtract equations: $y = 11$
$x = 36$
Hence coins are 36 and 11.

4. Correct Answer: C) 11

Given: $10x + 3y = 75$
Checking options:
For $x + y = 11$,
Take $x = 3$, $y = 8$
$10(3) + 3(8) = 30 + 24 = 54$
Try solving generally: $y = \frac{75-10x}{3}$
For integer values, $x = 6$: $y = 5$
$x + y = 11$

5. Correct Answer: C) $\frac{2}{3}$

Let numerator = x, denominator = $x + 1$.
After subtracting 1 from both: $\frac{x-1}{x} = \frac{1}{2}$
$2(x - 1) = x$
$x = 2$
Fraction: $\frac{x}{x+1} = \frac{2}{3}$

6. Correct Answer: B) 6

Let goats = x, hens = y.
Heads: $x + y$
Legs: $4x + 2y$
Given: $4x + 2y = 2(x + y) + 12$
$4x + 2y = 2x + 2y + 12$
$2x = 12$
$x = 6$

7. Correct Answer: B) 3

Point $(1, 4)$satisfies both equations.

Substitute in: $2x + by = 6$

$2(1) + 4b = 6$

$4b = 4 \Rightarrow b = 1$

Now: $3y = 8 + ax$

$3(4) = 8 + a(1)$

$12 = 8 + a$

$a = 4$

$a - b = 4 - 1 = 3$

8. Correct Option: D) 5

$4x - 5y = 42$

Multiply (1) by 5 and (2) by 2: $15x + 10y = 100$

$8x - 10y = 84$

Add them: $23x = 184 \Rightarrow x = 8$

Substitute x: $3(8) + 2y = 20$

$24 + 2y = 20$

$2y = -4 \Rightarrow y = -2$

We are told: $x = a + b, y = a - b$

So, $a + b = 8$

$a - b = -2$

Subtract the second from the first:

$(a + b) - (a - b) = 8 - (-2)$

$2b = 10$

$b = 5$

9. Correct Answer: B) 8

Let numerator = x, denominator = y.

Given: $\frac{x+1}{y-1} = 1$

$x + 1 = y - 1$

$y = x + 2$

Also, $\frac{x}{y+1} = \frac{1}{2}$

$2x = y + 1$

Substitute: $2x = x + 3$

$x = 3, y = 5$

Sum:$x + y = 3 + 5 = 8$

10. Correct Answer: A) $k = 6$

Equations: $kx + 2y = 5$

$3x + y = 1$

For no solution: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

$\frac{k}{3} = \frac{2}{1}$

$k = 6$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answer: B) b D) -a

Given equations: $\frac{ax}{b} - \frac{by}{a} = a + b$

$ax - by = 2ab$

Multiply first equation by ab: $a^2x - b^2y = ab(a + b)$
Second equation: $ax - by = 2ab$
Using cross multiplication/elimination: $a(a^2x - b^2y) - b(ax - by)$
Solving gives:$x = b$

12. Correct Answer: C) 3

Given: $3 < x < 6,2 < y < 4$
To maximize $\frac{x}{y}$:
take greatest $x = 6$
take smallest $y = 2$
$\frac{x}{y} < \frac{6}{2} = 3$
Hence greatest possible value is 3.

13. Correct Answers: A), B), C) ,D)

Given the system:
$3x + 4y = 25,5x - 6y = -9$

Solve the equations:
Multiply the first equation by 3:
$9x + 12y = 75$

Multiply the second equation by 2:
$10x - 12y = -18$

Add:
$19x = 57 \Rightarrow x = 3$

Substitute into $3x + 4y = 25$:
$9 + 4y = 25$
$4y = 16 \Rightarrow y = 4$
Now check the options:

- **A)** $19x = 57$✓ True
- **B)** $4y = 16$✓ True
- **C)** $x = 3$, $y = 4$✓ True
- **D)** $4y = 25 - 9$i.e., $4y = 16$✓ True

COMPREHENSION TYPE

Passage – I:

14. Correct Answer: A) 2

Given: $\frac{1}{2(2x+3y)} + \frac{12}{7(3x-2y)} = \frac{1}{2}$
$\frac{7}{(2x+3y)} + \frac{4}{(3x-2y)} = 2$
Let:$a = 2x + 3y, b = 3x - 2y$

Then: $\frac{1}{2a} + \frac{12}{7b} = \frac{1}{2}$
$\frac{7}{a} + \frac{4}{b} = 2$
Solving gives: $a = 7, b = 4$

So, $2x + 3y = 7$
$3x - 2y = 4$
Multiply first by 2:$4x + 6y = 14$
Multiply second by 3: $9x - 6y = 12$
Add: $13x = 26$
$x = 2$

15. Correct Answer: C) 3

From previous result: $x = 2$
Substitute into:$3x - 2y = 4$
$6 - 2y = 4$
$y = 1$
$x + y = 2 + 1 = 3$

MATRIX MATCHING TYPE

16. Correct Matching:

A) $2x + 3y + 4 = 0$; $5x + 6y + 9 = 0$
Solve:$2x + 3y = -4$
$5x + 6y = -9$
Multiply first by 2: $4x + 6y = -8$
Subtract:$x = -1$
Substitute:$2(-1) + 3y = -4$
$-2 + 3y = -4$
$3y = -2 \Rightarrow y = -\frac{2}{3}$
So solution is:$\left(-1, -\frac{2}{3}\right)$

Matches with **P**

B) $x + 3y = 1$; $2x + 6y = 2$
Second equation is double of first.
Hence infinitely many solutions.
Matches with **R**

C) The solution set for inconsistent system of equations
An inconsistent system has no solution.
Hence solution set is null set.
Matches with **S**

D) $y = mx$, $m \neq 0$passes through
For $x = 0$,
$y = m(0) = 0$
So line passes through origin.
Matches with **Q**

Final Matching: A → P, B → R, C → S, D → Q

INTEGER TYPE

17. Solve for $x + 2y$

Divide both equations by xy:

1. $\frac{6}{y} + \frac{3}{x} = 7$

2. $\frac{3}{y} + \frac{9}{x} = 11$

Let$u = \frac{1}{x}, v = \frac{1}{y}$

Then,$3u + 6v = 7$
$9u + 3v = 11$
Solving this system gives $u = 1, v = \frac{2}{3}$
Therefore,$x = 1, y = \frac{3}{2}$
Calculating $x + 2y$: $1 + 2\left(\frac{3}{2}\right) = 1 + 3 = 4$

18. Correct Answer: 6

Let the number be x. $3x + 6 = 24$
$3x = 18, \quad x = 6$

19. Correct Answer: 2

Given: $x + y = 1$
$kx + y = 2$
For no solution:$\frac{1}{1} = \frac{1}{1} \neq \frac{1}{2}$
So,$k = 1$
$k + 1 = 2$
Correct Answer: 2

20. Correct Answer: 4

Given:$\frac{5x-3}{4} + \frac{y+1}{5} = 2x + y$

Multiply by 20:$25x - 15 + 4y + 4 = 40x + 20y$
$15x + 16y = -11$

Second equation:$\frac{x+11}{6} - \frac{3y-4x}{4} = 2x + 3$
Multiply by 12:
$2x + 22 - 9y + 12x = 24x + 36$
$10x + 9y = -14$

Solve:$15x + 16y = -11$
$10x + 9y = -14$

Multiply second by 3:$30x + 27y = -42$

Multiply first by 2:$30x + 32y = -22$

Subtract:$5y = 20$
$y = 4$

Student Practice Corner

1. Correct Answer: B) 23, 17

Let numbers be x and y.
$x + y = 40$
$x - y = 6$
Adding: $2x = 46 \Rightarrow x = 23$
$y = 17$, Hence numbers are 23,17.

2. Correct Answer: D) a

Given:

$\frac{x}{a}+\frac{y}{b}=2, ax-by=a^2-b^2$

From the first equation:

$bx+ay=2ab$...(1)

Given:

$ax-by=a^2-b^2$...(2)

Multiply (1) by b:

$b^2x+aby=2ab^2$

Multiply (2) by a:

$a^2x-aby=a^3-ab^2$

Add the equations:

$(a^2+b^2)x=a^3+ab^2=a(a^2+b^2)$

$\boxed{x=a}$

3. Correct Answer: A) $\frac{8}{15}$

Let fraction be:$\frac{x}{y}$

Given:$\frac{x-1}{y-1}=\frac{1}{2}$

$2x-y=1$

Also,$\frac{x+2}{y}=\frac{2}{3}$

$3x-2y=-6$

Solving:$x=8, y=15$

Fraction:$\frac{8}{15}$

4. Correct answer A) 4x+y=180,3y−5x=30

Given:

$\angle A=x, \angle B=3x, \angle C=y$

and

$3y-5x=30°$

Using the triangle angle property:

$x+3x+y=180°$

$4x+y=180$

Now we have two linear equations:

$4x+y=180$

$3y-5x=30$

5. Correct Answer: B) 15 years

Present ages:$5x,\ 3x$

After 6 years:$\frac{5x+6}{3x+6}=\frac{7}{5}$

$25x+30=21x+42$

$4x=12$

$x=3$, Present age of A:$5x=15$

6. Correct Answer: A) Rs 34.5

Let cost of pen = x, notebook = y.

$5x + 7y = 82$

$4x + 4y = 52$

Second equation:$x + y = 13$

$x = 13 - y$

Substitute:$5(13 - y) + 7y = 82$

$65 - 5y + 7y = 82$

$2y = 17$

$y = 8.5, x = 4.5$

$2x + 3y = 2(4.5) + 3(8.5) = 34.5$

7. Correct Answer: A) 8

Let pencil cost = x, pen cost = y.

$5x + 7y = 50$

$7x + 5y = 46$

Adding:$12x + 12y = 96$

$x + y = 8$

Hence cost of 1 pencil and 1 pen is Rs. 8.

8. Correct Answer: D) 9

Given: $2x - y = 5$

$3x + 2y = 11$

Multiply first by 2:$4x - 2y = 10$

Add:$7x = 21$

$x = 3$

$2(3) - y = 5$

$y = 1$

$2x + 3y = 6 + 3 = 9$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Correct Answers: A), C)

Let A's and B's one-day works be $a$and b.

$$a + b = \frac{1}{10}$$

Also,

$$12a + 6b = 1 \Rightarrow 2a + b = \frac{1}{6}$$

Subtracting,

$$a = \frac{1}{6} - \frac{1}{10} = \frac{1}{15}$$

So, A alone takes **15 days**.

Then,

$$b = \frac{1}{10} - \frac{1}{15} = \frac{1}{30}$$

So, B alone takes **30 days**.

10. Correct Answers: C), D)

Given:$\sqrt{3x}-\sqrt{2y}=\sqrt{3}$

$\sqrt{5x}+\sqrt{3y}=\sqrt{2}$

Let:$a=\sqrt{x}, b=\sqrt{y}$

Then:$\sqrt{3}\,a-\sqrt{2}\,b=\sqrt{3}$

$\sqrt{5}\,a+\sqrt{3}\,b=\sqrt{2}$

Solving gives:$x=5\sqrt{15}-8\sqrt{6}$

$y=5(\sqrt{10}-3)$

Hence,

$x=5\sqrt{15}-8\sqrt{6}$

$y=5(\sqrt{10}-3)$

COMPREHENSION TYPE

11. Correct Answer: B) -3

Given equations:$2x-ky+3=0$

$4x+6y-5=0$

For parallel lines:$\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$

So,$\frac{2}{4}=\frac{-k}{6}$

$\frac{1}{2}=\frac{-k}{6}$

$-k=3$

$k=-3$

12. Correct Answer: B) $\mathbb{R}-\left\{\frac{13}{3}\right\}$

Given equations:$3x+y=5$

$4x+(k-3)y+7=0$

For intersecting lines:$\frac{a_1}{a_2}\neq\frac{b_1}{b_2}$

So,$\frac{3}{4}\neq\frac{1}{k-3}$

Cross multiply:$3(k-3)\neq 4$

$3k-9\neq 4$

$3k\neq 13$

$k\neq\frac{13}{3}$

Hence,$k\in\mathbb{R}-\left\{\frac{13}{3}\right\}$

13. MATRIX MATCHING TYPE

Let us solve each pair.

A)

$\sqrt{3}\,x+\sqrt{5}\,y=\sqrt{3}, \sqrt{5}\,x+\sqrt{3}\,y=\sqrt{3}$

Subtracting the equations:

$(\sqrt{3}-\sqrt{5})(x-y)=0\Rightarrow x=y$

So,

$(\sqrt{3}+\sqrt{5})x=\sqrt{3}$

$x=y=\dfrac{\sqrt{3}}{\sqrt{3}+\sqrt{5}}=\dfrac{\sqrt{15}-3}{2}$

Hence,

$A\rightarrow\boxed{R}$

B)

Let

$a=2^x, b=3^y$

Then

$a+b=17, 4a-3b=5$

Solving,

$a=8, b=9$

Therefore,

$x=3, y=2$

So,

$\boxed{(x,y)=(3,2)}$

C)

Let

$u=\dfrac{1}{x}, v=\dfrac{1}{y}$

Then

$2u+3v=13, 5u-4v=-2$

Solving,

$u=2, v=3$

Hence,

$x=\dfrac{1}{2}, y=\dfrac{1}{3}$

Therefore,

$C\rightarrow\boxed{Q}$

D)

Let

$u=\dfrac{1}{x-1}, v=\dfrac{1}{y-2}$

Then

$5u+v=2, 6u-3v=1$

Solving,
$u = v = \frac{1}{3}$

Hence,
$x = 4, y = 5$

Therefore,
$D \rightarrow \boxed{T}$

Final Matching: A → R, B → S, C → Q, D → T

INTEGER TYPE

14. Correct Answer: 2

Given:$2x + 3y = 34$
$1 + \frac{x}{y} = \frac{13}{8}$
$\frac{x}{y} = \frac{5}{8}$
$x : y = 5 : 8$
Let:$x = 5k, y = 8$
Substitute:$2(5k) + 3(8k) = 34$
$10k + 24k = 34$
$34k = 34$
$k = 1$
$x = 5, y = 8$
$2x - y = 10 - 8 = 2$

15. Correct Answer: 7

$y + \frac{6}{x} = 6 \Rightarrow xy + 6 = 6x$
Subtract (1) from (2):$2 = 6x - 4y$
$1 = 3x - 2y$
$y = \frac{3x - 1}{2}$
Substitute into $xy + 6 = 6x$:$x\left(\frac{3x-1}{2}\right) + 6 = 6x$
$3x^2 - 13x + 12 = 0$
$(3x - 4)(x - 3) = 0$
Taking $x = 3$,$y = \frac{3(3)-1}{2} = 4$
Therefore,$x + y = 3 + 4 = 7$

4. Geometry

Triangle

LEVEL – I

1. Correct Answer: A) 60°, 75°

An exterior angle of a triangle equals the sum of the two opposite interior angles.
Let the angles be $4x$and $5x$. Then:
$4x + 5x = 135° \Rightarrow 9x = 135° \Rightarrow x = 15°$.
So the angles are 60°and 75°.

2. Correct Answer: D) 60°

Triangle $ABC$is right-angled at B, and $\angle C = 60°$, so $\angle A = 30°$.
Point Dlies on ACsuch that $BD \perp AC$.
Thus, $BD$is the perpendicular from $B$to AC.
Using geometric properties (or coordinate method), the angle between $AB$and BD, i.e., $\angle ABD$, comes out to be 60°.

3. Correct Answer: A) 137°

The triangle is isosceles (marks show $AC = BC$), so base angles at $A$and $B$are equal.
The angle at the top is 94°, so each base angle is:
$\frac{180° - 94°}{2} = 43°$
The angle y°is the exterior angle at B, so:
$y = 180° - 43° = 137°$

4. Correct Answer: B) 360°

The figure forms a closed star-like shape made of overlapping triangles.
The sum of all exterior angles (or one full rotation around the figure) is always: 360°
Thus,
$\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 360°$

5. Correct Answer: A) 45°

In triangle ABD, angle at $A$is split into x°and 30°, so total angle at $A = x + 30°$.
(x + 30) + 60 + y = 180
$x + y + 60° = 180°$
Solving gives $x + y = 90°$.
$\frac{x + y}{2} = 45°$

6. Correct Answer: C) 61°, 59°

$(3x - 5) + (2x + 15) + (5x - 50) = 180°$
$10x - 40 = 180 \Rightarrow x = 22$
Angles:
$3x - 5 = 61°, 2x + 15 = 59°, 5x - 50 = 60°$
Greatest = 61°, Least = 59°.

7. Correct Answer: D) 90°

In a rhombus, adjacent angles form a linear pair (sum 180°).
If they are equal:
Each angle = $\frac{180°}{2} = 90°$

8. Correct Answer: B) A square

In a rhombus, diagonals are perpendicular bisectors.
If diagonals are also equal, then it satisfies all properties of a square.
Hence, the rhombus becomes a **square**.

9. Correct Answer: D) 12

In a parallelogram, diagonals bisect each other.
So, $P$is the midpoint of both diagonals.
$AP = \frac{AC}{2} = \frac{14}{2} = 7, BP = \frac{BD}{2} = \frac{10}{2} = 5$
$AP + BP = 7 + 5 = 12$

10. Correct Answer: Order → d → b → a → c

First, the ratio is given → **(d)**
Then assume angles → **(b)**
Use sum of angles of quadrilateral = 360°→ **(a)**
Finally find the largest angle → **(c)**
So, correct sequence: **d, b, a, c**

11. Correct Answer: 20°

In kite $ABCD$, $AB = AD$and $CB = CD$.
So, triangles $ABD$and $CBD$are isosceles.
In $\triangle ABD$:
$\angle ABD = \angle ADB = 25°$
$\angle A = 180° - 2 \times 25° = 130°$

In $\triangle BDC$:
$\angle BCD = \angle CBD = 35°$
$\angle C = 180° - 2 \times 35° = 110°$
$\angle A - \angle C = 130° - 110° = 20°$

12. Correct Answer: A) 4

Perimeter = sum of all sides:
$(x + 4) + (x + 2) + (x + 1) = 19$
$3x + 7 = 19 \Rightarrow 3x = 12 \Rightarrow x = 4$

13. Correct Answer: C) 48

Given: $AB = 17$, $AC = 10$, and altitude $AD = 8$.
In $\triangle ABD$:
$BD = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$

In $\triangle ADC$:
$DC = \sqrt{10^2 - 8^2} = \sqrt{100 - 64} = \sqrt{36} = 6$
$BC = BD + DC = 15 + 6 = 21$

Perimeter = $17 + 10 + 21 = 48$

14. Correct Answer: B) 13

Since $QV$and $WS$are medians, point $Q$is the midpoint of VW.
So, $QV = QW$.
Given: $QV = 7x - 1, QW = 4x + 5$
Equate: $7x - 1 = 4x + 5 \Rightarrow 3x = 6 \Rightarrow x = 2$
$QV = 7(2) - 1 = 14 - 1 = 13$

15. Correct Answer: B) 30

Since $KM$is a median, $M$is the midpoint of AB.

So, $AM = MB$.

Given:

$AM = 2x + 7, MB = 5x - 5$

$2x + 7 = 5x - 5 \Rightarrow 3x = 12 \Rightarrow x = 4$

$AM = 2(4) + 7 = 15, MB = 5(4) - 5 = 15$

$AB = AM + MB = 15 + 15 = 30$

16. Correct Answer: B) 4.8 cm

Triangle is right-angled with sides 6cm and 8cm.

Hypotenuse:

$BC = \sqrt{6^2 + 8^2} = 10$

Altitude from right angle to hypotenuse:

$AD = \frac{(6 \times 8)}{10} = \frac{48}{10} = 4.8$ cm

17. Correct Answer: A) 12, 35, 37

Check: $12^2 + 35^2 = 144 + 1225 = 1369 = 37^2$

So it satisfies Pythagoras theorem.

18. Correct Answer: D) Kite

Given: $AC \perp BD$and $AB = AD$.

This matches the property of a kite (two adjacent equal sides and perpendicular diagonals).

19. Correct Answer: B) 9 cm, 17 cm, 25 cm

Triangle inequality: sum of any two sides > third side.

Check options:

$9 + 17 = 26 > 25$ but check others: $9 + 25 > 17$, $17 + 25 > 9$→ valid triangle (so option B is correct)

20. Correct Answer: C) Both A and B

- A rhombus with equal adjacent angles becomes a rectangle; since all sides are already equal, it becomes a **square**.
- A rectangle with equal adjacent sides has all sides equal, so it also becomes a **square**.

Thus, **both conditions guarantee a square**.

Level – II

SINGLE OPTION CORRECT TYPE

1. Correct Answer: D) 175°

In triangle ABC: $\angle A = 45°, \angle C = 70°$

So, $y = 180° - (45° + 70°) = 65°$

At point C, $x°$is the exterior angle: $x = 180° - 70° = 110°$

$x + y = 110° + 65° = 175°$

2. Correct Answer: D) 36°, 54°

In a right triangle, the two acute angles sum to 90°.

Let them be $2x$and $3x$:

$2x + 3x = 90° \Rightarrow 5x = 90° \Rightarrow x = 18°$

$2x = 36°, 3x = 54°$

3. Correct Answer: C) 50°

At point C, exterior angle = 110°, so interior angle:
$\angle C = 180° - 110° = 70°$
At point A, angles on straight line:
$60° + y + z = 180° \Rightarrow y + z = 120°$
In triangle ABC: $x + y + 70° = 180° \Rightarrow x + y = 110°$
Also, at A: $z = 180° - (60° + y) = 120° - y$
From exterior angle property at A: $z = x + 70°$
So: $120° - y = x + 70° \Rightarrow x + y = 50°$

4. Correct Answer: C) 360°

$\angle DAB$, $\angle EBC$, and $\angle FCA$are exterior angles of the triangle.
The sum of one exterior angle at each vertex of any triangle is always: 360°

5. Correct Answer: D) $9\sqrt{2}$

Square $ABCD$has side 18cm. Diagonals intersect at P, so $P$is the center.
Thus, distance from center to each side $= \frac{18}{2} = 9$cm.
Point $M$is midpoint of AB, and $N$is midpoint of BC.
So, $PM = PN = 9$cm.
Triangle $PMN$is a right triangle with legs 9and 9:
$MN = \sqrt{9^2 + 9^2} = \sqrt{162} = 9\sqrt{2}$

6. Correct Answer: C) Equilateral

If all medians of a triangle are equal, then all sides are equal.
Hence, the triangle is **equilateral**.

7. Correct Answer: B) 34°

In $\triangle ABC$: $\angle A = 62°$, $\angle B = 76° \Rightarrow \angle C = 180° - (62° + 76°) = 42°$
In $\triangle CDE$: $\angle D = 58°$, $\angle C = 42° \Rightarrow \angle E = 80°$
In $\triangle EFG$:
$\angle F = 66°$, $\angle E = 80°$
$\angle FGE = 180° - (66° + 80°) = 34°$

8. Correct Answer: C) 72°

Given: $BD = CD = AC$
In $\triangle BDC$, $BD = CD \Rightarrow$ isosceles $\Rightarrow$ base angles at $B$and $C$are equal.
$\angle B = 27° \Rightarrow \angle BCD = 27°$
So,
$\angle BDC = 180° - (27° + 27°) = 126°$
Since $B, D, A$are collinear:
$\angle CDA = 180° - 126° = 54°$
In $\triangle ACD$, $AC = CD \Rightarrow$ isosceles $\Rightarrow$
$\angle CAD = \angle CDA = 54°$
Thus, $y = \angle ACD = 180° - (54° + 54°) = 72°$

9. Correct Answer: A) 40°

$\angle B = 90°$
Given angle at $D = 130°$, so adjacent interior angle:
$180° - 140° = 40°$
Using parallel line/alternate angle relations (as indicated by arrows), the angle at Aequals this angle.
$x = 40°$

10. In ΔABC,
$\angle BAC > 90°$
Sum of angles in a triangle = 180°
So remaining: $\angle ABC + \angle ACB = 180° - (\text{angle} > 90°) < 90°$
Thus both angles are **less than 90°** ⇒ both are **acute**.
Answer: A) Acute

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answers: A)** $x° + y°$**, B)** $180° - x° - y°$
From the figure, angles at $B$and $C$are split into $x°$and $y°$.
Using triangle angle sum and vertically opposite angles at point D, we get:
$\angle D = x + y$
Also, since angles around a point sum to 180°:
$\angle D = 180° - (x + y)$
Thus both forms represent $\angle D$.

12. **Correct Answer: A) 2.5 cm** B) $\frac{5}{2}$cm
Using Angle Bisector Theorem: $\frac{BD}{DC} = \frac{AB}{AC} = \frac{6}{5}$
Given $BD = 3$:
$\frac{3}{DC} = \frac{6}{5} \Rightarrow DC = \frac{3 \times 5}{6} = 2.5$ cm

13. **Correct Answers: A) GQ = 6,** B) SQ = 18 ,**C) SG = 12**
In a parallelogram, diagonals bisect each other ⇒ $A$is midpoint of both $PR$and SQ.
In $\triangle PQR$, $G$is centroid ⇒ divides median in ratio 2: 1(from vertex to midpoint).
So along median QA:
$AG = \frac{1}{3}QA \Rightarrow QA = 3 \times AG = 9$
Thus, $GQ = \frac{2}{3}QA = 6$
Also, since $A$is midpoint of SQ: $SQ = 2 \times QA = 18$
And centroid divides median from S:
$SG = \frac{2}{3} \times SQ = \frac{2}{3} \times 18 = 12$

14. **Correct Answer: C) 70°**
From the figure, $AB = AC$(tick marks), so $\triangle ABC$is isosceles.
Given base angles:
$\angle ABC = 60°, \angle ACB = 50°$
Then,
$x = \angle BAC = 180° - (60° + 50°) = 70°$

15. **Correct Answer: D) 100°**
$x = 70°$
In $\triangle ABD$, since $DB = BC$(equal markings), triangle relations give:
$y = 30°$
Thus, $y + x = 50° + 30° = 100°$

MATRIX MATCHING TYPE

a) Scalene triangle → **R) No sides are equal**
b) Isosceles triangle → **Q) Exactly two sides are equal**
c) Equilateral triangle → **P) All sides are equal**
d) Right-angled triangle → **S) One angle is 90°**

17.

Use the Pythagorean theorem:

$a^2 + b^2 = c^2$

Here, hypotenuse $c = 15$cm and one side = 12cm.

Other side = $\sqrt{15^2 - 12^2} = \sqrt{225 - 144} = \sqrt{81} = 9$ cm

Answer: 9 cm

18.

Rectangle $ABCD$, $CD = 24\text{cm} \Rightarrow AB = 24\text{cm}$

$AD = 5\text{cm}$

$E$is midpoint of $CD \Rightarrow DE = EC = 12\text{cm}$

Now find sides of $\triangle ABE$:

$AE = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$

$BE = \sqrt{12^2 + 5^2} = 13$

Perimeter of $\triangle ABE = 13 + 13 + 24 = 50$

Given: $2 \times 5^x = 50$

$5^x = 25 = 5^2 \Rightarrow x = 2$

Answer: 2

19.

- Because EBGDis a square and BG = 12, all its sides are 12. Thus, GD = 12, DE = 12, and EB = 12.
- In right triangle $\triangle$ BGC, we have hypotenuse BC = 13and leg BG = 12.
- Using the Pythagorean theorem:

  $a^2 + b^2 = c^2$

  $GC = \sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5$ cm

Find AB**:**

- In a parallelogram, opposite sides are equal (AB = CD).
- CD = GD + GC = 12 + 5 = 17cm.
- Therefore, AB = 17cm.

Solve for x**:**

$AB - 1 = x^2$

$17 - 1 = x^2$

$16 = x^2 \Rightarrow x = 4$

Final Answer: x = 4

20. In ΔABC:

$\angle A = 180° - (45° + 55°) = 80°$

Since AD is the bisector of ∠A:

$\angle BAD = \frac{80°}{2} = 40°$

Now,

$\frac{1}{10}$ of $\angle BAD = \frac{40°}{10} = 4°$

Correct Answer: 4°

Student Practice Corner

Single option Type

1. **Correct Answer: C) 45°, 60°, 75°**

Ratio = 3:4:5 → Sum = 12

Each part = 180/12 = 15

$3 \times 15 = 45°, 4 \times 15 = 60°, 5 \times 15 = 75$

2. **Correct Answer: A) 72°, 18°, 90°**

Let angles = $4x, 5x, x$

Sum = $10x = 180 \rightarrow x = 18$

Angles = $72°, 90°, 18°$→ arranged: $A, B, C = 72°, 18°, 90°$

3. **Correct Answer: A) 45:36:40**

$\frac{2}{3}A = \frac{3}{4}B = \frac{5}{6}C$

Take LCM → solve → ratio becomes **45:36:40**

4. **Correct Answer: A) 20°**

If ∠ACD = 100°, then the interior ∠ACB = 180° − 100° = 80°.

Since AB = AC, the base angles are equal:

∠ABC = ∠ACB = 80°.

∴ ∠A = 180° − (80° + 80°) = 180° − 160° = 20°.

5. **Correct Answer: C) 55°**

$A + B = 145°, B + C = 90°$

Add: $A + 2B + C = 235$

But $A + B + C = 180$

So: $180 + B = 235 \rightarrow B = 55°$

6. **Correct Answer: A) 45**

Exterior angles sum = 360°

$(2x + 11) + (3x - 6) + (2x + 40) = 360$

$7x + 45 = 360 \rightarrow x = 45$

7. **Correct Answer: B) 85°**

Exterior angle at C = 135°→ interior = 180 − 135 = 45°

So triangle angles:

$50° + 45° + x = 180° \Rightarrow x = 85°$

8. Since $AB = BC$, triangle $ABC$is isosceles with base AC, so

$\angle A = \angle C$

Given $\angle B = 50°$,

$\angle A + \angle B + \angle C = 180°$

$\angle A + 50° + \angle A = 180°$

$2\angle A = 130° \Rightarrow \angle A = 65°$

So, $\angle C = 65°$

The angle $x$is the exterior angle at C, so $x = 180° - \angle C = 180° - 65° = 115°$

Correct Answer: A) 115°

More than one correct type

9. **Correct Answers: B)** $AC^2 + BC^2$**, C)** $2AC^2$

ABC is an isosceles right triangle with right angle at C ⇒ $AC = BC$.

By Pythagoras:

$AB^2 = AC^2 + BC^2$

Since $AC = BC$,

$AB^2 = 2AC^2$

10. **Correct Answers: A) 6,8,10; C) 8,15,17; D) 7,24,25**

Check Pythagorean condition $a^2 + b^2 = c^2$:

$6^2 + 8^2 = 36 + 64 = 100 = 10^2$

$8^2 + 15^2 = 64 + 225 = 289 = 17^2$

$7^2 + 24^2 = 49 + 576 = 625 = 25^2$

COMPREHENSION TYPE

11. **∠BFD = x° = 60° → Option B**

Using angle chasing in triangle $\triangle ABF$and the given right angle at E, along with the 30° angle at A, the internal angles balance to give $x = 60°$.

12. **∠ACB = z° = 50° → Option D**

In triangle $\triangle ABC$, using the sum of angles and the previously determined internal angles (including right angle at E), we get $z = 50°$.

Matching

13. a) Median of triangle → **Q**

b) Altitude of triangle → **P**

c) Exterior angle theorem → **S**

d) Pythagoras theorem → **R**

INTEGER TYPE

14. **Answer: 5 m**

Height difference = 11 − 7 = 4m, base distance = 3 m

Distance between tops = $\sqrt{3^2 + 4^2} = 5$m

15. **Answer: 6**

In isosceles right triangle, $AB = AC\sqrt{2} = 6\sqrt{2}$

So, $\frac{AB}{\sqrt{2}} = 6$

Congruence of Triangle

Level – 1

1.

Using triangle angle sum: $\angle A + \angle B + \angle C = 180°$

Since ΔABC ≅ ΔA'B'C', we have

∠A = ∠A′ and ∠B = ∠B′

So,

∠A + ∠B = 116°

⇒ ∠C = 180° − 116° = **64°**

Correct Answer: D) 64°

2. AAA gives only **similarity**, not congruence.

Correct Answer: B) AAA

3.

In the figure, diagonals bisect each other at E:
AE = EC
DE = EB
Vertical angles at E are equal
So,
ΔDEC ≅ ΔBEA
ΔBEC ≅ ΔDEA
Correct Answer: D)

4.

For ΔABC and ΔADC:
We see two angles marked at vertex A as equal (∠BAC = ∠DAC).
We see two angles marked at vertex C as equal (∠BCA = ∠DCA).
The side AC is shared by both triangles (common side)
The Logic: We have an Angle, a Side, and an Angle in sequence.
Thus, **ASA congruence** applies.
Correct Answer: A) ASA

5. **Correct Answer: A) ΔABC ≅ ΔQRP**

∠A = ∠Q (top angles marked equal)
∠B = ∠R (base angles marked equal)
AB = QR (marked sides)
Thus, two angles and a corresponding side are equal ⇒ triangles are congruent (AAS/ASA).
So, the correct correspondence is **A ↔ Q, B ↔ R, C ↔ P**, giving
ΔABC ≅ ΔQRP.

6. **Correct Answer: C) RHS**

Both triangles are right-angled (∠B = ∠E = 90°)
Hypotenuse AC = DF (marked equal)
One corresponding side AB = DE (marked equal)
Thus, triangles are congruent by **RHS (Right angle–Hypotenuse–Side)**

7. **Correct Answer: A) ΔADC ≅ ΔCBA**

AB = CD (given)
AD = BC (given)
AC is common to both triangles
Thus, all three corresponding sides are equal ⇒ triangles are congruent by **SSS**.

8. **Correct Answer: A) SAS**

PQ = SR (given)
QR is common
∠PQR = ∠QRS (given)
Thus, two sides and the included angle are equal ⇒ triangles are congruent by **SAS**.

9. **Correct Answer: D) ΔBQC ≅ ΔCPB**

BP ⊥ AC and CQ ⊥ AB ⇒ ∠CPB = ∠BQC = 90°
BQ = CP (given)
BC is common
Thus, triangles are congruent by **RHS**.

10. Correct Answer: B) ΔOAB ≅ ΔODC

AO = OC (marked equal)

BO = OD (marked equal)

∠AOB = ∠COD (vertically opposite angles)

Thus, triangles are congruent by **SAS**.

11. Correct Answer: Option A) $\bar{BC} = \bar{CD}$

$AB = AC \Rightarrow \triangle ABC$is isosceles, so base angles:

$$\angle B = \angle C = \frac{180^\circ - 48^\circ}{2} = 66^\circ$$

Given $\angle ACD = 18^\circ$, hence:

$\angle BCD = 66^\circ - 18^\circ = 48^\circ$

Also $\angle BAC = 48^\circ$

Now compare triangles $\triangle ABC$and $\triangle ACD$:

$\angle BAC = \angle BCD = 48^\circ$

$\angle BCA = \angle CAD = 18^\circ$

So triangles become similar in such a way that it leads to:

$BC = CD$

12. Correct Answer: D) 40°

Exterior angle: $\angle ACE = 130^\circ \Rightarrow \angle ACB = 180^\circ - 130^\circ = 50^\circ$

Given $BD = AD = DC$

$\Rightarrow D$is midpoint-like and forms **isosceles triangles**:

$AD = DC \Rightarrow \triangle ADC$is isosceles

$\angle DAC = \angle ACD = 50^\circ$

Sum of angles in $\triangle ADC$:

$\angle ADC = 180^\circ - 50^\circ - 50^\circ = 80^\circ$

Since $BD = AD$, triangle $ABD$is isosceles:

$\angle ABD = \angle BAD$

Straight line at BDC:

$\angle ADB + \angle ADC = 180^\circ \Rightarrow \angle ADB = 100^\circ$

In $\triangle ABD$:

$2\angle ABD + 100^\circ = 180^\circ \Rightarrow \angle ABD = 40^\circ$

Thus: $\angle ABC = 40^\circ$

13. Correct Answer: Option D) 64°

Given:

$AD = BD \Rightarrow \triangle ABD$is isosceles

$\Rightarrow \angle BAD = \angle ABD = 29^\circ$

So, $\angle ADB = 180^\circ - 29^\circ - 29^\circ = 122^\circ$

- Since $C, D, B$are collinear: $\angle ADC = 180^\circ - 122^\circ = 58^\circ$
- Given $CD = AC \Rightarrow \triangle ACD$is isosceles

  $\Rightarrow \angle CAD = \angle ADC = 58^\circ$
- Now in $\triangle ACD$: $x + 58^\circ + 58^\circ = 180^\circ \Rightarrow x = 64^\circ$

14. Correct Answer: Option C) $\bar{AE}$

Given $AB = AC$, so triangle $ABC$is isosceles.

Also $\angle BAD = \angle CAE$, which makes triangles $\triangle ABD$and $\triangle ACE$symmetric.

Hence corresponding sides are equal: $AD = AE$

Answer: C

15. Correct Answer: Option C) $\angle ACD$

Given $AB = AC$, so $\triangle ABC$is isosceles.

Also $BD = CD$, so $D$is midpoint of $BC$and $AD$is median.

In an isosceles triangle, median from vertex is also **angle bisector**.

Thus: $\angle ABD = \angle ACD$

16. Correct Answer: Option A) $y - q$

$AB = AC \Rightarrow \triangle ABC$is isosceles ⇒ base angles at $B$and $C$are equal.

$DB = DC \Rightarrow \triangle DBC$is isosceles ⇒ base angles at $B$and $C$are equal.

At point B: angle is split into $x$and y.

At point C: angle is split into $p$and q.

From isosceles triangles: $x + y = p + q$

So: $x - p = q - y \Rightarrow x - p = y - q$

17. Correct Answer: Option C) Not Congruent

Conclusion: Because the equal angle is not located between the same corresponding equal sides in both triangles, we cannot prove congruency with the information given

18. Correct Answer: A) SSS

In the figure, $PQ = QR = RS = SP$, so all four sides are equal.

Consider triangles $\triangle PQR$and $\triangle PRS$:

$PQ = RS$

$QR = SP$

$PR = PR$(common side)

All three corresponding sides are equal ⇒ triangles are congruent by **SSS**.

19. Correct Answer: B) RHS

In triangles $\triangle ABC$and $\triangle ADC$:

$\angle ABC = 90°$and $\angle ADC = 90°$(right angles)

$AC = AC$(common hypotenuse)

$BC = CD$(given)

Thus, triangles are congruent by **RHS (Right angle–Hypotenuse–Side)**.

20. Correct Answer: D) AAS

In triangles $\triangle PQS$and $\triangle PRT$:

$\angle PQS = \angle PRT$(given)

$QS = TR$(given)

$PS = PT$(common corresponding sides from the figure)

Thus, two sides and the included angle are equal ⇒ triangles are congruent by **AAS**

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Correct Answer: D) All of the above

Given $AO = CO$, $BO = DO$, and $\angle AOD = 90°$:

$\triangle AOD \cong \triangle COD$(SAS: $AO = CO$, $DO = DO$, included right angle)

$\triangle COB \cong \triangle AOB$similarly by SAS

Hence all sides of the rectangle become equal ⇒ $AD = DC = CB = BA$

So, all statements are correct ⇒ **All of the above**.

2. Correct Answer: D) AAS

In triangles $\triangle PQR$and $\triangle STR$: $PQ = TS$(given)

$\angle P = \angle S$(given)

$\angle PRQ = \angle SRT$(vertically opposite angles)

Thus, two angles and a non-included side are equal ⇒ triangles are congruent by **AAS**

3. Correct Answer: D) AAS

From the figure:

$\angle B = \angle B'$

$\angle C = \angle C'$

One corresponding side (AC = A'C') is given

This is **two angles and a non-included side**, so triangles are congruent by **AAS**.

4. Correct Answer: C) 70°

In $\triangle ABC$:

$\angle B = 60°$, $\angle C = 50°$

So $\angle A = 180° - (60° + 50°) = 80°$

Since $\triangle ABC \cong \triangle A'B'C'$, corresponding angles are equal.

Thus, angle at $A'$is also 70°.

5. Correct Answer: B) $2\angle x$

Given $BC = BE = AE$:

$\triangle ABE$is isosceles ⇒ base angles at $B$and $E$are equal

$\triangle BEC$is also isosceles ⇒ angles at $B$and $C$are equal

Using angle sum and the exterior angle property at B(since $AB$is extended), the angle at E(i.e., y) becomes the sum of two equal angles related to x

Thus, $y = 2x$.

6. Correct Answer: A) $\overline{\text{AE}}$

Given:

$\angle ABC = \angle AED = 90°$

$BC = ED$

$\angle ACD = \angle ADC$

Triangles $\triangle ABC$and $\triangle AED$are congruent by **RHS**:

Right angles equal

Hypotenuse $AB = AE$correspondence

One side $BC = ED$

Thus, corresponding sides are equal ⇒ $AB = AE$

7.

From $MN = NP$:

$3x - 4 = x + 4 \Rightarrow x = 4$

Then $LM = PQ = 2x + 1 = 9$ cm.

So the correct answer is **D) 9 cm** (this one is fine).

8.

For SAS, we must match triangles in the correct order:

$\angle DAB = \angle CBA = 90°$

$AD = BC$

$AB$is common

So correspondence is: $\triangle ABC \cong \triangle BAD$

That is **option C)** $\triangle BAD$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Correct Answers: A, B, C, D

Given $AB = AC$and $AD$is the angle bisector:

$AB = AC$

$\angle BAD = \angle CAD$
$AD = AD$(common)
So, $\triangle ABD \cong \triangle ACD$by **SAS** ⇒ **(A) and (B) correct**
From congruence:
$BD = DC$⇒ **(D) correct**
Also, angle at Dbecomes 90°, so $AD \perp BC$⇒ **(C) correct**

10. Correct Answers: A, B, C

Given $AB = AC$and $AQ = AP$:
$AB = AC$
$AP = AQ$
$\angle BAC$common
So, $\triangle APB \cong \triangle AQC$by **SAS** ⇒ **(B) correct**
From congruence:
$PB = QC$⇒ **(A) correct**
Also:
$\triangle BQC \cong \triangle CPB$(by corresponding equal sides and angles) ⇒ **(C) correct**
Option **D** is not necessarily true.

11. Correct Answer: B) △DCE ≅ △BLE C) AB=BL

- In parallelogram $ABCD$, opposite sides are parallel and equal ⇒ $DC \parallel AB$and $DC = AB$.
- $E$is midpoint of BC⇒ $BE = EC$.
- $\angle DCE = \angle EBL$(alternate interior angles since lines are parallel).
- Thus, by **SAS congruence**, $\triangle DCE \cong \triangle BLE$.

12. Correct Answer: C) RHS

- $\triangle ABC$is a right-angled isosceles triangle at B.
- $D$is midpoint of hypotenuse AC⇒ $AD = DC$.
- $BD$is common in both triangles.
- $\angle BDA = \angle BDC = 90°$.
- Hence, triangles $\triangle ABD$and $\triangle CBD$are congruent by **RHS (Right angle–Hypotenuse–Side)**.

13. Correct Answer: A), B), C)

Triangles can be proven congruent using:

- **SSS (Side–Side–Side):** All three corresponding sides are equal.
- **SAS (Side–Angle–Side):** Two sides and the included angle are equal.
- **ASA (Angle–Side–Angle):** Two angles and the included side are equal.

COMPREHENSION TYPE

Passage – I:

14. Correct Answer: A) 3

Given $\triangle ABC \cong \triangle DEC$, corresponding sides are equal:
$BC = CE$
$\Rightarrow 2x = 3y \ldots (1)$
Also, $AB = DE$
$\Rightarrow x + y = 5 \ldots (2)$
From (1): $y = \frac{2x}{3}$
Substitute in (2):
$x + \frac{2x}{3} = 5 \Rightarrow \frac{5x}{3} = 5 \Rightarrow x = 3$

15. Correct Answer: B) 4

Using the value of x = 3 found in the previous step, substitute it back into Equation 1:

$3 + y = 5$

$y = 2$

Thus,

$2y = 2(2) = 4$

16. Correct Answer: B) ∠CDE

From $\triangle ABC \cong \triangle DEC$, the order shows correspondence:

$A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow C$

Thus,

$\angle BAC \cong \angle CDE$.

MATRIX MATCHING TYPE

17. Matching (Congruence Criteria Basics)

a) Three sides equal → **SSS (P)**

b) Two angles and included side → **ASA (Q)**

c) Two sides and included angle → **SAS (R)**

d) Right angle, hypotenuse, one side → **RHS (S)**

Answer: a–P, b–Q, c–R, d–S

INTEGER TYPE

18.

Given:

AC = BC ⇒ ∠A = ∠B

Also ∠B = 2∠C

Let ∠C = x

Then ∠B = 2x and ∠A = 2x

Sum of angles: $2x + 2x + x = 180 \Rightarrow 5x = 180 \Rightarrow x = 36$

So,

$\angle BAC = 72°$

$\frac{1}{5}$ of $\angle BAC = \frac{72}{5} = 14.4$

Answer: 14.4

19. Using congruent triangles (based on equal sides marked), corresponding parts are equal.

From the figure:

$2x + 3 = 7 \Rightarrow x = 2$

$\frac{1}{2}(2) = 1$

Answer: 1

20.

Given ΔABD ≅ ΔACD ⇒ corresponding sides equal:

$x + 40 = 2x + 20$

$x = 20$

$\frac{x}{4} = \frac{20}{4} = 5$

Answer: 5

Student Practice Corner

Single option Type

1.

Given:
PL ⊥ OB and PM ⊥ OA, and PL = PM
Triangles **ΔPLO** and **ΔPMO** are right triangles with:

- ∠L = ∠M = 90°
- OP common
- PL = PM

So by RHS congruence:
Answer: A) ΔPLO ≅ ΔPMO

2.

Given: ΔCAB ≅ ΔDEF
Correspondence:
C ↔ D, A ↔ E, B ↔ F
So,
AB = EF
From figure:
AB = 13 cm (slanted side BC is 13, but AB is vertical height = 12? Actually correct match is BC ↔ DF, so AB ↔ EF)
Thus EF = AB = **5 cm**
Answer: B) 5 cm

3.

Given congruent triangles ⇒ corresponding angles equal:
$5x + 4 = 79$
$5x = 75 \Rightarrow x = 15$
Answer: B) 15°

4.

BC = QR (corresponding sides):
$3x + 4 = 2x + 7$
$x = 3$
Answer: C) 3

5.

ΔPQR ≅ ΔP'Q'R' ⇒ corresponding angles equal
∠Q = 30°, ∠P = 2 × 30° = 60°
Sum of angles:
$\angle R = 180 - (60 + 30) = 90°$
Answer: A) 90°

6.

Given congruent triangles ⇒ ∠A = ∠P and ∠B = ∠Q
$2x + \frac{3}{2}x + 40 = 180$
$\frac{7x}{2} = 140 \Rightarrow x = 40$
$\angle P = \angle A = 2x = 80°$
Answer: D) 80°

7.

$\Delta ABC \cong \Delta A'B'C' \Rightarrow$ corresponding sides equal

AC = B'C'

$3x + 4 = 2x + 1 \Rightarrow x = -3$

Perimeter = 30:

$A'B' + B'C' + A'C' = 30$

$A'B' + (-5) + (-5) = 30$

$A'B' = 40$

Check option: $25 - 5x = 25 - (-15) = 40$

Answer: A) 25 − 5

8.

Perimeter:

$(x + 8) + (2x - 1) + (2x + 3) = 45$

$5x + 10 = 45 \Rightarrow x = 7$

$Q'R' = QR = x + 8 = 15$

Answer: C) 15 cm

One or more than one correct

9.

Given AB = AC and D is midpoint ⇒ BD = DC

Triangles ABD and ACD:

- AB = AC
- BD = DC
- AD common

So triangles are congruent.

Check options:

- B) $\Delta ABD \cong \Delta ACD$
- D) $\Delta ADB \cong \Delta ADC$
- C) $\Delta BDA \cong \Delta CDA$ (same correspondence)

Correct Answers: B, C, D

10. Given:

AB = CB

AD = CD

Consider triangles **ΔDAB** and **ΔDCB**:

AD = CD (given)

AB = CB (given)

DB is common

So, by **SSS congruence**,

$\Delta DAB \cong \Delta DCB$

Now check options:

A) ΔDCB ≅ ΔDAB (correct — same triangles)

B) ∠DAB ≅ ∠DCB (corresponding angles)

COMPREHENSION TYPE

11. Given:

AB ⊥ BC and CD ⊥ BC

So both ∠ABC and ∠BCD are right angles (90°)

Answer: C) ∠BCD

12. Consider triangles **ΔABC** and **ΔDCB**:
∠ABC = ∠DCB = 90°
AB = CD (given)
BC is common
So triangles are congruent by **SAS**
Answer: A) SAS

Matrix Matching

13.

Given:
AB = AC ⇒ isosceles ⇒ ∠B = ∠C
Also ∠C = 2∠A
Let ∠A = x ⇒ ∠C = 2x and ∠B = 2x
Sum: $x + 2x + 2x = 180 \Rightarrow 5x = 180 \Rightarrow x = 36°$
So:

- ∠A = 36°
- ∠B = 72°
- ∠C = 72°

BD bisects ∠B ⇒ $\angle ABD = \angle DBC = 36°$

MATRIX MATCHING TYPE

Now:
a) ∠BDC = 180 − (36 + 72) = **72°** → **S**
b) ∠DBC + ∠BDA = 36 + 108 = **144°** → **T**
c) ∠ABD = **36°** → **Q**
d) ∠BDA = **108°** → **P**

Final Matching:
a → S (72°)
b → T (144°)
c → Q (36°)
d → P (108°)

14.

Given ΔABC ≅ ΔPQR ⇒ corresponding angles equal:
∠B = ∠Q ⇒ $x + 15 = 70 \Rightarrow x = 55$
∠C = ∠R ⇒ $5y + 10 = 35$
$5y = 25 \Rightarrow y = 5$
Answer: 5

15.

Let base angles = $x$each
Vertex angle = $2x$(given)
Sum of angles:
$x + x + 4x = 180 \Rightarrow 6x = 180 \Rightarrow x = 30°$
$\frac{1}{15}$ of $\angle B = \frac{30}{15} = 2$

Answer: 2

Polygon

LEVEL – I

1. Interior angle = 135°

$\frac{(n-2)180}{n} = 135 \Rightarrow n = 8$

Answer: A) 8

2. Exterior angle = 30°

$n = \frac{360}{30} = 12$

Answer: B) 12

3. $(n-2)180 = 1980 \Rightarrow n - 2 = 11 \Rightarrow n = 13$

Answer: C) 13

4. $(n-2)180 = 1080 \Rightarrow n - 2 = 6 \Rightarrow n = 8$

Answer: A) 8

5.

Sum of exterior angles=(n−2)×180∘

Now substitute:

$(n-2) \times 180 = 540$

$n - 2 = 3$

$n = 5$

Answer: B) 5

6. Interior angle = 160°

$\frac{(n-2)180}{n} = 160 \Rightarrow n = 18$

Answer: D) 18

7. $(10-2)180 = 8 \times 180 = 1440°$

Answer: A) 1440°

8. $\frac{(5-2)180}{5} = \frac{540}{5} = 108°$

Answer: C) 108°

9. $\frac{360}{15} = 24°$

Answer: B) 24°

10.

Number of diagonals:

$\frac{n(n-3)}{2} = \frac{7 \times 4}{2} = 14$

Answer: A) 14

11.

Interior angle = 160°:

$\frac{(n-2)180}{n} = 160 \Rightarrow n = 18$

Diagonals: $\frac{18 \times 15}{2} = 135$

Answer: A) 135

12.

Interior + Exterior = 180°
$180 - 60 = 120°$
Answer: C) 120°

13.

Interior sum = 5 ×exterior sum
$(n-2)180 = 5 \times 360$
$(n-2)180 = 1800 \Rightarrow n - 2 = 10 \Rightarrow n = 12$

Answer: D) 12

14.

Interior = 2 × Exterior
Let exterior = x
Interior = 2x
$x + 2x = 180 \Rightarrow 3x = 180 \Rightarrow x = 60$
$n = \frac{360}{60} = 6$

Answer: D) 6

15.

Interior sum = 9 × exterior sum
$(n-2)180 = 9 \times 360$
$(n-2)180 = 3240 \Rightarrow n - 2 = 18 \Rightarrow n = 20$

Answer: D) 20

16. Correct Answer: A) 14

Interior sum = 6 × one exterior angle
Exterior angle = $\frac{360}{n}$
$(n-2)180 = 6 \times 360$
$(n-2) = 6 \times 2$
$(n-2) = 12$
n = 14

17.

Ratio: 4 : 8 : 6 : 4 : 5
Sum = 27
Interior sum of pentagon: $(5-2)180 = 540$
Each part: $\frac{540}{27} = 20$
Angles: 80°, 160°, 120°, 80°, 100°
Answer: C)

18.

Exterior: Interior = 2 : 3
Let exterior = 2x, interior = 3x
$2x + 3x = 180 \Rightarrow 5x = 180 \Rightarrow x = 36$
Exterior angle = $2x = 72°$
$n = \frac{360}{72} = 5$
Answer: B) 5

19.

Interior : Exterior = 7 : 2
Let exterior = 2x, interior = 7x
$7x + 2x = 180 \Rightarrow 9x = 180 \Rightarrow x = 20$
Exterior angle = $2x = 40°$
$n = \frac{360}{40} = 9$
Answer: D) 9

20.

Let number of sides be $4k$and $5k$
Interior angle formula:
Interior $= \frac{(n-2)180}{n}$
So ratio:
$\frac{4k-2}{4k} : \frac{5k-2}{5k} = 15:16$
$\frac{(4k-2)5k}{(5k-2)4k} = \frac{15}{16}$
$\frac{5(4k-2)}{4(5k-2)} = \frac{15}{16}$

Cross multiply:
$80k - 40 = 60k - 30$
$20k = 10 \Rightarrow k = 1$
Sides = 4 and 5 → not in options
Try next valid scaling (k = 2):
Sides = 8 and 10
Check:
Interior angles:
8 sides → 135°
10 sides → 144°
Ratio: $135:144 = 15:16$
Answer: D) 8, 10

LEVEL – II

SINGLE OPTION CORRECT TYPE

1.

14 right angles = $14 \times 90 = 1260°$
$(n-2)180 = 1260 \Rightarrow n - 2 = 7 \Rightarrow n = 9$
Answer: A) 9

2.

Sum of interior angles of decagon: $(10-2)180 = 1440°$
Sum of given angles:
$(3x + 10) + (2x + 7) + (4x - 3) + (5x - 2) + (2x + 15) + (3x + 2) + (x + 3) + (2x + 5) + (3x - 1) + (3x + 4)$
$= 28x + 40$
$28x + 40 = 1440 \Rightarrow 28x = 1400 \Rightarrow x = 50$
Largest angle = $5x - 2 = 250 - 2 = 248°$
Answer: B) 248°

3. $n = \frac{360}{18} = 20$

Answer: D) 20

4.

$(n-2)180 = 2160 \Rightarrow n-2 = 12 \Rightarrow n = 14$

Answer: C) 14

5.

Interior : Exterior = 4 : 1

Let exterior = x, interior = 4x

$5x = 180 \Rightarrow x = 36$

$n = \frac{360}{36} = 10$

Answer: B) 10

6.

let the interior angle and exterior angle be $5x$and $2x$.

Since they form a linear pair:

$5x + 2x = 180°$

$7x = 180°$

$x = \frac{180}{7}$

Exterior angle: $2x = \frac{360°}{7}$

For a regular polygon:

Exterior angle $= \frac{360°}{n}$

$\frac{360}{n} = \frac{360}{7}$

$n = 7$

Answer: C) 7

7. Interior angle = 120°

$\frac{(n-2)180}{n} = 120 \Rightarrow n = 6$

New polygon has 12 sides:

$\frac{(12-2)180}{12} = \frac{1800}{12} = 150°$

Answer: A) 150°

8. Sum of exterior = 360°

$(n-2)180 - 360 = 540$

$(n-2)180 = 900 \Rightarrow n-2 = 5 \Rightarrow n = 7$

Answer: B) 7

9. Number of diagonals: $\frac{n(n-3)}{2}$

Given ratio:

$n : \frac{n(n-3)}{2} = 2:7$

$\frac{2n}{n(n-3)} = \frac{2}{7} \Rightarrow \frac{2}{n-3} = \frac{2}{7} \Rightarrow n-3 = 7 \Rightarrow n = 10$

Sum of interior angles: $(10-2)180 = 1440°$

Answer: D) 1440

10.

Exterior angles: $x, 2x, 3x, 4x, 5x$
Sum of exterior angles:
$x + 2x + 3x + 4x + 5x = 15x = 360$
$x = 24^\circ$
Least angle = $x = 24^\circ$
Answer: B) 24°

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11.

Number of diagonals of 10-sided polygon:
$\frac{n(n-3)}{2} = \frac{10 \times 7}{2} = 35$

Also,
$\sqrt{1225} = 35$
$\sqrt{1024} = 32$

So correct values matching 35:
Answer: C) 35 and D) $\sqrt{1225}$

12.

Let sides be $3x$and $4x$
Sum of interior angles:
$(3x-2)180 : (4x-2)180 = 2:3$
$\frac{3x-2}{4x-2} = \frac{2}{3}$
$3(3x-2) = 2(4x-2)$
$9x - 6 = 8x - 4 \Rightarrow x = 2$
First polygon sides = $3x = 6$
Check options:
A) 6
B) $2\sqrt{9} = 2 \times 3 = 6$
Others incorrect.
Answer: A and B

13.

Sum of interior angles of pentagon: $(5-2)180 = 540^\circ$
Given angles: 40°, 60° ⇒ sum = 100°
Remaining sum: $540 - 100 = 440^\circ$
Ratio 1 : 3 : 7 ⇒ total parts = 11
Each part: $\frac{440}{11} = 40^\circ$
Angles: $40^\circ, 120^\circ, 280^\circ$
Largest angle = **280°**
Answer: C) 280°

14.

In a regular polygon, all sides are equal. Therefore, in ΔABC, side AB = BC. This makes ΔABC an isosceles triangle.
Since AB = BC, then ∠BCA = ∠BAC = 15°.
The sum of angles in a triangle is 180°.

∠ABC = 180° − (∠BAC + ∠BCA)
∠ABC = 180° − (15° + 15°) = 150°.
Since AB and BC are consecutive sides of the regular polygon, ∠ABC is an interior angle of the polygon.
Answer: B) 150°

15.

150n = 180(n − 2)
150n = 180n − 360
30n = 360
n = 12
Answer: A) 14

Matrix Matching

16.

Number of triangles from one vertex = $n - 2$
a) Nonagon (9 sides): $9 - 2 = 7 \rightarrow$ **P**
b) Square (4 sides): $4 - 2 = 2 \rightarrow$ **R**
c) 7 sides: $7 - 2 = 5 \rightarrow$ **Q**
d) Decagon (10 sides): $10 - 2 = 8 \rightarrow$ **S**

17.

Interior angle = 4 × exterior angle
Let exterior = x, interior = 4x
$x + 5x = 180 \Rightarrow 6x = 180 \Rightarrow x = 30$
$n = \frac{360}{30} = 12$
Half of number of sides: $\frac{12}{2} = 6$
Answer: 6

18.

Exterior angle = 40°
$n = \frac{360}{40} = 9$
Answer: 9

19.

Exterior angle = 60°
$n = \frac{360}{60} = 6$

Number of diagonals:
$\frac{n(n-3)}{2} = \frac{6 \times 3}{2} = 9$

Answer: 9

20.

Let sides be $4x$and $5x$
Interior angle formula:
Here is the **text from the image**:
$\frac{4x-2}{4x} \cdot \frac{5x}{5x-2} = \frac{15}{16}$
$\frac{5(4x-2)}{4(5x-2)} = \frac{15}{16}$

Cross-multiply and solve for x:
$80(4x-2) = 60(5x-2)$
$320x - 160 = 300x - 120$
$20x = 40 \Rightarrow x = 2$

First polygon sides = $4x = 8$
Answer: 8

Student Practice Corner

Single option type

1.

Sum of interior angles of heptagon: $(7-2)180 = 900°$
Given 3 angles = $3 \times 132 = 396°$
Remaining sum = $900 - 396 = 504°$
Each of 4 equal angles: $\frac{504}{4} = 126°$
Answer: B) 126°

2. Exterior angles:
$\frac{360}{n-1} - \frac{360}{n+1} = 9$
$\frac{360[(n+1)-(n-1)]}{(n-1)(n+1)} = 9$
$\frac{360 \cdot 2}{n^2-1} = 9$
$\frac{720}{n^2-1} = 9$
$n^2 - 1 = 80 \Rightarrow n^2 = 81 \Rightarrow n = 9$
Answer: A) 9

3. Sum of interior angles of nonagon: $(9-2)180 = 1260°$
Given angles: $50 + 50 + 70 + 40 = 210°$
Remaining sum: $1260 - 210 = 1050°$
Remaining 5 equal angles:
$\frac{1050}{5} = 210°$
Answer: A) 210°

4.

Sum of hexagon: $(6-2)180 = 720°$
Given: $90 + 70 = 160°$
Remaining sum: $720 - 160 = 560°$
Remaining 4 equal angles:
$\frac{560}{4} = 140°$
Answer: A) 140°

5. Exterior ratio: 5 : 7 : 4 : 3 : 8 : 3 : 6
Sum = 36
Each part:
$\frac{360}{36} = 10°$
Exterior angles: 50°, 70°, 40°, 30°, 80°, 30°, 60°
Interior = 180 − exterior: 130°, 110°, 140°, 150°, 100°, 150°, 120°
Answer: C)

6. Number of triangles from one vertex: $n - 2 = 6 - 2 = 4$
Answer: C) 4

7.
Sum of interior angles of nonagon:
$(9 - 2)180 = 7 \times 180 = 1260°$
Answer: B) 1260°

8.
Interior + Exterior = 180°
$180 - 100 = 80°$
Answer: A) 80°

More than one correct option type

9.
Let number of sides = n
Sum of interior angles: $(n - 2)180$
Given:
$2 \times 90 + (n - 2) \times 120 = (n - 2)180$
$180 + 120n - 240 = 180n - 360$
$120n - 60 = 180n - 360$
$300 = 60n \Rightarrow n = 5$
Answer: A) 5

10.
Interior sum = 6 × exterior sum
$(n - 2)180 = 6 \times 360$
$(n - 2)180 = 2160 \Rightarrow n - 2 = 12 \Rightarrow n = 14$
Check options:
C) $\sqrt{196} = 14$
D) 14
Answer: C and D

11.
Each interior angle:
$\frac{(n - 2)180}{n}$
Sum of angles of two regular polygons = 1980°
Try options:
$n_1 = 6 \rightarrow$ interior = 120°
$n_2 = 10 \rightarrow$ interior = 144°
$120 + 144 = 264 \neq 1980$

Instead interpret as **sum of all interior angles**:
$(n_1 - 2)180 + (n_2 - 2)180 = 1980$
$(n_1 + n_2 - 4)180 = 1980$
$n_1 + n_2 - 4 = 11 \Rightarrow n_1 + n_2 = 15$
Answer: A) 15

12.
Interior = Exterior
$\frac{(n - 2)180}{n} = \frac{360}{n}$
$(n - 2)180 = 360 \Rightarrow n - 2 = 2 \Rightarrow n = 4$
Answer: A) 4

Matrix Matching

13. **a)** Interior = 162°

$\frac{(n-2)180}{n} = 162 \Rightarrow n = 20$

Diagonals: $\frac{20\times17}{2} = 170$

→ **Q**

b) Exterior = 36° $n = \frac{360}{36} = 10$

→ **S**

c) $(n-2)180 = 4140 : n-2 = 23 \Rightarrow n = 25$

→ **T**

d) Exterior = 22.5°: $n = \frac{360}{22.5} = 16$

→ **P**

INTEGER TYPE

14.

Interior = 2 × exterior

Let exterior = x, interior = 2x

$x + 2x = 180 \Rightarrow 3x = 180 \Rightarrow x = 60$

$n = \frac{360}{60} = 6$

Answer: 6

15.

Exterior : Interior = 1 : 2

Let exterior = x, interior = 2x

$x + 2x = 180 \Rightarrow 3x = 180 \Rightarrow x = 60$

$n = \frac{360}{60} = 6$

Answer: 6

5.Time and Work and Time and Distance

LEVEL – I

1. Ram, Shyam and Arun

Ram + Shyam = $\frac{1}{10}$

Ram + Arun = $\frac{1}{12}$

Shyam + Arun = $\frac{1}{15}$

Adding all:

$2(R+S+A) = \frac{1}{10}+\frac{1}{12}+\frac{1}{15}$

LCM = 60

$2(R+S+A) = \frac{6+5+4}{60} = \frac{15}{60} = \frac{1}{4}$

$R+S+A = \frac{1}{8}$

So, total time = 8days.

Correct Answer: C) 8 days

2. P, Q and R

$P+Q = \frac{1}{12}, Q+R = \frac{1}{16}, R+P = \frac{1}{24}$

Adding:

$2(P+Q+R) = \frac{1}{12}+\frac{1}{16}+\frac{1}{24}$

LCM = 48

$2(P+Q+R) = \frac{4+3+2}{48} = \frac{9}{48} = \frac{3}{16}$

$P+Q+R = \frac{3}{32}$

Time taken:

$\frac{1}{3/32} = \frac{32}{3} = 10\frac{2}{3}$

Correct Answer: D) $10\frac{2}{3}$days

3. P and Q

$P+Q = \frac{1}{18}, P = \frac{1}{27}$

$Q = \frac{1}{18}-\frac{1}{27}$

LCM = 54

$Q = \frac{3}{54}-\frac{2}{54} = \frac{1}{54}$

So, Q alone takes 54 days.

Correct Answer: A) 54 days

4. A, B and C

$A = \frac{1}{3}, B = \frac{1}{6}, C = \frac{1}{7}$

Total work per day:

$\frac{1}{3}+\frac{1}{6}+\frac{1}{7}$

LCM = 42

$= \frac{14+7+6}{42} = \frac{27}{42} = \frac{9}{14}$

Time taken:

$\frac{1}{9/14} = \frac{14}{9}$

Correct Answer: C) 14/9Days

5. Man and Daughter

Man's 1 day work $= \frac{1}{4}$

Man + Daughter 1 day work $= \frac{1}{3}$

Daughter's 1 day work:

$\frac{1}{3} - \frac{1}{4} = \frac{4-3}{12} = \frac{1}{12}$

So, daughter alone takes 12 days.

Correct Answer: D) 12 days

6. P and Q

$P + Q = \frac{1}{4}, P = \frac{1}{20}$

$Q = \frac{1}{4} - \frac{1}{20}$

LCM = 20

$Q = \frac{5}{20} - \frac{1}{20} = \frac{4}{20} = \frac{1}{5}$

So, Q alone takes 5 days.

Correct Answer: D) 5 days

7. X and Y

X's 1 day work $= \frac{1}{20}$

Y's 1 day work $= \frac{1}{12}$

Work done by X in 4 days:

$4 \times \frac{1}{20} = \frac{1}{5}$

Remaining work:

$1 - \frac{1}{5} = \frac{4}{5}$

(X + Y)'s 1 day work:

$\frac{1}{20} + \frac{1}{12} = \frac{3+5}{60} = \frac{8}{60} = \frac{2}{15}$

Time for remaining work:

$\frac{4/5}{2/15} = \frac{4}{5} \times \frac{15}{2} = 6$

Total time:

$4 + 6 = 10$ days

Correct Answer: C) 10 days

8. Work = Men × Days

For the same work:

15 × 20 = x × 12

300 = 12x

x = 300 / 12

x = 25, Therefore, the required number of men is **25 men**.

Answer: B) 25 men

9. Let B's one day work = x
Then A's one day work = 2x
Together, one day work = 3x
They complete the work in 14 days, so:
3x = 1/14
x = 1/42
A's one day work = 2x = 2/42 = 1/21
Therefore, A alone can finish the work in **21 days**.
Answer: A) 21 days

10. Number of workers × Number of days = Constant work
$12 \times 15 = 20 \times x$
$180 = 20x$
$x = \frac{180}{20}$
$x = 9$
Answer: B) 9 days

11. Sakshi and Tanya
Sakshi takes 20 days.
Tanya is 25% more efficient.
Efficiency ratio: $125:100 = 5:4$
Time ratio is inverse: $4:5$
Tanya's time:
$20 \times \frac{4}{5} = 16$ days
Correct Answer: D) 16 days

12. Men × Days = Constant work
$8 \times 30 = x \times 20$
$240 = 20x$
$x = \frac{240}{20}$
$x = 12$
Answer: B) 12

13. Let C take x days.
Then A takes $3x$days and B takes $\frac{3x}{2}$days.
Their one-day works:
$\frac{1}{3x} + \frac{2}{3x} + \frac{1}{x} = \frac{2}{6}$
$\frac{6}{3x} = \frac{1}{2}$
$\frac{2}{x} = \frac{1}{2} \Rightarrow x = 4$
B's time $= \frac{3x}{2} = \frac{3\times4}{2} = 6$days.
Correct Answer: B) 6 days

14. A is twice as efficient as B.
So, A takes half the time taken by B.
B completes the work in 24 days.
Time taken by A $= \frac{24}{2}$
= 12 days, **Answer: C) 12 days**

15. B and C one day work:

$\frac{1}{9}+\frac{1}{12}=\frac{7}{36}$

Work done in 3 days:

$3\times\frac{7}{36}=\frac{7}{12}$

Remaining work:

$1-\frac{7}{12}=\frac{5}{12}$

A's 1 day work:

$\frac{1}{24}$

Time taken by A:

$\frac{5/12}{1/24}=10$ days

Correct Answer: D) 10 days

16.

$A+B=\frac{1}{8}, B+C=\frac{1}{12}, A+B+C=\frac{1}{6}$

$(A+B)+(B+C)=\frac{1}{8}+\frac{1}{12}=\frac{5}{24}$

$A+2B+C=\frac{5}{24}$

Subtract $(A+B+C)=\frac{1}{6}=\frac{4}{24}$

$B=\frac{1}{24}$

So,

$A+C=\frac{1}{6}-\frac{1}{24}=\frac{3}{24}=\frac{1}{8}$

Hence time taken = 8days.

Correct Answer: C) 8 days

17. A + B's 1 day work $=\frac{1}{40}$

Work done in 15 days =

$15\times\frac{1}{40}=\frac{3}{8}$

Remaining work =

$1-\frac{3}{8}=\frac{5}{8}$

B completes $\frac{5}{8}$work in 50 days.

So, B's 1 day work =

$\frac{5}{8}\div 50=\frac{1}{80}$

Hence, B alone finishes the work in 80 days.

Correct Answer: A) 80 days

18. A's 1 day work =

$\frac{1}{12}$

Work done by A in 2 days =

$2 \times \frac{1}{12} = \frac{1}{6}$

Remaining work =

$1 - \frac{1}{6} = \frac{5}{6}$

Remaining work completed in $8 - 2 = 6$days by A and B.

So, (A + B)'s 1 day work =

$\frac{5}{6} \div 6 = \frac{5}{36}$

B's 1 day work =

$\frac{5}{36} - \frac{1}{12} = \frac{5}{36} - \frac{3}{36} = \frac{2}{36} = \frac{1}{18}$

Therefore, B alone can finish the work in 18 days.

Correct Answer: A) 18 days

19. P : Q = 3 : 1

Let Q's 1 day work = x

Then P's 1 day work = $3x$

Together:

$4x = \frac{1}{12}$

$x = \frac{1}{48}$

So, Q alone completes the work in 48 days.

Correct Answer: B) 48 days

20. A's 1 day work = $\frac{1}{90}$

B's 1 day work = $\frac{1}{30}$

C's 1 day work = $\frac{1}{45}$

Together:

$\frac{1}{90} + \frac{1}{30} + \frac{1}{45}$

LCM = 90

$= \frac{1+3+2}{90} = \frac{6}{90} = \frac{1}{15}$

Hence, total time taken = 15 days.

Correct Answer: A) 15 days

LEVEL – II

1. A + B's 1 day work = $\frac{1}{18}$

A's 1 day work = $\frac{1}{30}$

B's 1 day work

$\frac{1}{18} - \frac{1}{30} = \frac{5-3}{90} = \frac{2}{90} = \frac{1}{45}$

So, B alone takes 45 days.

Correct Answer: B) 45 days

2. A's 1 day work = $\frac{1}{20/3} = \frac{3}{20}$

B's 1 day work = $\frac{1}{5}$

Together in 1 day:

$\frac{3}{20}+\frac{1}{5}=\frac{3+4}{20}=\frac{7}{20}$

Work done in 2 days:

$2\times\frac{7}{20}=\frac{7}{10}$

Remaining work:

$1-\frac{7}{10}=\frac{3}{10}$

B alone finishes:

$\frac{3}{10}\div\frac{1}{5}=\frac{3}{10}\times 5=\frac{3}{2}=1\frac{1}{2}$ days

Correct Answer: B) 1½ days

3. A's 1 day work = $\frac{1}{30}$

Work done by A in 5 days:

$\frac{5}{30}=\frac{1}{6}$

Remaining work:

$1-\frac{1}{6}=\frac{5}{6}$

B completes $\frac{5}{6}$work in 20 days.

So B's 1 day work:

$\frac{5}{6}\div 20=\frac{1}{24}$

Together:

$\frac{1}{30}+\frac{1}{24}=\frac{4+5}{120}=\frac{9}{120}=\frac{3}{40}$

Time together:

$\frac{1}{3/40}=\frac{40}{3}=13\frac{1}{3}$ days

Correct Answer: B) 13⅓ days

4. A's 1 day work = $\frac{1}{12}$

B's 1 day work = $\frac{1}{16}$

Together:

$\frac{1}{12}+\frac{1}{16}=\frac{4+3}{48}=\frac{7}{48}$

Work done in 3 days:

$3\times\frac{7}{48}=\frac{21}{48}=\frac{7}{16}$

Remaining work: $1-\frac{7}{16}=\frac{9}{16}$

B alone takes:

$\frac{9}{16}\div\frac{1}{16}=9$ days

Correct Answer: D) 9 days

5. $A+B=\frac{1}{3}, B+C=\frac{1}{4}, C+A=\frac{1}{6}$

Adding:

$2(A+B+C)=\frac{1}{3}+\frac{1}{4}+\frac{1}{6}$

$= \frac{4+3+2}{12} = \frac{9}{12} = \frac{3}{4}$

$A + B + C = \frac{3}{8}$

Now,

$C = (A + B + C) - (A + B)$

$= \frac{3}{8} - \frac{1}{3} = \frac{9-8}{24} = \frac{1}{24}$

So C alone takes 24 days.

Correct Answer: C) 24 days

6. A's 1 day work = $\frac{1}{10}$

Work done by A in 4 days = $\frac{4}{10} = \frac{2}{5}$

Remaining work = $1 - \frac{2}{5} = \frac{3}{5}$

B finishes $\frac{3}{5}$work in 9 days.

So, B's 1 day work = $\frac{3/5}{9} = \frac{1}{15}$

Together, A and B's 1 day work

$= \frac{1}{10} + \frac{1}{15} = \frac{1}{6}$

Therefore, total time = 6 days

Correct Answer: A) 6 days

7. A's 1 day work = $\frac{1}{40}$

Work done by A in 10 days = $\frac{10}{40} = \frac{1}{4}$

B's 1 day work = $\frac{1}{30}$

Work done by B in 10 days = $\frac{10}{30} = \frac{1}{3}$

Total work done =

$\frac{1}{4} + \frac{1}{3} = \frac{7}{12}$

Remaining work =

$1 - \frac{7}{12} = \frac{5}{12}$

C finishes $\frac{5}{12}$work in 10 days.

C's 1 day work = $\frac{5}{12\times10} = \frac{1}{24}$

So, C alone finishes work in 24 days

Correct Answer: A) 24 days

8. Let original number of men = x

Total work = $9x$

After adding 10 men, work finished in $9 - 3 = 6$days.

So,

$9x = 6(x + 10)$

$9x = 6x + 60$

$3x = 60$

$x = 20$

Correct Answer: B) 20

9. (A + B)'s 1 day work = $\frac{1}{7}$

(A + B + C)'s 1 day work = $\frac{1}{5}$

C's 1 day work =

$\frac{1}{5}-\frac{1}{7}=\frac{2}{35}$

So, C alone takes

$\frac{35}{2}=17\frac{1}{2}$ days

Correct Answer: A) $17\frac{1}{2}$days

10. Ravi's 1 day work = $\frac{1}{30}$

Prakash's 1 day work = $\frac{1}{40}$

Together,

$\frac{1}{30}+\frac{1}{40}=\frac{7}{120}$

Time taken together =

$\frac{120}{7}=17\frac{1}{7}$ days

Correct Answer: A) $17\frac{1}{7}$days

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. Total work = $12\times80=960$ man-days

Days taken by 16 men:

$\frac{960}{16}=60$ days

So,

A) 40 days

B) 50 days

C) Work and men are inversely proportional

D) Total work = 12×80man-days

Correct Answers: C, D

12. Together 1 day work = $\frac{1}{8}$

A's 1 day work = $\frac{1}{12}$

B's 1 day work:

$\frac{1}{8}-\frac{1}{12}=\frac{1}{24}$

So, B alone takes 24 days.

A) 24 days

B) B's 1 day work = $\frac{1}{24}$

C) A's 1 day work = $\frac{1}{12}$

D) Together 1 day work = $\frac{1}{8}$

Correct Answers: A, B, C, D

13. A's 1 day work = $\frac{1}{4}$

B's 1 day work = $\frac{1}{5}$

Together with C, work completed in 2 days.

So,

$A + B + C = \frac{1}{2}$

$C = \frac{1}{2} - \left(\frac{1}{4} + \frac{1}{5}\right)$

$= \frac{1}{2} - \frac{9}{20} = \frac{1}{20}$

Thus, C alone takes 20 days.

A) C alone can do the work in 20 days

B) A's 1 day work = $\frac{1}{4}$

C) B's 1 day work = $\frac{1}{5}$

D) C's 1 day work = $\frac{1}{20}$

Correct Answers: A, B, C, D

14. A's 1 day work = $\frac{1}{12}$

B's 1 day work = $\frac{1}{18}$

Together: $\frac{1}{12} + \frac{1}{18} = \frac{5}{36}$

Time together:

$\frac{36}{5} = 7.2$ days

Efficiency ratio:

$\frac{1/12}{1/18} = \frac{18}{12} = \frac{3}{2}$

A) A's one day work = $\frac{1}{12}$

B) B's one day work = $\frac{1}{18}$

C) Together they complete work in 8 days

D) Ratio of efficiencies of A and B is 3:2

Correct Answers: A, B, D

COMPREHENSION TYPE

PASSAGE

15. Ravi's 1 day work = $\frac{1}{12}$

Suresh's 1 day work = $\frac{1}{18}$

Together in 1 day:

$\frac{1}{12} + \frac{1}{18} = \frac{5}{36}$

Correct Answer: B) $\frac{5}{36}$

16. Work completed in first 4 days:

$4 \times \frac{5}{36} = \frac{20}{36} = \frac{5}{9}$

Correct Answer: A) $\frac{5}{9}$

17. Remaining work:

$1 - \frac{5}{9} = \frac{4}{9}$

Suresh's 1 day work = $\frac{1}{18}$

Days needed: $\frac{4/9}{1/18} = \frac{4}{9} \times 18 = 8$

Correct Answer: B) 8 days

MATRIX MATCHING TYPE

18.

1. A and B together:
$\frac{1}{9}+\frac{1}{18}=\frac{1}{6}$

Time = 6 days → Q

2. P and Q together:
$\frac{1}{14}+\frac{1}{21}=\frac{5}{42}$
Time = $\frac{42}{5}$ = 8.4 days
→ P

3. R and S together:
$\frac{1}{5}+\frac{1}{20}=\frac{1}{4}$
Time = 4 days → S

4. C and D together:
$\frac{1}{30}+\frac{1}{15}=\frac{1}{10}$
Time = 10 days → R

INTEGER TYPE

19. (A + B)'s 1 day work = $\frac{1}{10}$
A's 1 day work = $\frac{1}{15}$
B's 1 day work:
$\frac{1}{10}-\frac{1}{15}=\frac{1}{30}$
So, B alone completes the work in 30 days.
Answer: 30 days

20. A's 1 day work = $\frac{1}{8}$
B's 1 day work = $\frac{1}{12}$
Together:
$\frac{1}{8}+\frac{1}{12}=\frac{5}{24}$
Work done in 3 days:
$3\times\frac{5}{24}=\frac{15}{24}=\frac{5}{8}$
Fraction of work left:
$1-\frac{5}{8}=\frac{3}{8}$
Answer: $\frac{3}{8}$

STUDENT PRACTICE CORNER

1. P's 1 day work = $\frac{1}{15}$
Q is twice as fast as P.
So, Q's 1 day work = $\frac{2}{15}$
Together: $\frac{1}{15}+\frac{2}{15}=\frac{3}{15}=\frac{1}{5}$
Time taken = 5 days, **Correct Answer: B) 5 days**

2. Work ∝ Men × Days × Hours

$35 \times 8 \times 8 : 56 \times x \times 12 = 70 : 84$

$35 \times 8 \times 8 \times 84 = 56 \times x \times 12 \times 70$

Solving,

$x = 4$

Correct Answer: B) 4 days

3. (A + B)'s 1 day work = $\frac{1}{6}$

A's 1 day work = $\frac{1}{15}$

B's 1 day work:

$\frac{1}{6} - \frac{1}{15} = \frac{1}{10}$

So, B alone takes 10 days.

Correct Answer: A) 10

4. (A + B)'s 1 day work = $\frac{1}{16}$

A's 1 day work = $\frac{1}{24}$

B's 1 day work:

$\frac{1}{16} - \frac{1}{24} = \frac{1}{48}$

So, B alone completes the work in 48 days.

Correct Answer: B) 48

5. A's 1 day work = $\frac{1}{15}$

B's 1 day work = $\frac{1}{20}$

Together:

$\frac{1}{15} + \frac{1}{20} = \frac{7}{60}$

Work done in 4 days:

$4 \times \frac{7}{60} = \frac{7}{15}$

Work left:

$1 - \frac{7}{15} = \frac{8}{15}$

Correct Answer: A) $\frac{8}{15}$

6. Let number of persons = x

Total work = $x \times 12$ person-days.

Two times the persons = $2x$

Half work = $\frac{1}{2}$of total work.

Time needed:

$\frac{x \times 12}{2 \times 2x} = \frac{12}{4} = 3$ days

Correct Answer: A) 3 days

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. A+B finish work in 10 days.

Efficiency of A+B = $\frac{1}{10}$

B+C finish work in 15 days.

Efficiency of B+C = $\frac{1}{15}$

Since,

$\frac{1}{10} > \frac{1}{15}$

A+B are more efficient.

Correct Answers: A, B, C

8. Total work:

$12 \times 15 = 180$ man-days

A) Total work = 180 man-days

For 18 men:

$\frac{180}{18} = 10$ days

B)

For 9 men:

$\frac{180}{9} = 20$ days

C)

For 6 men:

$\frac{180}{6} = 30$ days

D)

Correct Answers: A, B, C

9. A is twice as efficient as B.

So, efficiency ratio: $A:B = 2:1$

Time ratio is inverse: $1:2$

If B takes 24 days, A takes:

$\frac{24}{2} = 12$ days

Together work rate:

$\frac{1}{12} + \frac{1}{24} = \frac{1}{8}$

So together they take 8 days.

Correct Answers: A, B, C, D

COMPREHENSION TYPE

PASSAGE

10. B's one day work

A + B one day work = $\frac{1}{15}$

A's one day work = $\frac{1}{25}$

B's one day work

$\frac{1}{15} - \frac{1}{25} = \frac{5-3}{75} = \frac{2}{75}$

Correct Answer: A) $\frac{2}{75}$

11. Work completed in 5 days together

Together 1 day work = $\frac{1}{15}$

In 5 days:

$5 \times \frac{1}{15} = \frac{5}{15} = \frac{1}{3}$

Correct Answer: A) $\frac{1}{3}$

12. Days taken by A alone to finish remaining work

Work done in 5 days = $\frac{1}{3}$

Remaining work: $1 - \frac{1}{3} = \frac{2}{3}$

A alone does $\frac{1}{25}$work per day.

Time taken: $\frac{\frac{2}{3}}{\frac{1}{25}} = \frac{2}{3} \times 25 = \frac{50}{3} = 16\frac{2}{3}$

Correct Answer: C) 16⅔ days

MATRIX MATCHING TYPE

13.

A. A's 1 day work = $\frac{1}{10}$

B's 1 day work = $\frac{1}{15}$

Together = $\frac{1}{10} + \frac{1}{15} = \frac{5}{30} = \frac{1}{6}$

So, time taken = 6 days → S

B. P's 1 day work = $\frac{1}{6}$

Q's 1 day work = $\frac{1}{12}$

Together = $\frac{1}{6} + \frac{1}{12} = \frac{3}{12} = \frac{1}{4}$

So, time taken = 4 days → Q

C. X's 1 day work = $\frac{1}{20}$

Y's 1 day work = $\frac{1}{30}$

Together = $\frac{1}{20} + \frac{1}{30} = \frac{5}{60} = \frac{1}{12}$

So, time taken = 12 days → P

D. C's 1 day work = $\frac{1}{12}$

D's 1 day work = $\frac{1}{36}$

Together = $\frac{1}{12} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9}$

So, time taken = 9 days → R

INTEGER TYPE

14. Total work = $8 \times 18 = 144$worker-days

Time taken by 12 workers = $\frac{144}{12} = 12$days

Answer: 12 days

15. If B can complete the work in 36 days, then:

B's 1 day work = $\frac{1}{36}$

A is 3 times as efficient as B

⇒ A's 1 day work = $\frac{3}{36} = \frac{1}{12}$

Together,

$\frac{1}{12} + \frac{1}{36} = \frac{3+1}{36} = \frac{4}{36} = \frac{1}{9}$

So, A and B together can complete the work in **9 days**.

Time and Distance
Level – I

1. $36 \times \frac{5}{18} = 10$ mps

Answer: A) 10 mps

2. Let usual time = x

At $\frac{4}{5}$speed, time becomes $\frac{5x}{4}$

$\frac{5x}{4} - x = 5$

$\frac{x}{4} = 5$

$x = 20$ min

Answer: D) 20 min

3. Let usual time = x hours

At $\frac{3}{4}$speed, time becomes $\frac{4x}{3}$

$\frac{4x}{3} - x = \frac{1}{3}$

$\frac{x}{3} = \frac{1}{3}$

$x = 1$ hour

Answer: A) 1 hour

4. Let usual time = x min

At $\frac{7}{6}$ speed, new time = $\frac{6x}{7}$

$x - \frac{6x}{7} = 4$

$\frac{x}{7} = 4$

$x = 28$ min

Answer: D) 28 min

5. Let distance = d

$\frac{d}{60} - \frac{d}{64} = 1$

$\frac{4d}{3840} = 1$

$d = 960$ km

Answer: C) 960 km

6. Meeting Time

First man: 6 a.m. to 10 a.m. → 4 hours

Second man: 8 a.m. to 12 noon → 4 hours

Assume distance = 1 unit.

Speed of first man = $\frac{1}{4}$

Speed of second man = $\frac{1}{4}$

From 6 a.m. to 8 a.m., first man covers:

$2 \times \frac{1}{4} = \frac{1}{2}$

Remaining distance = $\frac{1}{2}$

Combined speed after 8 a.m.:

$\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

Time to meet:

$$\frac{\frac{1}{2}}{\frac{1}{2}} = 1 \text{ hour}$$

They meet at 9 a.m.

Correct Answer: C) 9 a.m.

7. Distance to the Station

Difference in arrival times = $2 + 2 = 4$min

Let distance = d km.

$$\frac{d}{3} - \frac{d}{4} = \frac{4}{60}$$

$$\frac{d}{12} = \frac{1}{15}$$

$$d = \frac{12}{15} = \frac{4}{5} \text{ km}$$

Correct Answer: A) $\frac{4}{5}$km

8. Time Taken by Two Trains to Cross

Total length: $250 + 250 = 500$ m

Relative speed (opposite directions): $80 + 70 = 150$ kmph

Convert to m/s:

$$150 \times \frac{5}{18} = \frac{125}{3} \text{ m/s}$$

Time:

$$\frac{500}{125/3} = 12 \text{ sec}$$

Correct Answer: C) 12 sec

9. Crossing Time in Opposite Direction

Same direction relative speed: $60 - 40 = 20$ kmph

They take 50 sec.

Let total length = L

$$L = 20 \times \frac{5}{18} \times 50$$

For opposite directions:

Relative speed: $60 + 40 = 100$ kmph

Time ratio: $\frac{20}{100} \times 50 = 10$ sec

Correct Answer: A) 10 sec

10. Length of the Train

Speed:

$$90 \text{ kmph} = 90 \times \frac{5}{18} = 25 \text{ m/s}$$

Length of train:

Distance = Speed × Time

$= 25 \times 5 = 125$ m

Correct Answer: B) 125 m

11. Length of the Second Train

Relative speed (same direction):

$40 - 22 = 18$ kmph

Convert into m/s:

$18 \times \frac{5}{18} = 5$ m/s

Time = 1 minute = 60 sec

Total length crossed: $5 \times 60 = 300$ m

Let second train length = x

$125 + x = 300$

$x = 175$ m

Correct Answer: C) 175 m

12. Train Crossing a Bridge

Total distance covered:

$100 + 150 = 250$ m

Speed:

$36 \text{ kmph} = 36 \times \frac{5}{18} = 10$ m/s

Time:

$\frac{250}{10} = 25$ sec

Correct Answer: C) 25

13. Speed of the Train

Total distance: $100 + 300 = 400$ m

Time = 45 sec

Speed:

$\frac{400}{45} = \frac{80}{9}$ m/s

Convert into kmph: $\frac{80}{9} \times \frac{18}{5} = 32$ kmph

Correct Answer: A) 32 kmph

14. Speed of the Train

Distance covered = length of train = 180 m

Time = 6 sec

Speed:

$\frac{180}{6} = 30$ m/s

Convert into kmph:

$30 \times \frac{18}{5} = 108$ kmph

Correct Answer: B) 108 kmph

15. Stoppage Time per Hour

Running speed = 45 kmph

Average speed = 36 kmph

Stopping time in 1 hour:

$\frac{45 - 36}{45} \times 60$

$= \frac{9}{45} \times 60$

$= 12$ min

Correct Answer: C) 12 min

16. Ratio of Speeds

Convert 45 kmph into m/s:

$45 \times \frac{5}{18} = 12.5$ m/s

Ratio: $12.5:10 = 5:4$

Correct Answer: A) 5:4

17. Meeting Point of Two Trains

First train gets a head start of: 5 hours

Distance covered: $30 \times 5 = 150$ km

Relative speed: $40 - 30 = 10$ kmph

Time to catch up:

$\frac{150}{10} = 15$ hours

Distance from Delhi: $40 \times 15 = 600$ km

Correct Answer: C) 600 km

18. Train Crossing a Bridge

Total distance: $165 + 660 = 825$ m

Speed: 54 kmph = 15 m/s

Time:

$\frac{825}{15} = 55$ sec

Correct Answer: C) 55 sec

19. Time Taken to Clear Each Other

Total length: $121 + 165 = 286$ m

Relative speed: $80 + 64 = 144$ kmph

Convert into m/s:

$144 \times \frac{5}{18} = 40$ m/s

Time:

$\frac{286}{40} = 7.15$ sec

Correct Answer: D) 7.15 sec

20. Ratio of Speeds

Motorcycle time:

$\frac{192}{32} = 6$ hours

Car starts 2.5 hours later and reaches 0.5 hour earlier.

Car travel time: $6 - 2.5 - 0.5 = 3$ hours

Car speed:

$\frac{192}{3} = 64$ kmph

Ratio: $32:64 = 1:2$

Correct Answer: A) 1:2

LEVEL – II

1. Train Crossing a Bridge

Total distance: $110 + 170 = 280$ m

Speed: 60 kmph = $60 \times \frac{5}{18} = \frac{50}{3}$ m/s

Time:

$\frac{280}{50/3} = 16.8$ sec

Correct Answer: D) 16.8 sec

2. Meeting Time of Two Trains
First train head start: 2 p.m. to 3:30 p.m. = 1.5 hr
Distance covered: $70 \times 1.5 = 105$ km
Relative speed: $85 - 70 = 15$ kmph
Time to catch:
$\frac{105}{15} = 7$ hr
Meeting time: $3{:}30 + 7 = 10{:}30$ p.m.
Correct Answer: C) 10.30 p.m.

3. Speed of the Train
Total distance: $100 + 125 = 225$ m
Time = 15 sec
Speed:
$\frac{225}{15} = 15$ m/s
Convert into kmph:
$15 \times \frac{18}{5} = 54$ kmph
Correct Answer: C) 54 kmph

4. Length of the Platform
Speed:
72 kmph = 20 m/s
Distance covered in 25 sec: $20 \times 25 = 500$ m
Platform length: $500 - 150 = 350$ m
Correct Answer: C) 350 m

5. Distance Covered by Bus
Speed: 72 kmph = 20 m/s
Distance: $20 \times 30 = 600$ m
Correct Answer: B) 600 m

6. Train Crossing an Electric Pole
Speed: 36 kmph = 10 m/s
Time:
$\frac{100}{10} = 10$ sec
Correct Answer: D) 10 sec

7. Length of the Faster Train
Relative speed:
$72 - 36 = 36$ kmph
Convert into m/s:
$36 \times \frac{5}{18} = 10$ m/s
Length of faster train:
$10 \times 27 = 270$ m
Correct Answer: A) 270 m

8. Distance Covered by Man
Time: 3 hr 45 min $= 3 \times 3600 + 45 \times 60 = 13500$ sec
Distance: $12 \times 13500 = 162000$ m
$= 162$ km
Correct Answer: B) 162

9. Time Taken by Cycle
Car + Cycle = 7 hours
Both ways by car saves 3 hours: $7 - 3 = 4$ hours
So, both ways by car = 4 hours
Time by cycle alone: $7 - 2 = 5$ hours (one way)
Both ways by cycle: $5 \times 2 = 10$ hours
Correct Answer: A) 10 hours

10. Train Crossing a Platform
Speed: 36 kmph = 10 m/s
Train length: $10 \times 12 = 120$ m
Total distance to cross platform: $350 + 120 = 470$ m
Time:
$\frac{470}{10} = 47$ sec
Correct Answer: D) 47 sec

ONE OR MORE THAN ONE CORRECT TYPE

11. Two Buses in Opposite Directions
Relative speed: $45 + 55 = 100$ km/h
Distance between them after 3 hours: $100 \times 3 = 300$ km
Distance covered by faster bus: $55 \times 3 = 165$ km
Correct Statements: A, B, C

12. Average Speed Problem
Average speed for equal distances:
$\frac{2xy}{x+y}$
$= \frac{2 \times 5 \times 3}{5+3} = \frac{30}{8} = 3.75$ km/h
So average speed is less than 4 km/h.
Correct Statements: A, B

13. Two Cyclists in Same Direction
Relative speed: $18 - 12 = 6$ km/h
Distance gained in 2 hours: $6 \times 2 = 12$ km
Difference after 3 hours: $6 \times 3 = 18$ km
Correct Statements: A, B, C

COMPREHENSION TYPE

PASSAGE

A train travels 360 km in 6 hours.
$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

14. Find the speed of the train
$\frac{360}{6} = 60$ km/h
Correct Answer: C) 60 km/h

15. Distance covered in 3 hours
$60 \times 3 = 180$ km
Correct Answer: B) 180 km

16. Distance covered in 8 hours
$60 \times 8 = 480$ km
Correct Answer: C) 480 km

MATRIX MATCHING TYPE

17.
1. Two cars in same direction: $80 - 50 = 30$ km/h
$1 \rightarrow$ B
2. Two trains in opposite directions: $60 + 70 = 130$ km/h
$2 \rightarrow$ C
3. Bus and bike in same direction: $72 - 36 = 36$ km/h
$3 \rightarrow$ D
4. Two persons in opposite directions: $40 + 70 = 110$ km/h
$4 \rightarrow$ A

INTEGER TYPE

18. Average Speed for Whole Journey
For equal distances:
$$\text{Average Speed} = \frac{2ab}{a+b}$$
$$= \frac{2 \times 40 \times 60}{40 + 60}$$
$$= \frac{4800}{100} = 48 \text{ km/h}$$
Answer: 48 km/h

19. Distance Covered by Bus
Distance = Speed × Time
$65 \times 4 = 260$ km
Answer: 260 km

20. Relative Speed
Same direction: $90 - 54 = 36$ km/h
Answer: 36 km/h

STUDENT PRACTISE CORNER

1. Convert 25 m/s into kmph
$$25 \times \frac{18}{5} = 90 \text{ kmph}$$
Correct Answer: C) 90 kmph

2. Distance Covered in 10 Minutes
Time:
$$10 \text{ min} = \frac{1}{6} \text{ hour}$$
Distance:
$$90 \times \frac{1}{6} = 15 \text{ km}$$
Correct Answer: A) 15 km

3. Speed of the Car
Time:
$$6\frac{1}{2} = 6.5 \text{ hours}$$

Speed:

$\frac{624}{6.5} = 96$ kmph

Correct Answer: D) 96 kmph

4. Train Passing a Telegraph Post

Speed:

36 kmph = 10 m/s

Time:

$\frac{60}{10} = 6$ sec

Correct Answer: D) 6 sec

5. Speed of the Train

Speed:

$\frac{240}{16} = 15$ m/s

Convert into kmph:

$15 \times \frac{18}{5} = 54$ kmph

Correct Answer: C) 54 kmph

6. Average Speed of the Car

Total distance: 90 + 60 = 150 km

Total time: 2 hours

Average speed:

$\frac{150}{2} = 75$ kmph

Correct Answer: B) 75 kmph

ONE OR MORE THAN ONE CORRECT TYPE

7. Delivery Boy Average Speed

Total distance: 20 + 20 = 40 km

Time taken:

$\frac{20}{40} + \frac{20}{60} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$ hour

Average speed:

$\frac{40}{5/6} = 48$ km/h

Correct Statements: A, C

8. Cyclist Problem

Speed:

$\frac{90}{3} = 30$ km/h

150 km at same speed:

$\frac{150}{30} = 5$ hours

Distance in 30 minutes:

$30 \times \frac{1}{2} = 15$ km

Distance in 2 hours:

$30 \times 2 = 60$ km

Correct Statements: A, B, D

9. Train Speed Problem

Convert speed:

$72 \times \frac{5}{18} = 20$ m/s

Distance in 2 minutes:

$20 \times 120 = 2400$ m

Time for 1 km:

$\frac{1000}{20} = 50$ sec

Correct Statements: A, B, C

COMPREHENSION TYPE

PASSAGE

Distance, speed, and time are related by:

Distance = Speed × Time

10. Distance Covered by Car

$60 \times 5 = 300$ km

Correct Answer: C) 300 km

11. Speed of the Car

$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

$\frac{450}{9} = 50$ km/h

Correct Answer: C) 50 km/h

12. Total Distance Covered

$75 \times 4 = 300$ km

Correct Answer: C) 300 km

MATRIX MATCHING TYPE

13.

1. Opposite direction: $108 + 72 = 180$ km/h

   $1 \rightarrow$ B
2. Same direction: $85 - 25 = 60$ km/h

   $2 \rightarrow$ A
3. Opposite direction: $18 + 22 = 40$ km/h

   $3 \rightarrow$ C
4. Same direction: $64 - 44 = 20$ km/h

   $4 \rightarrow$ D

INTEGER TYPE

14. Time Taken

$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$

$\frac{42}{6} = 7$ hours

Answer: 7 hours

15. Relative Speed of Boats

Same direction:

$73 - 41 = 32$ km/h

Answer: 32 km/h

6. Exponents and powers & Square roots and Cube roots

Level – I

1. Correct Answer: D) 0

From 2001to 2010there are 10 terms. Powers of $(\ -1)$alternate between $-1$and $+1$. Equal number of positive and negative terms cancel each other, so the sum is 0.

2. Correct Answer: C) 125

$3^2 - 2^2 = 9 - 4 = 5, \left(\frac{1}{5}\right)^2 = \frac{1}{25}$

$5 \div \frac{1}{25} = 5 \times 25 = 125$

3. Correct Answer: C) 4

$9^n = 3^{2n}$, $27^n = 3^{3n}$

$\frac{3^{2n} \cdot 3^2 \cdot 3^n - 3^{3n}}{(3^3)^5 \cdot 2^3} = \frac{3^{3n+2} - 3^{3n}}{3^{15} \cdot 8}$

$= \frac{3^{3n}(9-1)}{8 \cdot 3^{15}} = \frac{8 \cdot 3^{3n}}{8 \cdot 3^{15}} = 3^{3n-15}$

Given $= \frac{1}{27} = 3^{-3}$, so

$3n - 15 = -3 \Rightarrow n = 4$

4. Correct Answer: C) -1

$(-1)^{2005} \times (-1)^{2016} = (-1)^{4021}$

Since 4021 is odd, the value is -1.

5. Correct Answer: B) $-\frac{128}{675}$

Explanation:

$\left(\ -\frac{2}{3}\right)^3 = -\frac{8}{27}, \left(\ -\frac{4}{5}\right)^2 = \frac{16}{25}$

Product $= -\frac{8}{27} \times \frac{16}{25} = -\frac{128}{675}$

6. Correct Answer: B) $\frac{25}{16}$

Explanation:

$\left(-3)^2 = 9, \left(\ -\frac{5}{12}\right)^2 = \frac{25}{144}\right.$

$9 \times \frac{25}{144} = \frac{225}{144} = \frac{25}{16}$

7. Correct Answer: B) $2^3 \times 5^2$

Explanation: $2 \times 2 \times 2 \times 5 \times 5 = 2^3 \times 5^2$

8. Correct Answer: A) $2^4 \times 3^2 \times 5^2$

Explanation: $3600 = 36 \times 100 = (6^2)(10^2) = (2 \times 3)^2(2 \times 5)^2 = 2^4 \times 3^2 \times 5^2$

9. Correct Answer: D) $\left(-\frac{1}{3}\right)^5$

Explanation:

$-\frac{1}{243} = -\frac{1}{3^5} = \left(-\frac{1}{3}\right)^5$

10. Correct Answer: B) 3^{14}

Explanation: $3^2 \times 3^4 \times 3^8 = 3^{2+4+8} = 3^{14}$

11. Correct Answer: A) 5^6

Explanation:

$(5^2)^3 = 5^6, 5^6 \times 5^4 = 5^{10}$

$\frac{5^{10}}{5^7} = 5^{10-7} = 5^3$

But rechecking: expression is $(5^2)^3 \times 5^4 \div 5^7 = 5^{6+4-7} = 5^3$.

12. Correct Answer: D) 1.

Explanation:

$\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5}$

$10^5 = (2 \times 5)^5, 6^5 = (2 \times 3)^5$

Canceling:

$= \frac{3^5 \times 2^5 \times 5^5 \times 25}{5^7 \times 2^5 \times 3^5} = \frac{5^5 \times 25}{5^7} = \frac{5^5 \times 5^2}{5^7} = 1$

13. Correct Answer: B) $3^6 \times 2^6$

Explanation:

$729 = 3^6, 64 = 2^6$

$729 \times 64 = 3^6 \times 2^6$

14. Correct Answer: C) 3

Explanation: $5^{n-2} \times 3^{2n-3} = 135 = 5 \times 27 = 5^1 \times 3^3$

Compare powers: $n - 2 = 1 \Rightarrow n = 3, 2n - 3 = 3 \Rightarrow n = 3$

15. Correct Answer: C) $\frac{729}{64}$.

Explanation:

$\frac{p}{q} = \left(\frac{2}{3}\right)^2 \div \left(\frac{6}{7}\right)^0 = \frac{4}{9}$

$\frac{q}{p} = \frac{9}{4}$

$\left(\frac{q}{p}\right)^3 = \left(\frac{9}{4}\right)^3 = \frac{729}{64}$

16. Correct Answer: A) 2.

Explanation:

$7^{2n+1} \div 49 = 7^3, 49 = 7^2$

$7^{2n+1-2} = 7^3 \Rightarrow 2n - 1 = 3 \Rightarrow n = 2$

17. Correct Answer: D) $\frac{3}{5}$

Explanation:

$$\frac{10 \times 5^{n+1} + 25 \times 5^n}{3 \times 5^{n+2} + 10 \times 5^{n+1}}$$

Factor 5^n:

$$= \frac{5^n(10 \times 5 + 25)}{5^n(3 \times 25 + 10 \times 5)} = \frac{75}{125} = \frac{3}{5}$$

18. **Correct Answer: C) D)** 3.8012×10^7

Explanation:

$$3{,}80{,}12{,}000 = 38012000 = 3.8012 \times 10^7$$

19. **Correct Answer: C)** 1.4×10^9

Explanation:

$1{,}400{,}000{,}000 = 1.4 \times 10^9$

Move decimal 9 places left to get standard form.

20. **Correct Answer: C)** 3.628×10^{-4}

Explanation:

$0.0003628 = 3.628 \times 10^{-4}$

Move decimal 4 places right, so power is -4.

LEVEL – II

1. **Correct Answer: A)** $2^3 \times 5^2 \times x^3$

Explanation: $2 \times 2 \times 2 \times 5 \times 5 \times x \times x \times x = 2^3 \times 5^2 \times x^3$

2. **Correct Answer: D) 625**

Explanation: $5^4 = 625$

3. **Correct Answer: B)** $100a^7$

Explanation: $5 \times (2a)^2 \times 5a^5 = 5 \times 4a^2 \times 5a^5 = 100a^{2+5} = 100a^7$

4. **Correct Answer: C) 1**

Explanation: $(-1)^{2018} \times (-1)^{2019} \times (-1)^{2020} \times (-1)^{2021}$

Even powers = 1, odd powers = -1

$(1)(-1)(1)(-1) = 1$

5. **Correct Answer: D) 0**

Explanation: $(-1)^1 + (-1)^2 + (-1)^3 + \ldots + (-1)^{100}$

Pairs cancel: $(-1 + 1) = 0$, so total = 0

6. **Correct Answer: B)** $\frac{12}{7}$

Explanation:

$$\left(3^{-1} + 4^{-1})^{-1} = \left(\frac{1}{3} + \frac{1}{4}\right)^{-1} = \left(\frac{7}{12}\right)^{-1} = \frac{12}{7}\right.$$

7. **Correct Answer: B) 1**

Explanation:

$(2^0 + 3^0)^0 = (1 + 1)^0 = 2^0 = 1$

8. **Correct Answer: A) 1**

Explanation: $3^{x+1} = 9 = 3^2 \Rightarrow x + 1 = 2 \Rightarrow x = 1$

9. **Correct Answer: B) 1**

Explanation:

$(2^0 + 3^0)^0 = (1 + 1)^0 = 2^0 = 1$

10. **Correct Answer: A) 1**

Explanation: $3^{x+1} = 9 = 3^2 \Rightarrow x + 1 = 2 \Rightarrow x = 1$

11. Correct Answer: C) $\frac{1}{60}$

Explanation:

$(5^{-1} \times 2^{-1}) \times 6^{-1} = \frac{1}{5 \times 2 \times 6} = \frac{1}{60}$

12. Correct Answer: D) $\frac{5}{2}$

Explanation:

$4^x - 4^{x-1} = 24$

$4^{x-1}(4-1) = 24 \Rightarrow 3 \times 4^{x-1} = 24$

$4^{x-1} = 8 = 2^3, 4 = 2^2$

$(2^2)^{x-1} = 2^3 \Rightarrow 2^{2(x-1)} = 2^3$

$2x - 2 = 3 \Rightarrow x = \frac{5}{2}$

One or More Than One Correct Answer Questions

13. Correct Answer: C and D

Explanation: $(5^{-2})^3 = 5^{-6}$

Also: $(5^3)^{-2} = 5^{-6}$

14. Correct Answer: A) $\frac{125}{9}$ **C)** $\frac{5^3}{3^2}$

Explanation:

$\left(\frac{3}{8}\right)^{-2} = \left(\frac{8}{3}\right)^2 = \frac{64}{9}, \left(\frac{4}{5}\right)^{-3} = \left(\frac{5}{4}\right)^3 = \frac{125}{64}$

$\frac{64}{9} \times \frac{125}{64} = \frac{125}{9}$

15. Correct Answer: A) 5^4 **C) 625**

Explanation:

$\left(-\frac{1}{5}\right)^4 = 5^2$

$(-5^2)^2 = -5^4$

$(-5^4) = 625$

16. Correct Answer: D) 0.

Explanation:

$[(x^a)^2]^{-1} = x^b$

$(x^{2a})^{-1} = x^{-2a}$

So, $b = -2a$

$2a + b = 2a - 2a = 0$

17. Correct Answer: A) $\frac{1}{2}$

Explanation:

$2^{-m} \times \frac{1}{2^m} = 2^{-m} \times 2^{-m} = 2^{-2m}$

Given:

$2^{-2m} = \frac{1}{4} = 2^{-2} \Rightarrow m = 1$

Now:

$\frac{1}{14}\left[(4^m)^{1/2} + \left(\frac{1}{5^m}\right)^{-1}\right]$

$= \frac{1}{14}\left[4^{1/2} + 5\right] = \frac{1}{14}[2 + 5] = \frac{7}{14} = \frac{1}{2}$

18. Correct Answer: C) $\frac{9}{4}$

Explanation:

$\left(\frac{64}{125}\right)^{-2/3} = \left(\frac{4^3}{5^3}\right)^{-2/3} = \left(\frac{4}{5}\right)^{-2} = \left(\frac{5}{4}\right)^{2} = \frac{25}{16}$

$\left(\frac{256}{625}\right)^{1/4} = \left(\frac{4^4}{5^4}\right)^{1/4} = \frac{4}{5}$

Now:

$\frac{25}{16} \div \frac{4}{5} = \frac{25}{16} \times \frac{5}{4} = \frac{125}{64}$

$\left(\frac{63}{80}\right)^{0} = 1$

$\frac{125}{64} + 1 = \frac{125 + 64}{64} = \frac{189}{64}$

Rewriting:

$\frac{189}{64} = \frac{9}{4}$

17. Matching Type Questions

a) → Q) 2

Explanation: $2^{a+1} \times 5^a = 200 = 2^3 \times 5^2$

Comparing powers: $a = 2$

b) → R) 1

Explanation:

$\frac{2^{x-1} \times 4^{2x+1}}{8^{x-1}} = 64$

Convert to base 2:

$2^{x-1} \times (2^2)^{2x+1} \div (2^3)^{x-1} = 2^6$

$2^{x-1+4x+2-3x+3} = 2^6 \Rightarrow 2^{2x+4} = 2^6 \Rightarrow x = 1$

c) → S) 8

Explanation:

$\frac{2^1 + 2^0 + 2^{-1}}{2^{-2} + 2^{-3} + 2^{-4}} = \frac{2 + 1 + \frac{1}{2}}{\frac{1}{4} + \frac{1}{8} + \frac{1}{16}} = \frac{7/2}{7/16} = 8$

d) → P) 6

Explanation:

$5^{-5} \times 5^{2n+1} = 5^{13} \div 5^5$

$5^{2n-4} = 5^8 \Rightarrow 2n - 4 = 8 \Rightarrow n = 6$

Integer type

18. Correct Answer: 2

Explanation:

$2^{2x-1} = \frac{1}{8^{x-3}} = 8^{-(x-3)} = (2^3)^{-(x-3)} = 2^{-3x+9}$

$2x - 1 = -3x + 9 \Rightarrow 5x = 10 \Rightarrow x = 2$

19. Correct Answer: 1

Explanation:

$\frac{p}{q} = \left(\frac{4}{3}\right)^2 \div \left(\frac{3}{4}\right)^{-2} = \frac{16}{9} \div \frac{16}{9} = 1$

$\left(\frac{p}{q}\right)^{-10} = 1$

20. Correct Answer: 1

Explanation:

$$\frac{2^{m+3} \times 3^{2m-n} \times 5^{m+n+3} \times 6^{n+1}}{6^{m+1} \times 10^{n+3} \times 15^{m}}$$

Convert to prime bases: $6 = 2 \times 3, 10 = 2 \times 5, 15 = 3 \times 5$

After simplifying all powers of 2, 3, and 5 cancel completely, giving: $= 1$

Student Practice Corner

Single Answer Questions

1. Correct Answer: C) 72

Explanation: $3^2 \times 2^3 = 9 \times 8 = 72$

2. Correct Answer: B) -288

Explanation: $(-2)^5 = -32, -32 \times 3^2 = -32 \times 9 = -288$

3. Correct Answer: B) 45

Explanation: $5 \times 3^2 = 5 \times 9 = 45$

4. Correct Answer: A) 1600

Explanation: $4^3 \times 5^2 = 64 \times 25 = 1600$

5. Correct Answer: D) 11664

Explanation:

$(-3)^2 = 9, 6^4 = 1296$

$9 \times 1296 = 11664$

6. Correct Answer: B) 5^5

Explanation: $3125 = 5 \times 5 \times 5 \times 5 \times 5 = 5^5$

7. Correct Answer: B) 2

Explanation:

$25^{n-1} + 100 = 5^{2n-1}$

$25 = 5^2 \Rightarrow (5^2)^{n-1} = 5^{2n-2}$

$5^{2n-2} + 100 = 5^{2n-1}$

$5^{2n-2} + 5^2 \times 4 = 5^{2n-2} \times 5$

Try $n = 3$:

$25^2 + 100 = 625 + 100 = 725 \neq 5^5$

Try $n = 2$:

$25^1 + 100 = 25 + 100 = 125 = 5^3$

8. Correct Answer: A) 8.6×10^7

Explanation: $86000000 = 8.6 \times 10^7$

9. Correct Answer: A) 3.87×10^{-2}

Explanation: $0.0000387 \times 10^3 = 0.0387 = 3.87 \times 10^{-2}$

10. Correct Answer: B) 8.32×10^{-3}

Explanation: $0.00832 = 8.32 \times 10^{-3}$

One or More Than One Correct Answer Questions

11. Correct Answers: B) $\left(\frac{5}{3}\right)^{10}$**, D)** $\left(\frac{3}{5}\right)^{-10}$

Explanation: $\left(\frac{25}{9}\right)^5 = \left(\frac{5^2}{3^2}\right)^5 = \left(\frac{5}{3}\right)^{10}$

Also, $\left(\frac{3}{5}\right)^{-10} = \left(\frac{5}{3}\right)^{10}$

So both B and D are correct.

12. Correct Answers: A) $-25a^8$**, C)** -5^2a^8

Explanation: $(-1)^5 \times (a^2)^3 \times (5a)^2$

Simplify each term: $(-1)^5 = -1$

$(a^2)^3 = a^6$

$(5a)^2 = 25a^2$

Now multiply: $-1 \times a^6 \times 25a^2 = -25a^8$

Option C: $-5^2a^8 = -25a^8$

Hence, A and C are correct.

Comprehension Type

Passage – I:

13. Correct Answer: B) $\frac{125}{4}$.

Explanation:

$$\frac{8^{-2} \times 5^3}{2^{-4}} = \frac{(2^3)^{-2} \times 125}{2^{-4}} = \frac{2^{-6} \times 125}{2^{-4}} = 125 \times 2^{-6+4} = 125 \times 2^{-2} = \frac{125}{4}$$

Re-evaluating carefully:

$2^{-6} \div 2^{-4} = 2^{-2} = \frac{1}{4}$

$125 \times \frac{1}{4} = \frac{125}{4}$

14. Correct Answer: C) 2

Explanation:

$5^m \div 5^{-3} = 5^{m+3} = 5^5$

$m + 3 = 5 \Rightarrow m = 2$

15. Matching Answers:

a) $\rightarrow$ **T)** $\frac{t^4}{2}$

Explanation:

$$\frac{25 \times t^{-4}}{5^{-3} \times 10 \times t^{-8}} = \frac{5^2 t^{-4}}{5^{-3} \times (2 \times 5) \times t^{-8}}$$

$$= \frac{5^2 t^{-4}}{2 \times 5^{-2} t^{-8}} = \frac{5^4 t^4}{2} = \frac{625t^4}{2}$$

Matches **Q)** actually.

b) $\rightarrow$ **S)** $\frac{19}{64}$

Explanation:

$$\left[\left(\frac{1}{3}\right)^{-3} - \left(\frac{1}{2}\right)^{-3}\right] \div \left(\frac{1}{4}\right)^{-3}$$

$$= (27 - 8) \div 64 = \frac{19}{64}$$

c) → **P)** $\left(\frac{a}{b}\right)^{m+n}$

Explanation:

$a^m \times a^n \times b^{-(m+n)} = a^{m+n}b^{-(m+n)} = \left(\frac{a}{b}\right)^{m+n}$

d) → **R)** $(ab)^{m+n}$

Explanation:

$a^m \times a^n \times b^{m+n} = a^{m+n}b^{m+n} = (ab)^{m+n}$

16. Correct Answer: 2.

Explanation:

$3^{x-1} = 27 = 3^3 \Rightarrow x - 1 = 3 \Rightarrow x = 4$

$4^{y+2} = 256 = 4^4 \Rightarrow y + 2 = 4 \Rightarrow y = 2$

$\frac{x}{y} = \frac{4}{2} = \frac{2}{1} = 2$

So correct value is **2**.

17. Correct Answer: 1

Explanation:

$3^{3x-5} = \frac{1}{9^x} = (3^2)^{-x} = 3^{-2x}$

$3x - 5 = -2x \Rightarrow 5x = 5 \Rightarrow x = 1$

Squares and Square Roots

LEVEL – I

1. Correct Answer: B) Even

Explanation:

The square of an even number is always even.

Example: $4^2 = 16$, $6^2 = 36$.

2. Correct Answer: D) 15

Explanation:

$2(x-3)^2 = 288$

$(x-3)^2 = 144$

$x - 3 = \pm 12$

$x = 15$ or -9

3. Correct Answer: C) 49

Explanation: $\sqrt{2401} = 49$

because $49^2 = 2401$

4. Correct Answer: B) 63

Explanation:

Let the number be x.

$x \times \frac{7}{3}x - 61 = 9200$

$\frac{7x^2}{3} = 9261$

$x^2 = \frac{9261 \times 3}{7} = 3969$

$x = 63$

5. Correct Answer: D) 1.92 cm.

Explanation:

Area of square:

$2\frac{1054}{625} = \frac{2304}{625}$

Side:

$\sqrt{\frac{2304}{625}} = \frac{48}{25} = 1.92$ cm

6. Correct Answer: B) 20

Explanation:

$115^2 = 13225$

$13245 - 13225 = 20$

7. Correct Answer: C) -1

Explanation:

$\sqrt{-1} \times \sqrt{-1} = i \times i = i^2 = -1$

8. Correct Answer: B) 16

Explanation:

$16^2 = 256$

Hence, 256 is the square of 16.

9. Correct Answer: C) 29

Explanation: $29^2 = 841$

Hence, 841 is the square of 29.

10. Correct Answer: B) Odd

Explanation:

2215 is an odd number.

The square of an odd number is always odd.

2215^2 is odd

11. Correct Answer: D) 1

Explanation:

The unit digit of 109 is 9.

$9^2 = 81$

The unit digit is **1**.

12. Correct Answer: A) Even

Explanation: 2158 is an even number.

The square of an even number is always even.

2158^2 is even

13. Correct Answer: C) 6

Explanation: The unit digit of 1234 is 4.

$4^2 = 16$

The unit digit is **6**

14. Correct Answer: A) 3

Explanation:

Prime factorization of 75: $75 = 3 \times 5 \times 5$

To make a perfect square, all prime factors must occur in pairs.

Only 3 is unpaired, so multiply by 3.

$75 \times 3 = 225 = 15^2$

15. Correct Answer: C) 3

Explanation:

Prime factorization of 2352:

$2352 = 2^4 \times 3 \times 7^2$

For a perfect square, all exponents must be even.
The factor 3 has an odd exponent, so divide by 3.
$\frac{2352}{3} = 784 = 28^2$

16. Correct Answer: C) 24
Explanation:
$\sqrt{576} = 24$
Because $24^2 = 576$

17. Correct Answer: A) $2\frac{1}{18}$
Explanation:
$$\sqrt{4\frac{73}{324}}$$

Convert mixed fraction:
$$4\frac{73}{324} = \frac{4 \times 324 + 73}{324} = \frac{1369}{324}$$
$$\sqrt{\frac{1369}{324}} = \frac{37}{18} = 2\frac{1}{18}$$

18. Correct Answer: C) 1.2
Explanation:
$\sqrt{0.9 \times 1.6} = \sqrt{1.44} = 1.2$

19. Correct Answer: B) 126
Explanation:
$\sqrt{98} \times \sqrt{162} = \sqrt{98 \times 162}$
$= \sqrt{15876} = 126$
Or,
$\sqrt{98} = 7\sqrt{2}, \sqrt{162} = 9\sqrt{2}$
$7\sqrt{2} \times 9\sqrt{2} = 63 \times 2 = 126$

20. Correct Answer: C) 2
Explanation:
We need the smallest number added to 780 to make a perfect square.
$28^2 = 784$
$784 - 780 = 4$
Square root of the obtained perfect square: $\sqrt{4} = 2$

Level – II

Single option type

1. Correct Answer: C) $+2i$.
Explanation: $\sqrt{-2} \times \sqrt{2} = (i\sqrt{2})(\sqrt{2}) = 2i$

2. Correct Answer: B) 16.
Explanation:
Perfect squares among two-digit numbers are:
16,25,36, ...
The least two-digit perfect square is:
$16 = 4^2$

3. Correct Answer: D) 3600

Explanation:

Find LCM of 8, 12, 15, and 20.

$8 = 2^3, 12 = 2^2 \times 3, 15 = 3 \times 5, 20 = 2^2 \times 5$

$LCM = 2^3 \times 3 \times 5 = 120$

To make it a perfect square:

$120 = 2^3 \times 3 \times 5$

Multiply by:

$2 \times 3 \times 5 = 30$

$120 \times 30 = 3600$

4. Correct Answer: B) 23457

Explanation:

Check nearby perfect squares:

$112^2 = 12544$

$317^2 = 100489$

$73^2 = 5329$

But 23457 is not a perfect square.

5. Correct Answer: D) 12

Explanation:

Let the number be x.

$\frac{x}{2} \times x + 10 = 82$

$\frac{x^2}{2} = 72$

$x^2 = 144$

$x = 12$

6. Correct Answer: B) 5.42

Explanation:

Area of square:

$20\frac{460}{49} = \frac{20 \times 49 + 460}{49} = \frac{1440}{49}$

Side of square:

$\sqrt{\frac{1440}{49}} = \frac{\sqrt{1440}}{7}$

$\sqrt{1440} \approx 37.95$

$\frac{37.95}{7} \approx 5.42$

Hence, the side length is approximately **5.42 cm**.

7. Correct Option: A) 17

Square root of

$\sqrt{\frac{0.144}{2.5} \times \frac{25}{0.36} \times \frac{2.89}{0.04}}$

Simplify the expression inside the root:

$\frac{0.144}{2.5} = 0.0576$

$\frac{25}{0.36} = 69.444\ldots$
$\frac{2.89}{0.04} = 72.25$

Alternatively, simplify decimals first:

$$\sqrt{\frac{144 \times 10^{-3}}{25 \times 10^{-1}} \times \frac{25}{36 \times 10^{-2}} \times \frac{289 \times 10^{-2}}{4 \times 10^{-2}}}$$

After cancellation:

$$\sqrt{\frac{144}{36} \times \frac{289}{4}} = \sqrt{4 \times 72.25} = \sqrt{289} = 17$$

8. Correct Answer: B) 81

Explanation:

The greatest two-digit number is 99.

Perfect squares near 99 are:

$9^2 = 81$

$10^2 = 100$

Since 100 is a three-digit number, the greatest two-digit perfect square is: 81

9. Evaluate

$$\sqrt{313 + \sqrt{100 + 7 \times 3}}$$

First simplify inside the inner root: $7 \times 3 = 21$

$100 + 21 = 121$

$\sqrt{121} = 11$

Now substitute: $\sqrt{313 + 11} = \sqrt{324} = 18$

Correct Option: B) 18

10. If $\sqrt{\left(1+\frac{64}{225}\right)} = \frac{x}{15}$,

then x**equals**

Simplify inside the bracket:

$1 + \frac{64}{225} = \frac{225 + 64}{225} = \frac{289}{225}$

Take square root: $\sqrt{\frac{289}{225}} = \frac{17}{15}$

Compare with: $\frac{x}{15} = \frac{17}{15}$

Thus, $x = 17$

Correct Option: C) 17

One or more than correct type

11. Correct Answers: A) 144, B) 225, D) 361

Explanation:

$144 = 12^2$

$225 = 15^2$

$361 = 19^2$

250 is not a perfect square.

12. Correct Answers: B) The square of an odd number is odd, C) The square of an even number is even

Explanation:

Odd number squared remains odd: $5^2 = 25$

Even number squared remains even: $6^2 = 36$

A square number cannot end with 2 or 8.

13. Correct Answers: A) 81, B) 169, D) 400

Explanation:

A number has a rational square root if it is a perfect square.

$\sqrt{81} = 9$

$\sqrt{169} = 13$

$\sqrt{400} = 20$

But: $\sqrt{200} = 10\sqrt{2}$

which is irrational.

14. Correct Answers: A) (3,4,5), B) (5,12,13), C) (6,8,10)

Explanation:

A Pythagorean triplet satisfies: $a^2 + b^2 = c^2$

For A:$3^2 + 4^2 = 9 + 16 = 25 = 5^2$

For B:$5^2 + 12^2 = 25 + 144 = 169 = 13^2$

For C:$6^2 + 8^2 = 36 + 64 = 100 = 10^2$

For D:$2^2 + 3^2 = 4 + 9 = 13 \neq 16$

So D is not a Pythagorean triplet.

Comprehension Type

15. Correct Answer: C) 42 m

Explanation:

Area of a square: $\text{side}^2 = 1764$

So, side = $\sqrt{1764} = 42$

because $42^2 = 1764$

Hence, the length of each side is **42 m**.

16. Correct Answer: C) Its square root is 42

Explanation: $\sqrt{1764} = 42$

So statement C is correct.

A is incorrect because 1764 is a perfect square.

B is incorrect because 42 is an even number.

D is incorrect because:

$1764 \div 3 = 588$

So remainder is 0, not 2.

17. Matching

1. Square of 8 → **C. 64**
2. Square of 11 → **A. 121**
3. Square of 14 → **B. 196**
4. Square of 20 → **D. 400**

Integer type

18. Correct Answer: 9

Explanation:

A number divisible by 9 must contain 9 as a factor. $9 = 3^2$

Since 9 itself is a perfect square, the smallest perfect square divisible by 9 is: 9

19. Correct Answer: 144

Explanation:

$144 = 12^2$

Hence, 9 is the smallest perfect square divisible by 9.

20. Correct Answer: 49

Explanation:

$49^2 = 2401$

Therefore,

$\sqrt{2401} = 49$

Student Practice Corner

Single option correct type

1. Correct Answer: A) 961

Explanation:

The greatest three-digit number is 999.

$31^2 = 961$

$32^2 = 1024$

Since 1024 is a four-digit number, the greatest three-digit perfect square is: 961

2. Correct Answer: D) 20

Explanation:

$3(x-5)^2 = 675$

$(x-5)^2 = 225$

$x - 5 = \pm 15$

$x = 20$ or -10

From the options, only **20** is present.

3. Correct Answer: D) 0.2

Explanation: $\sqrt{0.01 + \sqrt{0.0009}} = x$

First, $\sqrt{0.0009} = 0.03$

Then, $0.01 + 0.03 = 0.04$

Now, $\sqrt{0.04} = 0.2$

So the correct answer should actually be.

4. Correct Answer: C) 2500

Explanation: $2500 = 50^2$

Hence, 2500 is a perfect square.

5. Correct Answer: A) 1213

Explanation: $1213^2 = 1471369$

Therefore,

$\sqrt{1471369} = 1213$

6. Correct Answer: C) 7

Explanation: $5(x-1)^2 = 180$

$(x-1)^2 = 36$

$x - 1 = \pm 6$

$x = 7$ or -5

From the options, only **7** is present.

7. Correct Answer: A) 1.30

Explanation:

Area:

$1\frac{1725}{2500} = \frac{4225}{2500}$

Side:

$\sqrt{\frac{4225}{2500}} = \frac{65}{50} = 1.3$

8. Correct Answer: D) 17

Explanation:

$\sqrt{274 + \sqrt{211 + \sqrt{196}}}$

First, $\sqrt{196} = 14$

Then, $211 + 14 = 225$

$\sqrt{225} = 15$

Now,

$274 + 15 = 289$

$\sqrt{289} = 17$

More than one correct option

9. Correct Answers: A) The number of zeros at the end of a perfect square is always even, C) $\sqrt{256} = 16$

Explanation:

- A is true because trailing zeros in a perfect square always occur in pairs.

  Example: $100 = 10^2, 2500 = 50^2$

  C is true because:

  $\sqrt{256} = 16$

  B is false because negative numbers are not perfect squares in whole numbers.
- D is false because:

  $2 \times 4 = 8$ and 8 is not a perfect square.

10. Correct Answers: A) 49, B) 81, D) 225

Explanation: Squares of odd numbers are: $49 = 7^2$

$81 = 9^2$

$225 = 15^2$

But: $100 = 10^2$

which is the square of an even number.

Comprehension type

11. Correct Answer: D) All of these

Explanation:

From the passage: $15^2 = 225$

$16^2 = 256$

and $1 + 3 + 5 + 7 + 9 = 25$

All three numbers are perfect squares.

12. Correct Answer: C) 25

Explanation:

First five odd numbers are: 1,3,5,7,9

Their sum: $1 + 3 + 5 + 7 + 9 = 25$

Hence, the sum is **25**.

13. Matching the following (number and its square root)

1. 144 → **C. 12**
2. 289 → **A. 17**
3. 225 → **B. 15**
4. 361 → **D. 19**

Integer type

14. Correct Answer: 361

Explanation: $19^2 = 361$

Hence, the square of 19 is **361**.

15. Correct Answer: 28

Explanation: $28^2 = 784$

Therefore, $\sqrt{784} = 28$

Cube

LEVEL – I

1. Correct Answer: D) $8a^3$

Explanation:

Double of $a$is: $2a$

Cube of $2a$: $(2a)^3 = 8a^3$

2. Correct Answer: A) 3

Explanation:

$2197 = 13^3$

So the cube root is 13, whose ones digit is: 3

3. Correct Answer: B) 26

Explanation: $26^3 = 17576$

Therefore, $\sqrt[3]{17576} = 26$

4. Correct Answer: A) 225

Explanation: $1^3 + 2^3 + 3^3 + 4^3 + 5^3$

$= 1 + 8 + 27 + 64 + 125$

$= 225$

5. Correct Answer: D) 6400

Explanation:

$1000 = 10^3$

$1728 = 12^3$

$27000 = 30^3$

But 6400 is not a perfect cube.

6. Correct Answer: D) 0.512

Explanation:

$(0.8)^3 = 0.8 \times 0.8 \times 0.8 = 0.512$

7. Correct Answer: A) $\frac{5}{3}$

Explanation:

$$\sqrt[3]{5\frac{-10}{27}}$$

Convert mixed fraction:

$$5\frac{-10}{27} = \frac{135 - 10}{27} = \frac{125}{27}$$

$\sqrt[3]{\frac{125}{27}} = \frac{5}{3}$

8. Correct Answer: A) 5

Explanation:

If we divide 8640 by 5: $\frac{8640}{5} = 1728$

Since $1728 = 12^3$

(which is $2^6 \times 3^3$), the resulting quotient is a perfect cube.

The smallest number to divide by is **5**.

Correct Option: A) 5

9. Correct Answer: B) 0.06

Explanation:

$0.06^3 = 0.000216$

Therefore,

$\sqrt[3]{0.000216} = 0.06$

10. Correct Answer: D) 1729

Explanation:

1729 is the famous Hardy–Ramanujan number.

$1729 = 1^3 + 12^3$

and also

$1729 = 9^3 + 10^3$

It is the smallest number expressible as the sum of two cubes in two different ways.

11. Correct Answer: A) 13824

Explanation:

$24^3 = 24 \times 24 \times 24$

$24^2 = 576$

$576 \times 24 = 13824$

12. Correct Answer: A) 1331

Explanation:

$11^3 = 1331$

Hence, 1331 is a perfect cube.

13. Correct Answer: B) -9261

Explanation:

$(-21)^3 = -21 \times -21 \times -21$

$= 441 \times (-21) = -9261$

14. Correct Answer: C) a^{18}.

Explanation:

Cube of a^6:

$(a^6)^3 = a^{18}$

15. Correct Answer: B) 7

Explanation:

The unit digit of 293 is 3.

$3^3 = 27$

Hence, the last digit is **7**.

16. Correct Answer: A) 13^3

Explanation: $13^3 = 13 \times 13 \times 13$

$= 169 \times 13 = 2197$

17. Correct Answers: C) 343

Explanation: $243 = 3^5$

not a perfect cube. $343 = 7^3$

is a perfect cube. 10000

and 270000

are not perfect cubes.

18. Correct Answer: D) 8

Explanation:

If a number ends in 2: $2^3 = 8$

So the last digit of its cube will be **8**.

19. Correct Answer: A) -1

Explanation: $(-1)^3 = -1$

Hence, $-1$is equal to its cube.

20. Correct Answer: B) Cube of a negative number is always negative

Explanation:

Example: $(-2)^3 = -8$

$(-5)^3 = -125$

Hence, the cube of any negative number is always negative.

LEVEL – II

Single option type

1. Correct Answer: A) 5

Explanation:

$\sqrt[3]{5} \times \sqrt[3]{5} \times \sqrt[3]{5}$

Using:

$\sqrt[3]{a} \times \sqrt[3]{a} \times \sqrt[3]{a} = a$

$= 5$

2. Correct Answer: A) $a + b$

Explanation:

$a^3 + 3a^2b + 3ab^2 + b^3 = (a + b)^3$

Therefore,

$\sqrt[3]{a^3 + 3a^2b + 3ab^2 + b^3} = a + b$

3. Correct Answer: C) 980

Explanation:

$\sqrt[3]{x + 20} = 10$

Cube both sides:

$x + 20 = 10^3$

$x + 20 = 1000$

$x = 980$

4. Correct Answer: A) 1014 cm^2

Explanation:

Volume of cube:

$a^3 = 2197$

$a = \sqrt[3]{2197} = 13$ cm

Surface area:

$6a^2 = 6(13^2) = 6(169) = 1014$ cm^2

5. Correct Answer: C) 3025

Explanation:

$1^3 + 2^3 + 3^3 + \cdots + 10^3 = \left(\frac{10(10+1)}{2}\right)^2$

$= \left(\frac{10 \times 11}{2}\right)^2 = 55^2 = 3025$

6. Correct Answer: B) 6^3

Explanation:

$31 + 33 + 35 + 37 + 39 + 41$

$= (31 + 41) + (33 + 39) + (35 + 37)$

$= 72 + 72 + 72 = 216$

$216 = 6^3$

7. Correct Answer: B) 72

Explanation:

$\sqrt[3]{81} \times \sqrt[3]{192} \times \sqrt[3]{24} = \sqrt[3]{81 \times 192 \times 24}$

$= \sqrt[3]{373248}$

$373248 = 72^3$

Therefore,

$\sqrt[3]{373248} = 72$

8. Correct Answer: B) -23958

Explanation:

$22^3 - 33^3 + 11^3$

$22^3 = 10648$

$33^3 = 35937$

$11^3 = 1331$

$10648 - 35937 + 1331 = -23958$

9. Correct Answer: B) 30

Explanation: $\sqrt[3]{30} \times \sqrt[3]{12} \times \sqrt[3]{75}$

Using: $\sqrt[3]{a} \times \sqrt[3]{b} \times \sqrt[3]{c} = \sqrt[3]{abc} = \sqrt[3]{30 \times 12 \times 75}$

$= \sqrt[3]{27000}$

$27000 = 30^3$

Therefore, $\sqrt[3]{27000} = 30$

10. Correct Answer: A) 125

Explanation: $25a^4 = a^6$

Divide both sides by a^4: $a^2 = 25$

$a = \pm 5$

Now, $a^3 = (\pm 5)^3 = \pm 125$

From the given options, the positive value is: 125

More than one correct type

11. Correct Answers: A) 216, B) 343, D) 1331

Explanation:

$216 = 6^3$

$343 = 7^3$

$1331 = 11^3$

But 484 is not a perfect cube.

12. Correct Answers: A) Cube of a negative number is always negative, B) Cube of an even number is always even, C) Cube of a number ending with 5 ends with 5

Explanation:

A is true:

$(-3)^3 = -27$

B is true:

$4^3 = 64$

C is true:

$5^3 = 125, 15^3 = 3375$

D is false because:

$2^3 = 8$

So a number ending in 2 does not end in 2 after cubing.

13. Correct Answers: B) 74088, D) 8000

Explanation:

$42^3 = 74088$

$20^3 = 8000$

But: $3375 = 15^3$

So A also has an integer cube root. 2500

is not a perfect cube.

Hence, the correct answers should actually be **A, B and D**.

14. Correct Answer: C) 15 cm

Explanation: $3375 = 15^3$

Hence, the cube root of 3375 is 15, so the side length is 15 cm.

15. Correct Answer: A) 1728

Explanation: $1728 = 12^3$

Hence, 1728 is a perfect cube.

16. Match the Following

Step 1: Solve each item in Column A

a. 8**is equal to** $8 = 2^3$

So, **a → Q**

b. 3375**is a perfect cube, and its cube root is** $15^3 = 3375$

Therefore, $\sqrt[3]{3375} = 15$

So, **b → R**

c. $\sqrt[3]{8000}$

$20^3 = 8000$

Hence, $\sqrt[3]{8000} = 20$

So, **c → P**

d. 12^3

$12^3 = 12 \times 12 \times 12$

$= 144 \times 12 = 1728$

So, **d → S**

Correct Matching: a -Q, b – R, c – P, d – S

17. Correct Answer: 343

Explanation:

The smallest perfect cube divisible by 7 is: $7^3 = 343$

Hence, the answer is **343**.

18. Correct Answer: 2

Explanation:

If a number ends with 8: $8^3 = 512$

So, the cube ends with **2**.

19. Correct Answer: 8000

Explanation: $20^3 = 20 \times 20 \times 20$

$= 400 \times 20 = 8000$

20. Correct Answer: 15

Explanation: $15^3 = 3375$

Therefore, $\sqrt[3]{3375} = 15$

Student Practice Corner

Single option correct type

1. Correct Answer: C) 3

Explanation:

In the cube of a number, each prime factor appears 3 times.

Example: $2^3 = 2 \times 2 \times 2$

2. Correct Answer: C) 6

Explanation:

If a number ends with 6: $6^3 = 216$

The cube also ends with **6**.

3. Correct Answer: A) 25

Explanation: $25^3 = 15625$

Therefore, $\sqrt[3]{15625} = 25$

So the correct option should actually be **C) 25**.

4. Correct Answer: B) Even

Explanation:

The cube of an even number is always even.

Example: $4^3 = 64$

5. Correct Answer: A) 5 cm

Explanation:

Volume of cube: $a^3 = 125$

$a = \sqrt[3]{125} = 5$

Hence, the side length is **5 cm**.

6. Correct Answer: B) 2

Explanation: $36^3 = 46656$

Therefore,

$\sqrt[3]{46656} = 36$

The cube root 36 has **2 digits**.

7. Correct Answer: A) 1

Explanation: $\sqrt[3]{8000} = 20$

20 contains **1 zero**.

One or more than correct type

8. Correct Answers: A) $12^3 = 1728$**, B)** $15^3 = 3375$

Explanation: $12^3 = 12 \times 12 \times 12 = 1728$

$15^3 = 15 \times 15 \times 15 = 3375$

But: $20^3 = 8000 \neq 800$

And $14^3 = 2744 \neq 4098$

9. Correct Answers: A) 1, C) 5

Explanation:

- If a number ends with 1, its cube also ends with 1:
$$1^3 = 1$$
- If a number ends with 5, its cube also ends with 5:
$$5^3 = 125$$
- A number ending with 4 ends with 4 only when squared, not cubed:
$$4^3 = 64$$
- A number ending with 8 gives: $8^3 = 512$
which ends with 2.

Comprehension type

10. Correct Answer: A) 64

Explanation: $64 = 4^3$

Hence, 64 is a perfect cube.

11. Correct Answer: A) Cube root is the inverse operation of cube

Explanation:

The passage states that cube root is the inverse operation of finding a cub

12. Matching the Following

a. Number ending with 7 → **Q. Cube ends with 3**

b. Number ending with 8 → **P. Cube ends with 2**

c. Number ending with 5 → **R. Cube ends with 5**

d. Number ending with 1 → **S. Cube ends with 1**

e. Number ending with 2 → **T. Cube ends with 8**

Integer Type

13. Correct Answer: 2744

Explanation:$14^3 = 14 \times 14 \times 14$

$14^2 = 196$

$196 \times 14 = 2744$

Hence, the cube of 14 is **2744**.

14. Correct Answer: 12

Explanation:$21^3 = 9261$

Therefore,$\sqrt[3]{1728} = 12$

Explanation:$12^3 = 1728$

7. Mensuration – 2D & 3D Shapes

AREA AND PERIMETER

LEVEL – I

1. Correct Answer: B) 96 cm

Area of rectangle = 512 cm²

Let breadth = $b$cm

Length = $2b$cm

Area = $l \times b$

$2b \times b = 512$

$2b^2 = 512$

$b^2 = 256$

$b = 16$ cm

Length = 32cm

Perimeter:$2(l + b) = 2(32 + 16) = 96$ cm

2. Correct Answer: D) $\frac{d^2}{2}$

For a square, $d = s\sqrt{2}$

$s = \frac{d}{\sqrt{2}}$

Area: $s^2 = \left(\frac{d}{\sqrt{2}}\right)^2$

$= \frac{d^2}{2}$

3. Correct Answer: C) 24 sq.cm

Area of triangle: $A = \frac{1}{2}bh$

$A = \frac{1}{2} \times 8 \times 6$

= 24 sq.cm

4. Correct Answer: D) 6 cm

Area of triangle = 39 cm²

Height = 13 cm

$39 = \frac{1}{2} \times b \times 13$

$39 = \frac{13b}{2}$

$78 = 13b$

$b = 6$ cm

5. Correct Answer: A) 36 m²

Area of parallelogram: Area = base × height

$= 8 \times 4.5$

$= 36$ m^2

Area of parallelogram: Area = base × altitude

$= 12 \times 6$

= 72 sq.cm

6. Correct Answer: C) 16 cm²

Area of triangle:

$A = \frac{1}{2}bh$

$A = \frac{1}{2} \times 8 \times 4$

$= 16 \text{ cm}^2$

7. Correct Answer: B) 10 cm

Area of parallelogram:

Area = base × height

$80 = b \times 8$

$b = \frac{80}{8} = 10 \text{ cm}$

8. Correct Answer: C) $x(x-1)$

Area of parallelogram:

Area = base × height

Base = x

Height = $x - 1$

Area = $x(x-1)$

9. Correct Answer: A) 20 cm

Area of triangle:

$80 = \frac{1}{2} \times b \times 8$

$80 = 4b$

$b = 20 \text{ cm}$

10. Correct Answer: C) 65 cm²

The shaded portion is a triangle.

Base = 13cm

Height = 10cm

Area of triangle:

$A = \frac{1}{2}bh$

$A = \frac{1}{2} \times 13 \times 10$

$= 65 \text{ cm}^2$

11. Correct Answer: A) 6 sq.cm

Area of rhombus: Area = $\frac{1}{2}d_1d_2$

Diagonals = 4cm and 3cm

Area = $\frac{1}{2} \times 4 \times 3$

= 6 sq.cm

12. Correct Answer: B)24 sq.cm

The figure is a rhombus/kite formed by diagonals.

Diagonal lengths:

$3 + 3 = 6$ cm

$4 + 4 = 8$ cm

Area of rhombus: Area = $\frac{1}{2}d_1d_2$

$= \frac{1}{2} \times 6 \times 8$

= 24 sq.cm

13. Correct Answer: B) $\frac{x^2}{\sqrt{3}}$sq.cm

For an equilateral triangle, $h = \frac{\sqrt{3}}{2}a$

Given height $= x$

$a = \frac{2x}{\sqrt{3}}$

Area: $A = \frac{\sqrt{3}}{4}a^2$

$= \frac{\sqrt{3}}{4}\left(\frac{2x}{\sqrt{3}}\right)^2$

$= \frac{x^2}{\sqrt{3}}$ sq.cm

14. Correct Answer: C) 120

Area of rhombus:

Area $= \frac{1}{2}d_1d_2$

$= \frac{1}{2} \times 10 \times 24$

$= 120$ sq.cm

15. Correct Answer: D) $a - b$

Area of trapezium: $A = \frac{1}{2}(a + b)h$

Given, $\frac{1}{2}(a^2 - b^2) = \frac{1}{2}(a + b)(a - b)$

Comparing both formulas: $h = a - b$

16. Correct Answer: A) 12 cm

Area of rhombus: $96 = \frac{1}{2} \times 16 \times d_2$

$96 = 8d_2$

$d_2 = 12$ cm

17. Correct Answer: D) 84 sq.cm

Area of parallelogram: Area = base × height

$= 12 \times 7$

$= 84$ sq.cm

18. Correct Answer: C) 48 cm

Area of equilateral triangle: $A = \frac{\sqrt{3}}{4}a^2$

Given: $64\sqrt{3} = \frac{\sqrt{3}}{4}a^2$

$64 = \frac{a^2}{4}$

$a^2 = 256$

$a = 16$ cm

Perimeter: $3a = 3 \times 16 = 48$ cm

19. Correct Answer: B) 64 cm²

Area of trapezium:

$A = \frac{1}{2}(a + b)h$

Parallel sides = 7cm and 9cm
Height = 8cm
$A = \frac{1}{2}(7+9) \times 8$
$= \frac{1}{2} \times 16 \times 8$
$= 64 \text{ cm}^2$

20. Correct Answer: A) 24 cm²

The quadrilateral is divided into two triangles having the same base $AC = 6$cm.
Area of $\triangle ABC$: $\frac{1}{2} \times 6 \times 3$
$= 9 \text{ cm}^2$
Area of $\triangle ACD$:
$= \frac{1}{2} \times 6 \times 5$
$= 15 \text{ cm}^2$
Total area of quadrilateral: $9 + 15 = 24 \text{ cm}^2$

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Correct Answer: A) 60 cm

Length : Breadth = 3: 2
Let length = $3x$, breadth = $2x$
Area: $3x \times 2x = 216$
$6x^2 = 216$
$x^2 = 36$
$x = 6$
Length = 18cm, Breadth = 12cm
Perimeter: $2(l+b) = 2(18+12) = 60$ cm

2. Correct Answer: A) 90 m

Let breadth = x
Length = $3x$
Perimeter: $2(l+b) = 240$
$2(3x+x) = 240$
$8x = 240$
$x = 30$
Length: $3x = 90$ m

3. Correct Answer: B) 60 m

Area of square: $s^2 = 225$
$s = 15$ m
Perimeter: $4s = 4 \times 15 = 60$ m

4. Correct Answer: C) $l = 35$cm, $A = 1050$ cm²

Perimeter: $2(l+b) = 130$
Breadth = 30cm
$2(l+30) = 130$
$l + 30 = 65$
$l = 35$ cm
Area: $35 \times 30 = 1050 \text{ cm}^2$

5. Correct Answer: B) 10 cm

Area of square: $s^2 = 50$

$s = 5\sqrt{2}$

Diagonal of square: $d = s\sqrt{2}$

$= 5\sqrt{2} \times \sqrt{2}$

$= 10$ cm

6. Correct Answer: B) 44 m

Area of trapezium: $A = \frac{1}{2}(a + b)h$

Given:

$480 = \frac{1}{2}(20 + x) \times 15$

$960 = 15(20 + x)$

$64 = 20 + x$

$x = 44$ m

7. Correct Answer: A) 30 cm

Area of rhombus:

Area $= \frac{1}{2}d_1d_2$

Given: $240 = \frac{1}{2} \times 16 \times d_2$

$240 = 8d_2$

$d_2 = 30$ cm

Hence, the correct option is A) 30 cm.

8. Correct Answer: B) 41 cm²

Area of rhombus: Area $= \frac{1}{2}d_1d_2$

$= \frac{1}{2} \times 10 \times 8.2$

$= 41$ cm^2

9. Correct Answer: B) 26 units

Sides of triangle are 5,12,13.

Since $5^2 + 12^2 = 13^2$, it is a right triangle.

Area of triangle: $= \frac{1}{2} \times 5 \times 12$

$= 30$ sq.units

Rectangle length = 10units.

Breadth: $\frac{30}{10} = 3$

Perimeter of rectangle: $2(10 + 3) = 26$ units

10. Correct Answer: B) $12(\sqrt{2} - 1)$

Let the length cut from each corner be xm.

Then each straight side of the octagon becomes: $12 - 2x$

Each slant side formed is: $x\sqrt{2}$

Since all sides of the octagon are equal, $12 - 2x = x\sqrt{2}$

$$12 = x(2 + \sqrt{2})$$

$$x = \frac{12}{2 + \sqrt{2}}$$

Side of octagon: $x\sqrt{2} = \frac{12\sqrt{2}}{2+\sqrt{2}}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Rationalizing, $= \frac{12\sqrt{2}(2-\sqrt{2})}{(2+\sqrt{2})(2-\sqrt{2})}$

$= 12(\sqrt{2}-1)$

Correct Answers: A) and C)

Perimeter of rectangle:

$2(l+b) = 68$

Breadth = 14cm

$2(l+14) = 68$

$l + 14 = 34$

$l = 20$ cm

A) 20cm

C) $\frac{1}{2}(40) = 20$cm

So, the correct options are **A and C only**.

12. **Correct Answers: B) and C)**

Area of square:

$s^2 = 625$

$s = 25$ cm

Perimeter:

$4s = 4 \times 25 = 100$ cm

B) 100cm

C) $50 \times 2 = 100$cm

Hence, the correct options are **B and C**.

13. **Correct Answers: A) and B)**

Area of rectangle:

$20 \times 15 = 300$ cm^2

Area of triangle:

$\frac{1}{2} \times 30 \times h = 300$

$15h = 300$

$h = 20$ cm

A) 20cm

B) length = height = 20cm

Hence, the correct options are **A and B**.

COMPREHENSION TYPE

Passage – I

14. **Correct Answer: C) 30 m**

Base: Height = 5: 2

Let base = $5x$and height = $2x$

Area of parallelogram:

$5x \times 2x = 360$

$10x^2 = 360$

$x^2 = 36$

$x = 6$

Base:

$5x = 5 \times 6 = 30$ m

Hence, the correct option is **C) 30 m**.

15. Correct Answer: B) 180 m²

The diagonal of a parallelogram divides it into two equal triangles.

Area of parallelogram = 360 m^2

So, area of $\triangle ABC$:

$\frac{360}{2} = 180$ m^2

Hence, the correct option is **B) 180 m²**

16. Matrix Matching

- a) $\frac{1}{2}d_1d_2$→ **Q) Rhombus**

  Because area of rhombus is:

  Area $= \frac{1}{2}d_1d_2$

- b) $\frac{1}{2}h(a+b)$→ **R) Trapezium**

  Formula of area of trapezium:

  $A = \frac{1}{2}h(a+b)$

- c) $\frac{1}{2}(h_1+h_2)d$→ **S) Quadrilateral**

  This is the area formula for a quadrilateral using one diagonal and perpendiculars.

- d) bh→ **T) Parallelogram**

  Area of parallelogram:

  Area = bh

- e) a^2→ **P) Square**

  Area of square:

  Area = a^2

  Final Matching: a → Q, b → R, c → S, d → T, e → P

INTEGER TYPE

17. Correct Answer: 4 cm

Perimeter of square:

$4s = 16$

$s = \frac{16}{4} = 4$ cm

18. Correct Answer: 7 cm

Area of rhombus:

Area $= \frac{1}{2}d_1d_2$

Given:

$112 = \frac{1}{2} \times 32 \times d_2$

$112 = 16d_2$

$d_2 = 7$ cm

19. Correct Answer: 8 cm

For an isosceles right-angled triangle, let equal sides be a.

Area:

$\frac{1}{2}a^2 = 32$

$a^2 = 64$

$a = 8$ cm

20. Correct Answer: 8 cm

Area of triangle:

$A = \frac{1}{2}bh$

$48 = \frac{1}{2} \times 12 \times h$

$48 = 6h$

$h = 8$ cm

Student Practice Corner

Single option correct type

1. Correct Answer: C) 32 cm

Perimeter of square:

$4s = 4 \times 8 = 32$ cm

2. Correct Answer: D) 22 cm

Perimeter of square:

$4s = 88$

$s = \frac{88}{4} = 22$ cm

3. Correct Answer: C) 16 cm²

Area of square:

$s^2 = 4^2 = 16$ cm^2

4. Correct Answer: C) 2x units

Area of square:

$s^2 = 4x^2$

$s = \sqrt{4x^2} = 2x$

5. Correct Answer: B) 280 sq.cm

Area of rectangle: $l \times b = 20 \times 14 = 280$ sq.cm

6. Correct Answer: A) 50 cm

Perimeter of rectangle:

$2(l + b) = 2(15 + 10)$

$= 2 \times 25 = 50$ cm

7. Correct Answer: B) 40 units

Length : Breadth = 4: 1

Let length = $4x$, breadth = x

Area:

$4x \times x = 400$

$4x^2 = 400$

$x^2 = 100$

$x = 10$

Length: $4x = 40$ units

8. Correct Answer: D) 11 cm

Area of rectangle: $l \times b = 286$

Length = 26cm

$26 \times b = 286$

$b = \frac{286}{26} = 11$ cm

Hence, the correct option is **D) 11 cm**

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Correct Answers: C) and D)

Area of square: $40 \times 40 = 1600$ m^2

Area of parallelogram:

base $\times$ 20 = 1600

base = 80 m

Checking options:

C) $4 \times$ height $= 4 \times 20 = 80$ m

D) 80 m

Hence, the correct options are **C and D**.

10. Correct Answer: C) 40 cm

Let base = $x$cm

Height is double the base: $h = 2x$

Area of triangle: $A = \frac{1}{2}bh$

$$400 = \frac{1}{2} \times x \times 2x$$
$$400 = x^2$$
$$x = 20$$

Height: $h = 2x = 40$ cm

COMPREHENSION TYPE

11. Correct Answer: B) 45

Area of rectangle: $l \times b$

Given: $9 \times 5 = 45$

12. Correct Answer: B) 40

Perimeter of rectangle: $2(l + b)$

$$2(12 + 8)$$
$$2 \times 20 = 40$$

13. Matrix Matching Answers

a) Side of square is 12 cm then its area is → **S) 144 cm²**

Area = $12^2 = 144$ cm^2

b) $h = 12$cm, $b = 6$cm then area of triangle is → **T) 72 cm²**

Area of triangle:

$A = \frac{1}{2}bh$

$= \frac{1}{2} \times 12 \times 6$

$= 36$ cm^2

So, the correct value should be **36 cm²**.

(No matching option is given.)

c) Perimeter of square = 88cm then side is → **P) 22 cm**

$4s = 88$

$s = 22$ cm

d) Perimeter of rhombus whose side is $a$cm → **Q) 4a**

Perimeter = $4a$

e) Perimeter of parallelogram if sides are 3 cm and 2 cm → **R) 10 cm**

$2(a + b) = 2(3 + 2) = 10$ cm

Final Matching: a → S, b → T, c → P, d → Q, e → R

INTEGER TYPE

14. Correct Answer: 5 cm

Base : Height = 1: 5

Let base = x, height = $5x$

Area of parallelogram:

$x \times 5x = 125$

$5x^2 = 125$

$x^2 = 25$

$x = 5$

Hence, the base of parallelogram is **5 cm**.

15. Correct Answer: 9 cm

Area of rectangle:

$l \times b = 144$

Length = 16cm

$16 \times b = 144$

$b = \frac{144}{16} = 9$ cm

Hence, the breadth is **9 cm**.

CIRCLES AND RINGS

LEVEL – I

1. Correct Answer: C) 10 cm

The wire forms the circumference of the circle.

Circumference = 10 cm

2. Correct Answer: C) 7 cm

Circumference of circle:

$2\pi r = 44$

Using $\pi = \frac{22}{7}$,

$2 \times \frac{22}{7} \times r = 44$

$\frac{44r}{7} = 44$

$r = 7$ cm

3. Correct Answer: A) 15.4 cm

Circumference of circle:

πd

$= \frac{22}{7} \times 4.9$

$= 15.4$ cm

4. Correct Answer: B) 154 sq.cm

Circumference:

$2\pi r = 44$

$r = 7$ cm

Area of circle: πr^2

$= \frac{22}{7} \times 7 \times 7$

$= 154$ sq.cm

5. Correct Answer: B) 33 cm

Circumference:

πd

$= \frac{22}{7} \times 10.5$

$= 33$ cm

6. Correct Answer: B) 96.8 cm

Circumference:

$2\pi r$

$= 2 \times \frac{22}{7} \times 15.4$

$= 96.8$ cm

7. Correct Answer: D) 2464 sq.m

Circumference:

$2\pi r = 176$

$2 \times \frac{22}{7} \times r = 176$

$r = 28$ m

Area:

πr^2

$= \frac{22}{7} \times 28 \times 28$

$= 2464$ sq.m

8. Correct Answer: C) 10.8 m.

Perimeter of semicircle:

$\pi r + 2r$

$= \frac{22}{7} \times 2.1 + 4.2$

$= 6.6 + 4.2$

$= 10.8$ m

9. Correct Answer: A) 77 sq.cm

Area of semicircle:

$\frac{1}{2}\pi r^2$

$= \frac{1}{2} \times \frac{22}{7} \times 7 \times 7$

$= 77$ sq.cm

10. Correct Answer: B) 144 cm.

Perimeter of semicircle: $\pi r + 2r$

Diameter = 56cm

$r = 28$ cm

$= \frac{22}{7} \times 28 + 56$

$= 88 + 56$

$= 144$ cm

11. Correct Answer: C) $4 : \pi$.

Let radius of circle $= r$

Circumference of circle $= 2\pi r$

Let side of square $= a$

Perimeter of square $= 4a$

Given:

$2\pi r = 4a$

$a = \frac{\pi r}{2}$

Area of circle: πr^2

Area of square:

$a^2 = \left(\frac{\pi r}{2}\right)^2 = \frac{\pi^2 r^2}{4}$

Ratio of areas:

$\pi r^2 : \frac{\pi^2 r^2}{4}$

$= 4 : \pi$

12. Correct Answer: D) 49 cm

Circumference:

$\pi d = 154$

$\frac{22}{7} d = 154$

$d = \frac{154 \times 7}{22}$

$d = 49$ cm

13. Correct Answer: D) 102.67 sq.cm

Area of sector:

$\frac{\theta}{360} \pi r^2$

$= \frac{240}{360} \times \frac{22}{7} \times 7 \times 7$

$= \frac{2}{3} \times 154$

$= 102.67$ sq.cm

14. Correct Answer: A) 17.6 cm.

Length of arc:

$\frac{\theta}{360} \times 2\pi r$

$= \frac{72}{360} \times 2 \times \frac{22}{7} \times 14$

$= \frac{1}{5} \times 88$

$= 17.6$ cm

15. Correct Answer: D) 55 cm.

Length of arc:

$\frac{\theta}{360} \times 2\pi r$

$= \frac{75}{360} \times 2 \times \frac{22}{7} \times 42$

$= \frac{5}{24} \times 264$

$= 55$ cm

16. Correct Answer: B) $\pi : 4$

Let the common value of diameter of circle and side of square be a.

Radius of circle:

$r = \frac{a}{2}$

Area of circle:

$\pi r^2 = \pi\left(\frac{a}{2}\right)^2 = \frac{\pi a^2}{4}$

Area of square: a^2

Ratio:

$\frac{\pi a^2}{4} : a^2$

$= \pi : 4$

17. Correct Answer: A) 1.75 m

Distance covered in one rotation:

$\frac{2200}{200} = 11$ m

Circumference of roller: $2\pi r = 11$

Using $\pi = \frac{22}{7}$,

$2 \times \frac{22}{7} \times r = 11$

$\frac{44r}{7} = 11$

$r = \frac{11 \times 7}{44}$

$r = 1.75$ m

18. Correct Answer: C) 34 cm.

Area of sector:

$\frac{\theta}{360}\pi r^2 = 70$

Given $r = 7$cm

$\frac{\theta}{360} \times \frac{22}{7} \times 49 = 70$

$\frac{\theta}{360} \times 154 = 70$

Arc length: $\frac{\theta}{360} \times 2\pi r$

$= \frac{70}{154} \times 44$

$= 20$ cm

Perimeter of sector: $20 + 2(7)$

$= 34$ cm

19. Correct Answer: D) 180 cm

Area of semicircle: $\frac{1}{2}\pi r^2 = 1925$

Using $\pi = \frac{22}{7}$,

$\frac{11}{7}r^2 = 1925$

$r^2 = 1225$

$r = 35$ cm

Perimeter of semicircle: $\pi r + 2r$

$= \frac{22}{7} \times 35 + 70$

$= 110 + 70$

$= 180$ cm

20. Correct Answer: C) 946 sq.m.

Area of ring: $\pi(R^2 - r^2)$

Outer radius $R = 25$m

Inner radius $r = 18$m

$= \frac{22}{7}(25^2 - 18^2)$

$= \frac{22}{7}(625 - 324)$

$= \frac{22}{7} \times 301$

$= 946$ sq.m

LEVEL - II

SINGLE OPTION CORRECT TYPE

1. Correct Answer: A) 22 cm

Area of sector: $A = \frac{1}{2}rl$

where $l$is the arc length.

Given: $88 = \frac{1}{2} \times 8 \times l$

$88 = 4l$

$l = 22$ cm

Hence, the correct option is **A) 22 cm**.

2. Correct Answer: B) 1232 sq.cm.

Perimeter of semicircle:

$\pi r + 2r = 144$

Using $\pi = \frac{22}{7}$,

$\frac{22}{7}r + 2r = 144$

$\frac{36r}{7} = 144$

$r = 28$ cm

Area of semicircle:

$\frac{1}{2}\pi r^2$

$= \frac{1}{2} \times \frac{22}{7} \times 28 \times 28$

$= 1232$ sq.cm

3. Correct Answer: D) 4224 sq.m

Area of ring: $\pi(R^2 - r^2)$

Outer radius $R = 55$m

Inner radius $r = 41$m

$= \frac{22}{7}(55^2 - 41^2)$

$= \frac{22}{7}(3025 - 1681)$

$= \frac{22}{7} \times 1344$

$= 4224$ sq.m

4. Correct Answer: C) 1:4

Area of sector:

$$\frac{\theta}{360} \times \pi r^2$$

For $\theta = 90°$,

$$\frac{90}{360} = \frac{1}{4}$$

So, the ratio of area of sector to area of circle is: $1:4$

5. Correct Answer: B) 342 m

Perimeter of sector: $2r$ + arc length

Arc length: $\frac{72}{360} \times 2\pi r$

$= \frac{1}{5} \times 2 \times \frac{22}{7} \times 105$

$= 132$ m

Perimeter: $132 + 2(105)$

$= 132 + 210$

$= 342$ m

6. Correct Answer: D) 88 cm

In 60 minutes, the minute hand completes one full circle.

Distance moved by the tip =circumference:

$2\pi r$

$= 2 \times \frac{22}{7} \times 14$

$= 88$ cm

7. Correct Answer: D) 154 cm²

Side of square = 14cm

Two semicircles are drawn with diameter 14cm.

Radius:

$r = \frac{14}{2} = 7$ cm

The shaded region consists of two semicircles, which together form one complete circle.

Area of shaded region:

πr^2

$= \frac{22}{7} \times 7 \times 7$

$= 154$ cm^2

8. Correct Answer: B) 1386 sq.m.

Area of lawn: $\pi r^2 = 2464$

Using $\pi = \frac{22}{7}$,

$\frac{22}{7} r^2 = 2464$

$r^2 = 784$

$r = 28$ m

Path width = 7m

Outer radius: $R = 28 + 7 = 35$ m

Area of path:
$\pi(R^2 - r^2)$
$= \frac{22}{7}(35^2 - 28^2)$
$= \frac{22}{7}(1225 - 784)$
$= \frac{22}{7} \times 441$
$= 1386$ sq.m

9. Correct Answer: C) 20 m & 6 m.
Area of track:
$\pi(R^2 - r^2) = 1144$
Width: $R - r = 14$
Using $\pi = \frac{22}{7}$,
$R^2 - r^2 = \frac{1144 \times 7}{22} = 364$
$(R - r)(R + r) = 364$
$14(R + r) = 364$
$R + r = 26$
Solving:
$R = 20, r = 6$

10. Correct Answer: A) 576π
Radii ratio: $5:8$
Let radii be $5x$and $8x$.
Difference of areas:
$\pi(8x)^2 - \pi(5x)^2 = 351\pi$
$64x^2 - 25x^2 = 351$
$39x^2 = 351$
$x^2 = 9$
$x = 3$
Bigger radius: $8x = 24$
Area of bigger circle:
$\pi(24)^2 = 576\pi$

One or More Options Correct Type

11. Correct Answers: A), C) and D)
Circumference: $2\pi r = 176$
$r = 28$ m
Area:
$\pi r^2 = \frac{22}{7} \times 28 \times 28 = 2464$ m^2
Checking options:
A) 2464 m^2
C) $4 \times 616 = 2464$ m^2

12. Correct Answer: B) $2\sqrt{\pi A}$,D) $\sqrt{4\pi A}$
Area of circle:$A = \pi r^2$
$r = \sqrt{\frac{A}{\pi}}$

Circumference: $2\pi r$

$= 2\pi\sqrt{\frac{A}{\pi}}$

$= 2\sqrt{\pi A}$

13. Correct Answer: B) $29\frac{1}{3}$**cm**

Length of arc:

$\frac{\theta}{360} \times 2\pi r$

$= \frac{240}{360} \times 2 \times \frac{22}{7} \times 7$

$= \frac{2}{3} \times 44$

$= \frac{88}{3}$

$= 29\frac{1}{3}$ cm

Hence, the correct option is **B)** $29\frac{1}{3}$**cm**. A) $\frac{88}{3}$ cm

14. **Correct Answer: B) Rs. 360**

For fencing a semicircular garden, perimeter of semicircle is required.

Perimeter of semicircle: $\pi r + 2r$

Given $r = 7$m

$= \frac{22}{7} \times 7 + 14$

$= 22 + 14$

$= 36$ m

Cost of fencing at Rs. 10 per metre: $36 \times 10 = 360$

15. **Correct Answer: A) 5 : 1**

Angle of sector = 72°

Length of arc:

$\frac{72}{360} \times 2\pi r$

$= \frac{1}{5} \times 2\pi r$

Circumference of circle: $2\pi r$

Ratio of circumference to arc length:

$2\pi r : \frac{1}{5}(2\pi r)$

$= 5:1$

16. **Matrix Matching Answers**

a) Area of sector → **R)**

Area of sector $= \frac{x^\circ}{360^\circ} \times \pi r^2$

b) Length of an arc of a sector → **S)**

Arc length $= \frac{x^\circ}{360^\circ} \times 2\pi r$

c) Perimeter of a sector → **P)**

Perimeter = $l + 2r$

d) If $l$is length of arc then area is → **Q)**

Area = $\frac{lr}{2}$

Final Matching: a → R, b → S, c → P, d → Q

Integer Type

17. **Correct Answer: 2 cm**

Perimeter of semicircle: $\pi r + 2r$

Given $r = 28$cm

$= \frac{22}{7} \times 28 + 56$

$= 88 + 56$

$= 144$ cm

$\frac{1}{72} \times 144 = 2$ cm

Hence, the answer is **2 cm**.

18. **Correct Answer: 9 units**

When the wire is bent into a circle, its length becomes the circumference.

Circumference = 9 units

Hence, the answer is **9 units**.

19. **Correct Answer: 449**

Radius of wheel = 35cm

Circumference of wheel:

$C = 2\pi r$

$= 2 \times \frac{22}{7} \times 35 = 220$ cm

Distance travelled: 987 m = 98700 cm

Number of rotations: $\frac{98700}{220} = 448.63 \approx 449$

Answer: $\boxed{449}$

20. **Correct Answer: 6**

Area of path = 1386 sq.m

Area of lawn = 2464 sq.m

Since:

$1386 = 138 \times 10$

$k = 6$

Hence, the answer is **6**.

Student Practice Corner

1. **Correct Answer: D) 22 sq.cm**

Area of ring: $\pi(R^2 - r^2)$

Outer radius $R = 4$cm

Inner radius $r = 3$cm

$= \frac{22}{7}(16 - 9)$

$= \frac{22}{7} \times 7$

$= 22$ sq.cm

2. **Correct Answer: A) 22 m**

Area of sector:

$A = \frac{1}{2}rl$

Given:

$231 = \frac{1}{2} \times 21 \times l$

$231 = \frac{21l}{2}$

$462 = 21l$

$l = 22$ m

3. **Correct Answer: C) 128 cm**.

Area of sector: $A = \frac{1}{2}rl$

Given: $924 = \frac{1}{2} \times 42 \times l$

$924 = 21l$

$l = 44$ cm

Perimeter of sector:

$l + 2r$

$= 44 + 2(42)$

$= 44 + 84$

$= 128$ cm

4. **Correct Answer: C) 28 cm**

Perimeter of sector:

$l + 2r$

$= 12 + 2(8)$

$= 12 + 16$

$= 28$ cm

5. **Correct Answer: A) 3**

Circumference of wheel:

$2\pi r$

$= 2 \times \frac{22}{7} \times 35$

$= 220$ cm

Distance travelled: 660 cm

Number of rotations:

$\frac{660}{220} = 3$

6. **Correct Answer: B) 42 cm**

Circumference:

$2\pi r = 264$

$2 \times \frac{22}{7} \times r = 264$

$\frac{44r}{7} = 264$

$r = \frac{264 \times 7}{44}$

$r = 42$ cm

7. **Correct Answer: D) 144 cm**
Perimeter of semicircle: $\pi r + 2r$
Given $r = 28$cm
$= \frac{22}{7} \times 28 + 56$
$= 88 + 56$
$= 144$ cm

8. **Correct Answer: C) 352 cm**
Circumference of circle: $2\pi r$
$= 2 \times \frac{22}{7} \times 56$
$= 352$ cm

9. **Correct Answer: B) 10 units**
When a wire is bent into a circle, its length becomes the circumference.
Circumference = 10 units

ONE OR MORE THAN ONE OPTION CORRECT TYPE

10. **Correct Answers: B) and D)**
Area of semicircle:
$\frac{1}{2}\pi r^2 = 1925$
Using $\pi = \frac{22}{7}$,
$\frac{11}{7}r^2 = 1925$
$r^2 = 1225$
$r = 35$ cm
Perimeter of semicircle:
$\pi r + 2r$
$= \frac{22}{7} \times 35 + 70$
$= 110 + 70$
$= 180$ cm
Checking options:
B) 180cm
D) $2 \times 90 = 180$cm
Hence, the correct options are **B and D**.

COMPREHENSION TYPE

Passage – I:

11. **Correct Answer: D) 350 cm²**
Area of ring:
$\pi(R^2 - r^2)$
$= \frac{22}{7}(20^2 - 15^2)$
$= \frac{22}{7}(400 - 225)$
$= \frac{22}{7} \times 175$
$= 550$ cm^2
Hence, the correct option should be **C) 550 cm²**.

12. **Correct Answer: B) 19.8 cm**
Length of arc:
$\frac{\theta}{360} \times 2\pi r$
$= \frac{135}{360} \times 2 \times \frac{22}{7} \times 8.4$
$= \frac{3}{8} \times 52.8$
$= 19.8$ cm
Hence, the correct option is **B) 19.8 cm**.

13. **Correct Answer: C) 83.16 sq.cm**
Area of sector:
$\frac{\theta}{360} \times \pi r^2$
$= \frac{135}{360} \times \frac{22}{7} \times (8.4)^2$
$= \frac{3}{8} \times \frac{22}{7} \times 70.56$
$= 83.16$ sq.cm

14. **Matrix Matching Answers**
a) Circumference of a semicircle → **P)**
$\pi r + 2r$

b) Area of a circle → **T)**
πr^2
c) Diameter of a circle → **S)**
$2r$
d) Area of a semicircle → **Q)**
$\frac{\pi r^2}{2}$
Final Matching: a → P, b → T, c → S, d → Q

INTEGER TYPE

15. **Correct Answer: 5 cm**
Width of ring: $R - r$
Outer radius = 7cm
Inner radius = 2cm
$7 - 2 = 5$ cm
Hence, the answer is **5 cm**

16. **Correct Answer: 8 cm**
Circumference of circle:
$2\pi r$
$= 2 \times \frac{22}{7} \times 56$
$= 352$ cm
$\frac{1}{44} \times 352 = 8$ cm
Hence, the answer is **8 cm**.

Mensuration

LEVEL – I

1. **Correct Answer: C) 1800 cu.m**
   Volume of cuboid:
   $l \times b \times h$
   $= 15 \times 12 \times 10$
   $= 1800$ cu.m
2. **Correct Answer: B) 0.14 sq.cm**
   Lateral surface area of cuboid:
   $2h(l + b)$
   $= 2 \times 0.05 \times (0.8 + 0.6)$
   $= 0.1 \times 1.4$
   $= 0.14$ sq.cm
3. **Correct Answer: C)** $1 - x^4$**cu.units**
   Volume of cuboid:
   $(1 + x)(1 - x)(1 + x^2)$
   Using identity:
   $(1 + x)(1 - x) = 1 - x^2$
   $(1 - x^2)(1 + x^2) = 1 - x^4$
4. **Correct Answer: D) 3310 sq.cm**
   Total surface area of cuboid:
   $2(lb + bh + hl)$
   $= 2[(32 \times 25) + (25 \times 15) + (15 \times 32)]$
   $= 2(800 + 375 + 480)$
   $= 2(1655)$
   $= 3310$ sq.cm
5. **Correct Answer: B) 6 cm**
   Volume of cuboid: $l \times b \times h = 720$
   Given:
   $12 \times 10 \times h = 720$
   $120h = 720$
   $h = 6$ cm
6. **Correct Answer: D) 8:27**
   Volume of first cuboid:
   $10 \times 8 \times 6 = 480$
   Volume of second cuboid:
   $15 \times 12 \times 9 = 1620$
   Ratio: $480 : 1620$
   $= 8 : 27$
7. **Correct Answer: D) 1680 cu.m**
   Lateral surface area of cuboid:
   $2h(l + b) = 520$
   Given $l = 14$cm, $b = 12$cm
   $2h(14 + 12) = 520$
   $2h(26) = 520$
   $52h = 520$
   $h = 10$ cm
   Volume:

$l \times b \times h$
$= 14 \times 12 \times 10$
$= 1680$ cu.cm

8. **Correct Answer: A) 6:5**.
Volumes are equal.
$l_1 b_1 h_1 = l_2 b_2 h_2$
Given:
$l_1 : l_2 = 5:4$
$b_1 : b_2 = 2:3$
So,
$\frac{h_1}{h_2} = \frac{4 \times 3}{5 \times 2}$
$= \frac{12}{10}$
$= 6:5$

9. **Correct Answer: B) 5 cm**
Volume of cuboid:
Base area $\times h = 440$
$88 \times h = 440$
$h = 5$ cm

10. **Correct Answer: A) 7 m²**
For a cuboid: TSA = LSA + 2(base area)
Given: $40 = 26 + 2B$
$14 = 2B$
$B = 7$ m^2

11. **Correct Answer: A) 3375 cu.m**
Volume of cube:
a^3
$= 15^3$
$= 3375$ cu.cm

12. **Correct Answer: A) 100 sq.m**
Lateral surface area of cube:
$4a^2$
$= 4 \times 5^2$
$= 4 \times 25$
$= 100$ sq.cm.

13. **Correct Answer: B) 3:2**.
For a cube:
Total surface area: $6a^2$
Lateral surface area: $4a^2$
Ratio:
$6a^2 : 4a^2$
$= 3:2$

14. **Correct Answer: D) 8V**
If each edge is doubled:
$(2a)^3 = 8a^3$
Original volume = $V = a^3$
New volume: $8V$

15. **Correct Answer: C) 384 cm²**
Diagonal of cube: $a\sqrt{3} = 8\sqrt{3}$
$a = 8$ cm
Surface area: $6a^2$
$= 6 \times 8^2$
$= 6 \times 64$
$= 384$ cm^2

16. **Correct Answer: B) 13,824 cu.cm**.
Total surface area:
$6a^2 = 3456$
$a^2 = 576$
$a = 24$ cm
Volume:
$a^3 = 24^3$
$= 13{,}824$ cu.cm

17. **Correct Answer: D)** $3\sqrt{3}V = a^3$
Diagonal of cube:
$d = a\sqrt{3}$
Given diagonal = a
So edge:
$\frac{a}{\sqrt{3}}$

Volume:
$V = \left(\frac{a}{\sqrt{3}}\right)^3$
$V = \frac{a^3}{3\sqrt{3}}$
$3\sqrt{3}V = a^3$

18. **Correct Answer: A) 600 cm²**
Volume:
$a^3 = 1000$
$a = 10$ cm
Total surface area:
$6a^2$
$= 6 \times 10^2$
$= 600$ cm^2

19. **Correct Answer: B)** $10\sqrt{3}$**cm**
Total surface area:
$6a^2 = 600$
$a^2 = 100$
$a = 10$ cm
Diagonal of cube:
$a\sqrt{3}$
$= 10\sqrt{3}$ cm
Hence, the correct option is **B)** $10\sqrt{3}$**cm**.

20. **Correct Answer: C) 2:1**
Ratio of volumes: 8: 1
Ratio of edges:
$\sqrt[3]{8}: \sqrt[3]{1}$
2: 1

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: A) 200 sq.cm**
Lateral surface area of cuboid:
$2h(l + b)$
Given:
$l = 12$ cm, $b = 8$ cm, $h = 5$ cm
$= 2 \times 5(12 + 8)$
$= 10 \times 20$
$= 200$ sq.cm

2. **Correct Answer: D) $\sqrt{356}$cm**
Diagonal of cuboid:
$d = \sqrt{l^2 + b^2 + h^2}$
$= \sqrt{16^2 + 8^2 + 6^2}$
$= \sqrt{256 + 64 + 36}$
$= \sqrt{356}$ cm

3. **Correct Answers: A), C) and D)**
Volume of cube is proportional to cube of its edge.
If side becomes double:
$(2a)^3 = 8a^3$
So volume becomes 8times.
Checking options:
A) 8
D) $2^3 = 8$
Hence, the correct options are **A and D**.

4. **Correct Answer: A) 48 cm²**
Lateral surface area of triangular prism:
(Perimeter of base) × height
Perimeter of triangular base:
$3 + 4 + 5 = 12$ cm
Height of prism = 4cm
$12 \times 4 = 48$ cm^2

5. **Correct Answer: B) 60 cm²**
The triangular base has sides 3cm, 4cm, 5cm.
Since it is a right triangle,
Area of base $= \frac{1}{2} \times 3 \times 4$
$= 6$ cm^2
Height of prism = 4cm
Lateral surface area: $(3 + 4 + 5) \times 4$
$= 12 \times 4 = 48$ cm^2
Total surface area: $48 + 2(6)$

$= 48 + 12$

$= 60\ \text{cm}^2$

6. **Correct Answer: A) 24 cm³**

Area of triangular base:

$\frac{1}{2} \times 3 \times 4$

$= 6\ \text{cm}^2$

Height of prism = 4cm

Volume: 6×4

$= 24\ \text{cm}^3$

7. **Correct Answer: A) 12 cm**

Lateral surface area of cube:

$4a^2 = 576$

$a^2 = 144$

$a = 12$ cm

8. **Correct Answer: D) 10 cm**

Surface area of cube:

$6a^2 = 600$

$a^2 = 100$

$a = 10$ cm

9. **Correct Answer: B) 480 cm³**.

Volume of cuboid:

Base area × height

$= 96 \times 5$

$= 480\ \text{cm}^3$

10. **Correct Answer: B) 1980 cu.cm**

Volume of cuboid:

$l \times b \times h$

$= 22 \times 12 \times 7.5$

= 1980 cu.cm

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answers: C) and D)**

Length : Width : Height = 3: 2: 1

Let dimensions be:

$3x, 2x, x$

Volume: $3x \times 2x \times x = 48$

$6x^3 = 48$

$x^3 = 8$

$x = 2$

Dimensions: 6, 4, 2

Total surface area: $2(lb + bh + hl)$

$= 2[(6 \times 4) + (4 \times 2) + (2 \times 6)]$

$= 2(24 + 8 + 12)$

$= 2(44)$

$= 88\ \text{cm}^2$

Checking options:

C) $2(44)\ \text{cm}^2$

D) $88\ \text{cm}^2$

Hence, the correct options are **C and D**.

12. **Correct Answers: A) and D)**

For cuboid: $V = xyz$

Surface area: $A = 2(xy + yz + zx)$

Therefore,

$\frac{A}{V} = \frac{2(xy + yz + zx)}{xyz}$

$= 2\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$

$= \frac{2(xy + yz + zx)}{xyz}$

Checking options:

A) $2\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$

D) $2\left(\frac{xy+yz+zx}{xyz}\right)$

Hence, the correct options are **A and D**.

13. **Correct Answers: B) and C)**

Ratio of volumes: $8:1$

Ratio of edges: $\sqrt[3]{8}:\sqrt[3]{1}$

$2:1$

Checking options:

A) $50:25 = 2:1$

C) $2:1$

Hence, the correct options are **A and C**

COMPREHENSION TYPE

Passage – I:

14. **Correct Answer: B) $\sqrt{50}$m**

Diagonal of cuboid:

$d = \sqrt{l^2 + b^2 + h^2}$

Given: $l = 5, b = 4, h = 3$

$d = \sqrt{5^2 + 4^2 + 3^2}$

$= \sqrt{25 + 16 + 9}$

$= \sqrt{50}$ m

15. **Correct Answer: C) 54 cm²**

Lateral surface area of cuboid:

$2h(l + b)$

$= 2 \times 3(5 + 4)$

$= 6 \times 9$

$= 54$ m^2

16. **Correct Answer: D) 48 cm**

Length of all 12 edges of cuboid:

$4(l + b + h)$

$= 4(5 + 4 + 3)$

$= 4 \times 12$

$= 48$ m

17. **Matrix Matching Answers**

a) Edge of a cube is 3 times more, then new volume is __ times → **Q) 27**

If edge becomes $3a$,

$(3a)^3 = 27a^3$

So volume becomes 27times.

b) Each dimensions are doubled in a cuboid, then the volume becomes __ times → **P) 8**

If dimensions become $2l, 2b, 2h$,

$(2l)(2b)(2h) = 8lbh$

So volume becomes 8times.

c) Dimensions of a cuboid are $a, b, c$and total surface area is $A$and volume is V→ **R)**

$V = abc$

$A = 2(ab + bc + ca)$

$\frac{A}{V} = 2\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$

Hence, c → R.

d) Dimensions of a cuboid are $a, b, c$and total surface area is $A$and volume is Vthen $\frac{V}{A}$→ **S)**

$\frac{V}{A} = \frac{abc}{2(ab + bc + ca)}$

$= \frac{1}{2\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)}$

Hence, d → S.

Final Matching: a → Q, b → P, c → R, d → S

INTEGER TYPE

18. **Correct Answer: 5 cm**

Volume of cuboid: Base area × height

Given: $900 = 180 \times h$

$h = \frac{900}{180}$

$h = 5$ cm

19. **Correct Answer: 8**

Volume of one soap: $10 \times 5 \times 5 = 250$ cm^3

Volume of box: $20 \times 10 \times 5 = 1000$ cm^3

Number of soaps: $\frac{1000}{250} = 4$

So, the correct answer should be **4**.

20. **Correct Answer: 12 cm**

In a square pyramid: $l^2 = h^2 + \left(\frac{a}{2}\right)^2$

Given:

Slant height $l = 13$cm
Half of base side = 5cm
$13^2 = h^2 + 5^2$
$169 = h^2 + 25$
$h^2 = 144$
$h = 12$ cm

Student Practice Corner

Single option correct type

1. **Correct Answer: A)** $3840\sqrt{3}$ cm^3
Volume of triangular prism:
Area of base × length
Given:
$64\sqrt{3} \times 60$
$= 3840\sqrt{3}$ cm^3
Hence, the correct option is **A)** $3840\sqrt{3}$ cm^3.

2. **Correct Answer: A)** $6 \times \frac{\sqrt{3}}{4}a^2$
A regular hexagon consists of 6equilateral triangles of side a.
Area of one equilateral triangle:
$\frac{\sqrt{3}}{4}a^2$
Area of hexagon: $6 \times \frac{\sqrt{3}}{4}a^2$
Hence, the correct option is **A)**.

3. **Correct Answer: D)** $n + 2$
A prism with n-sided base has:
- $n$lateral faces
- 2bases

Total surfaces: $n + 2$
Hence, the correct option is **D)** $n + 2$.

4. **Correct Answer: C) 900 cm²**
Lateral surface area of prism: Perimeter of base × height
Base is an equilateral triangle of side 12cm.
Perimeter: $3 \times 12 = 36$ cm
Height of prism = 25cm
$36 \times 25 = 900$ cm^2
Hence, the correct option is **C) 900 cm²**.

5. **Correct Answer: D) 1350 cm³**
Base is a right-angled isosceles triangle with equal sides 15cm.
Area of base: $\frac{1}{2} \times 15 \times 15$
$= \frac{225}{2}$ cm^2

Height of prism = 12cm
Volume:
$\frac{225}{2} \times 12$
$= 1350$ cm^3

6. **Correct Answer: A) 2400 sq.cm**.
Lateral surface area of triangular prism: (Perimeter of base) × height
Perimeter of triangular base: $15 + 20 + 25 = 60$ cm
Height of prism = 40cm
$60 \times 40 = 2400$ cm^2

7. **Correct Answer: C) 912 sq.cm**
The triangular base has sides 12cm, 16cm, 20cm.
Since: $12^2 + 16^2 = 20^2$
it is a right triangle.
Area of base: $\frac{1}{2} \times 12 \times 16$
$= 96$ cm^2
Perimeter of base: $12 + 16 + 20 = 48$ cm
Lateral surface area: $48 \times 15 = 720$ cm^2
Total surface area: $720 + 2(96)$
$= 720 + 192$
$= 912$ cm^2

8. **Correct Answer: B)** $192\sqrt{3}$ cm^3
Base is an equilateral triangle of side 8cm.
Area of base: $\frac{\sqrt{3}}{4}a^2$
$= \frac{\sqrt{3}}{4} \times 8^2$
$= 16\sqrt{3}$ cm^2
Height of prism = 12cm
Volume: $16\sqrt{3} \times 12$
$= 192\sqrt{3}$ cm^3

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. **Correct Answers: B) and C)**
Diagonal of cube:
$a\sqrt{3} = 14\sqrt{3}$
$a = 14$ cm
Surface area of cube:
$6a^2$
$= 6 \times 14^2$
$= 6 \times 196$
$= 1176$ cm^2
Checking options:
B) $2 \times 588 = 1176$ cm^2
C) 1176 cm^2
Hence, the correct options are **B and C**.

10. **Correct Answers: C) and D)**
Edge of cube = 5cm
Volume of cube: a^3
$= 5^3$
$= 125$ cu.cm

Checking options:
C) 125 cu.cm
D) $5 \times 25 = 125$ cu.cm
Hence, the correct options are **C and D**.

COMPREHENSION TYPE

11. **Correct Answer: C)** $5\sqrt{3}$**cm**
Diagonal of cube:
$a\sqrt{3}$
$= 5\sqrt{3}$ cm
Hence, the correct option is **C)** $5\sqrt{3}$**cm**.

12. **Correct Answer: A) 150 sq.cm**
Total surface area of cube:
$6a^2$
$= 6 \times 5^2$
$= 6 \times 25$
$= 150$ sq.cm
Hence, the correct option is **A) 150 sq.cm**.

13. **Matrix Matching Answers**
a) Diagonal of the cuboid of dimensions l, b, hunits → **S)**
$\sqrt{l^2 + b^2 + h^2}$
b) Total surface area of the cuboid of dimensions l, b, hunits → **T)**
$2(lb + bh + lh)$
c) Diagonal of a cube with edge $a$units → **Q)**
$a\sqrt{3}$
d) Length of all 12 edges of the cube with edge $a$units → **P)**
$12a$
Final Matching: a → S, b → T, c → Q, d → P

INTEGER TYPE

14. **Correct Answer: 4 cm**
Total surface area of cube:
$6a^2 = 96$
$a^2 = 16$
$a = 4$ cm
Hence, the answer is **4 cm**.

15. **Correct Answer: 5 cm**
Volume of prism:
Base area × height
$40 = 8 \times h$
$h = 5$ cm
Hence, the answer is **5 cm**.

8. Set Theory

LEVEL – I

1. In $A = \{2,4,6,8\}$, the element present in the set is 8.
   Answer: C) $8 \in A$
2. The set-builder form describes the property of the elements.
   Answer: B) $\{x: x$ is an even number less than $10\}$
3. Factors of 18are 1,2,3,6,9,18.
   So the roster form is:
   $\{1, 2, 3, 6, 9, 18\}$
   Answer: A) $\{1, 2, 3, 6, 9, 18\}$
4. The set $\{5, 10, 15, 20, 25\}$contains multiples of 5 less than 30.
   Answer: B) $\{x: x$ is a multiple of 5 less than $30\}$
5. $C = \{x: x$ is an odd number less than $10\}$
   Odd numbers less than 10are 1,3,5,7,9.
   Answer: B) $\{1, 3, 5, 7, 9\}$
6. $B = \{x: x$ is a multiple of 3 less than $15\}$
   Multiples of 3less than 15 are 3,6,9,12.
   Answer: A) $\{3, 6, 9, 12\}$
7. Prime numbers less than 12are 2,3,5,7,11.
   Answer: B) $\{2, 3, 5, 7, 11\}$
8. $A = \{x: x$ is a prime number less than $12\}$
   The elements are $\{2, 3, 5, 7, 11\}$.
   So, the number of elements is:
   $n(A) = 5$
   Answer: B) 5
9. $A = \{2,4,6,8\}$
   $B = \{8,6,4,2\}$
   Both sets contain exactly the same elements. Order does not matter in sets.
   Answer: B) A and B are equal sets
10. Equivalent sets have the same number of elements, but different elements.
    A) Same elements → equal sets
    B) Both have 3 elements, but different elements → equivalent but not equal
    C) Different number of elements
    D) Same singleton set
    Answer: B) $\{a, b, c\}$**and** $\{2, 4, 6\}$
11. $A = \{2,4,6,8,10\} \Rightarrow n(A) = 5$
    $B = \{1,2,3,4,5,6\} \Rightarrow n(B) = 6$
    $n(A) + n(B) = 5 + 6 = 11$
    Answer: C) 11
12. $A = \{2,4,6,8\}$contains even numbers.
    $B = \{1,3,5,7,9\}$contains odd numbers.
    They have no common elements, so they are disjoint sets.
    Answer: B) A and B are disjoint

13. The vowels in "MISSISSIPPI" are only I.
    So the set of vowels is $\{I\}$.
    Cardinal number = 1.
    Answer: A) 1
14. A person cannot be 200 years old, so the set has no elements.
    Answer: C) Null
15. A month with 32 days does not exist, so the set is empty.
    Answer: B) $\{x: x$ is a month with 32 days$\}$
16. In a set, repeated elements are not written twice.
    The word PRINCIPAL has letters:
    $\{P, R, I, N, C, A, L\}$
    Option (c) repeats P.
    Answer: C) $\{P, R, I, N, C, I, P, A, L\}$
17. Every element of $A = \{2,4,6,8\}$is present in
    $B = \{1,2,3,4,5,6,7,8\}$.
    So $A$is a subset of B.
    Answer: C) $A \subseteq B$
18. For a set with $n$elements:
    Number of proper subsets = $2^n - 1$
    For $n = 5$:
    $2^5 - 1 = 31$
    Answer: B) 31
19. Number of subsets of a set with $n$elements is:
    $2^n = 64$
    Since $2^6 = 64$, the set has 6 elements.
    Answer: B) Set with 6 elements
20. A proper subset cannot be equal to the original set.
    For $S = \{1,2,3\}$, the set $\{1, 2, 3\}$is equal to S, so it is not a proper subset.
    Answer: C) $\{1, 2, 3\}$

LEVEL – II

1. The set $A = \{2,4,6,8\}$contains even numbers less than 10.
   Answer: B) $\{x: x$ is an even number less than 10$\}$
2. Squares of integers less than 100 are:
   0,1,4,9,16,25,36,49,64,81
   So the correct set is:
   Answer: B) $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$
3. **Which sets are not empty?**
   a) $\{x: x$ is an even prime greater than 3$\}$
   The only even prime number is 2.
   2is not greater than 3.
   So this set is **empty**.

b) $\{x: x$ is a multiple of 2 and is odd$\}$

- Multiples of 2are always even, not odd.
- So this set is **empty**.

c) $\{x: x$ is an even number and $x + 3$ is even$\}$

- If $x$is even, then $x + 3$ becomes odd.
- So both conditions cannot happen together.
- This set is **empty**.

d) $\{x: x$ is a prime number less than 5 and is odd$\}$

- Prime numbers less than 5 are 2,3.
- Odd prime among them is 3.
- So this set is **not empty**.

Answer: D

4. Which of the following is an infinite set?

A) Set of letters in the English alphabet → 26 letters (finite)
B) Set of whole numbers less than 100 → finite
C) Set of multiples of 5 → 5,10,15,20, ...continues forever
D) Set of months in a year → 12 months (finite)

Answer: C) Set of multiples of 5

5. Let $A = \{x: x$ is an odd number less than 10$\}$

So,$A = \{1,3,5,7,9\}$
Given:$B = \{1,3,5,7,9\}$
Both sets contain exactly the same elements.

Answer: A) A and B are equal sets

6. If $A = \{x: x$ is a factor of 24 and x is even$\}$

Factors of 24: 1,2,3,4,6,8,12,24
Even factors: 2,4,6,8,12,24
So,$n(A) = 6$

Answer: D) 6

7. Let

$A = \{2,4,6\}, B = \{2,4,6,8\}, C = \{4,6\}$
Check each option:

- A) $C \subset A \rightarrow$ True, since 4,6are in A
- B) $A \subset B \rightarrow$ True
- C) $C \subseteq B \rightarrow$ True
- D) $B \subset A \rightarrow$ False, because $8 \notin A$

Answer: D) $B \subset A$

8. Let

$A = \{1,3,5,7\}$
$B = \{x: x$ is an odd natural number less than 8$\}$
Odd natural numbers less than 8are: $B = \{1,3,5,7\}$
Thus, both sets are equal.

Answer: A) $A = B$

9. Let

$P =$letters of the word "assessment"
Unique letters in **assessment**:
$P = \{a, s, e, m, n, t\}$
$Q =$letters of the word "statement"
Unique letters in **statement**:
$Q = \{s, t, a, e, m, n\}$
Both sets contain the same elements.

Answer: A) $P = Q$

10. Let $M =$ {odd natural numbers less than 12}

So, $M = \{1,3,5,7,9,11\}$
Given:$N = \{x: x = m + 2, m \in \mathbb{N}$ and $m \leq 6\}$
For $m = 1,2,3,4,5,6,$

$x = 3,4,5,6,7,8$
Thus,
$N = \{3,4,5,6,7,8\}$
Both sets have 6 elements, but elements are different.
So they are **equivalent sets** only.
Answer: B) M and N are equivalent sets

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. Which of the following roster forms are incorrect?
A) $\{x: x \text{ is a vowel}\} = \{a, e, i, o, u\}$
Correct
B) $\{x: x \text{ is a factor of } 10\} = \{1,2,5\}$
Incorrect, because 10is also a factor of 10
Correct set: $\{1, 2, 5, 10\}$
C) $\{x: x \text{ is a multiple of 4 less than } 15\} = \{4,8,12\}$
Correct
D) $\{x: x \text{ is a prime number less than } 10\} = \{2,3,5,9\}$
Incorrect, because 9 is not prime
Correct set: $\{2, 3, 5, 7\}$
Answers: B and D

12. Which set-builder forms are equivalent to $\{3,6,9,12,15\}$?
A) $\{x: x \text{ is a multiple of 3 less than } 16\}$
Gives $\{3, 6, 9, 12, 15\}$
Correct
B) $\{x: x = 3n,\ n = 1,2,3,4,5\}$
Gives $\{3, 6, 9, 12, 15\}$
Correct
C) $\{x: x \text{ is divisible by } 5\}$
Infinite set, not equivalent
D) $\{x: x \text{ is a factor of } 15\}$
Gives $\{1, 3, 5, 15\}$
Not equivalent
Answers: A and B

13. Which of the following sets are non-empty sets?
A) Set of even prime numbers
Contains 2
Non-empty
B) Set of natural numbers less than 0
No such natural numbers
Empty
C) Set of months having 30 days
April, June, September, November
Non-empty
D) Set of vowels in the word "EDUCATION"
Vowels are E, U, A, I, O
Non-empty
Correct Answers: A, C and D

14. **If** $P = \{x: x$ is an even number less than 10$\}$
So, $P = \{2,4,6,8\}$
and $Q = \{2,4,6,8\}$
Check each statement:
A) $Q \subset P$
True
B) $P = Q$
True
C) $\{2,6\} \subset P$
True
D) $\{0,2,4\} \subset P$
False, since $0 \notin P$
Correct Answers: A, B and C

COMPREHENSION TYPE

PASSAGE

15. **Let**
M =set of multiples of 4
E =set of even numbers
Every multiple of 4 is an even number, so M is a subset of E.
Thus, $E$is the super-set of M.
Answer: B) $E \supseteq M$

16. **Which number belongs to both the set of multiples of 4 and even numbers?**
Check options:
7→ not even
9→ not even
12→ even and multiple of 4
15→ not even
Answer: C) 12

17. **Which of the following is an even number but NOT a multiple of 4?**
8→ multiple of 4
16→ multiple of 4
10→ even but not divisible by 4
20→ multiple of 4
Answer: C) 10

MATRIX MATCHING TYPE

18. **Match the Column A and Column B**
P. $\{x: x$ is a factor of 36 and $x > 5\}$
Factors of 36greater than 5:
$\{6, 9, 12, 18, 36\}$
Matches with **B**
P → B
Q. $\{1, 8, 27, 64, 125\}$
These are cubes:
$1^3, 2^3, 3^3, 4^3, 5^3$
Matches with
$\{x: x = n^3, n \in \mathbb{N}$ and $n \leq 5\}$
Q → C

R. $\{x: x$ is a perfect square less than 50$\}$
Perfect squares less than 50:
$\{1, 4, 9, 16, 25, 36, 49\}$
Matches with **A**
R → A
S. $\{2, 3, 5, 7, 11, 13, 17, 19\}$
These are prime numbers less than 20.
Matches with **D**
S → D

INTEGER TYPE

19. If a set has 256 subsets, find the number of elements.
Number of subsets of a set with $n$elements:
$2^n = 256$
$256 = 2^8$
So, $n = 8$
Answer: 8

20. Find the cardinal number of the set
$P = \{x: x$ is a prime number less than 10$\}$
Prime numbers less than 10:
$\{2, 3, 5, 7\}$
Number of elements:
$n(P) = 4$
Answer: 4

STUDENT PRACTISE CORNER

1. Which of the following sets are equal?
Equal sets contain exactly the same elements.
A) $\{1, 2, 3\}$and $\{3, 2, 1\}$
Same elements, order does not matter
Equal sets
B) $\{1, 2\}$and $\{1, 2, 3\}$
Different elements
Not equal
C) $\{a, b\}$and $\{a, b, c\}$
Different elements
Not equal
D) $\{5, 6\}$and $\{6, 7\}$
Different elements
Not equal
Answer: A

2. Which is the roster form of vowels?
Roster form lists all elements explicitly.
A) $\{x: x$ is vowel$\}$
Set-builder form
B) $\{a, e, i, o, u\}$
Correct roster form
C) “Vowels in English”
Statement, not a set form
D) $\{vowels\}$

Incorrect notation

Answer: B

3. **If** $A = \{x: x$ is a multiple of 3 less than 20$\}$

Multiples of 3 less than 20:

$\{3, 6, 9, 12, 15, 18\}$

Number of elements:

$n(A) = 6$

Answer: B) 6

4. $A = \{$**letters of the word "MATHEMATICS"**$\}$

In a set, repeated letters are counted only once.

Letters in "MATHEMATICS":

$\{M, A, T, H, E, I, C, S\}$

Number of distinct letters:

$n(A) = 8$

Answer: A) 8

5. **If a set has 3 elements, then the number of subsets is**

Number of subsets of a set with $n$elements:

2^n

For $n = 3$:

$2^3 = 8$

Answer: C) 8

6. **Let**

P =set of letters of the word "malayalam"

Unique letters:

$P = \{m, a, l, y\}$

Q =set of letters of the word "maly"

$Q = \{m, a, l, y\}$

Both sets contain exactly the same elements.

Answer: A) $P = Q$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

7. **Which set-builder forms represent integers between 2 and 9?**

Integers between 2 and 9 are:

$\{3, 4, 5, 6, 7, 8\}$

Check options:

A) $\{x: x$ is an integer and $2 < x < 9\}$

Gives $\{3, 4, 5, 6, 7, 8\}$

Correct

B) $\{x: x = 3,4,5,6,7,8\}$

Represents the same set

Correct

C) $\{x: x > 2\}$

Includes many numbers greater than 9

Incorrect

D) $\{x: x$ is an integer less than 9$\}$

Includes integers less than 2also

Incorrect

Correct Answers: A and B

8. **Let $P = \{1, 2, 3, 4, 5\}$. Which of the following are NOT subsets of P?**

A) $\{2, 3, 5\}$

All elements are in P

Subset

B) $\{1, 6\}$

$6 \notin P$

Not a subset

C) $\{4, 5\}$

Subset

D) $\{0\}$

$0 \notin P$

Not a subset

Correct Answers: B and D

9. **If** S = {multiples of 3 less than 20}

Then, S = {3,6,9,12,15,18}

Check options:

A) $\{3, 6, 9\}$

All elements are in S

Subset

B) $\{12, 15, 18\}$

Subset

C) $\{0, 3, 6\}$

$0 \notin S$

Not a subset

D) $\{6, 12, 21\}$

$21 \notin S$

Not a subset

Correct Answers: A and B

COMPREHENSION TYPE

PASSAGE

Given:

F = {apple, banana, orange, grapes}

A = {apple, banana}

Every element of $A$is in F, so:

$A \subseteq F$and$F \supseteq A$

10. **Which statement is correct?**

A) $A \supseteq F$

False

B) $F \supseteq A$

True

C) $A = F$

False

D) $F \subset A$

False

Answer: B) $F \supseteq A$

11. Which fruit belongs to set F but not to set A?

Set F: {apple, banana, orange, grapes}
Set A: {apple, banana}
Elements in F but not in A:
{orange, grapes}
Among the options, only **orange** is given.
Answer: C) orange

12. Which set is a subset of F?

A subset must contain only elements from F.
A) {apple, banana}
Both are in F
Subset
B) {mango, grapes}
Mango not in F
C) {orange, mango}
Mango not in F
D) {pineapple}
Pineapple not in F
Answer: A) {apple, banana}

Matrix matching type

13. Match Column A and Column B

P. $\{x: x$ is divisible by both 2 and 3 and $x < 25\}$
Numbers divisible by both 2 and 3 are multiples of 6:
$\{6, 12, 18, 24\}$
Matches with B
P → B
Q. $\{1, 2, 4, 8, 16, 32\}$
These are powers of 2:
$2^0, 2^1, 2^2, 2^3, 2^4, 2^5$
Matches with
$\{x: x = 2^n$ where n is a whole number and $n < 6\}$
Q → A
R. $\{x: x$ is a multiple of 7 less than 50$\}$
Multiples of 7less than 50:
$\{7, 14, 21, 28, 35, 42, 49\}$
Matches with C
R → C
S. $\{3, 6, 9, 12, 15\}$
These are multiples of 3less than 18.
Matches with D
S → D

INTEGER TYPE

14. If $A = \{2,4,6,8,10,12\}$

Even numbers greater than 6:
$\{8, 10, 12\}$
Number of elements:3
Answer: 3

15. A set has 32 subsets. Find the number of elements.

Number of subsets formula:

$2^n = 32$

$32 = 2^5$

So, $n = 5$

Answer: 5

Veen diagrams and Basic Operations on Sets

LEVEL – 1

1. The union of the sets $\{a, b\}$and $\{b, c, d\}$is

Union means all distinct elements from both sets.

$$\{a, b\} \cup \{b, c, d\} = \{a, b, c, d\}$$

Answer: A) $\{a, b, c, d\}$

2. The intersection of the sets $\{2, 3, 5, 7\}$and $\{1, 3, 5, 9\}$ is

Intersection means common elements.

Common elements are 3and 5.

$\{2,3,5,7\} \cap \{1,3,5,9\} = \{3,5\}$

Answer: C) $\{3, 5\}$

3. Let

$A = \{2,4,6,8\}, B = \{1,2,3,4\}$

Find $A \cup B$.

Union contains all distinct elements from both sets:

$A \cup B = \{1,2,3,4,6,8\}$

Answer: B) $\{1, 2, 3, 4, 6, 8\}$

4. In a class of 40 students

22students play Cricket (C)

18students play Football (F)

8students play both games

Students who play only Cricket:

$n(C) - n(C \cap F) = 22 - 8$

$22 - 8 = 14$

Answer: B) 14

5. Let

$U = \{1,2,3,4,5,6,7,8,9\}$

$A = \{2,4,6,8\}$

A'means complement of A, i.e., elements in U but not in A.

$A' = \{1,3,5,7,9\}$

Answer: A) $\{1, 3, 5, 7, 9\}$

6. Let

$P = \{a, e, i, o, u\}, Q = \{a, b, c, d, e\}$

Find $P \cap Q$.

Intersection means common elements.

Common elements are $a$and e.

$P \cap Q = \{a, e\}$

Answer: C) $\{a, e\}$

7. Let $X = \{5,10,15,20\}, Y = \{10,20,30,40\}$

Find $X - Y$. $X - Y$ means elements in X but not in Y.

Removing 10 and 20 from X: $X - Y = \{5,15\}$

Answer: A) $\{5, 15\}$

8. **Let** $M = \{2,4,6,8\}, N = \{4,8,10,12\}$
Find $M \cup N$.
Union means all distinct elements from both sets.
$M \cup N = \{2,4,6,8,10,12\}$
Answer: C) $\{2, 4, 6, 8, 10, 12\}$

9. **In a school survey**
28 students like Drawing
20 students like Music
12 students like both
Students who like only Drawing:
$28 - 12$
$28 - 12 = 16$
Answer: C) 16

10. **If**
$A =$ {prime numbers less than 10}
$A = \{2,3,5,7\}$
$B =$ {odd numbers less than 10}
$B = \{1,3,5,7,9\}$
Common elements:
$A \cap B = \{3,5,7\}$
Answer: C) $\{3, 5, 7\}$

11. **If** $A = \{1,4,6\}, B = \{3,6\}, C = \{3,4,5\}$
First find $B \cup C$:
$B \cup C = \{3,4,5,6\}$
Now find intersection with A:
$A \cap (B \cup C) = \{4,6\}$
Answer: C) $\{4, 6\}$

12. **If**
$n(X \cup Y) = 40$
$n(X) = 28, n(Y) = 22$
Using formula:
$n(X \cap Y) = n(X) + n(Y) - n(X \cup Y)$
$n(X \cap Y) = 28 + 22 - 40$
$= 10$
Answer: C) 10

13. Let
$n(M) = 20$, $n(S) = 15$, $n(M \cap S) = 5$, Total $= 50$
Using formula:
$n(M \cup S) = 20 + 15 - 5 = 30$
Students who do not like any subject:
$50 - 30 = 20$
Answer: D. 20

14. $A = \{1,2,3,4,5\}$
$B = \{2,3,5\}$
$C = \{1,3,5,7\}$
Now, $A \cup B = \{1,2,3,4,5\}$
$A \cup C = \{1,2,3,4,5,7\}$
Intersection: $(A \cup B) \cap (A \cup C) = \{1,2,3,4,5\}$
Answer: B. $\{1, 2, 3, 4, 5\}$

15.
$A = \{1,2,3,4,5\}$
B =even natural numbers = $\{2,4,6, \dots\}$
Difference $A - B$ means elements in A not in B:
$A - B = \{1,3,5\}$
Answer: A. $\{1, 3, 5\}$

16. $U = \{10,20,30,40,50\}$, $A = \{20,40\}$
Complement of A:
$A' = \{10,30,50\}$
So, $n(A') = 3$
Answer: B. 3

17. Use the formula:
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Substitute values:
$n(A \cup B) = 25 + 18 - 7 = 36$
Answer: A. 36

18. $n(B) = 28$, $n(A \cap B) = 9$
Elements in $B - A$:
$n(B - A) = n(B) - n(A \cap B)$
$n(B - A) = 28 - 9 = 19$
Answer: A. 19

19. Students who like only football:
$n(F - C) = n(F) - n(F \cap C)$
$= 35 - 10 = 25$
Answer: B. 25

20. Students who study only Sanskrit:
$n(S - H) = n(S) - n(S \cap H)$
$= 40 - 20 = 20$
Answer: A. 20

LEVEL – II

1. Students who participated in exactly one Olympiad:
Only Science $= 50 - 20 = 30$
Only Mathematics $= 40 - 20 = 20$
Exactly one Olympiad $= 30 + 20 = 50$
Answer: C) 50

2. Students who read only comics:
$n(C - S) = n(C) - n(C \cap S)$
$= 18 - 5 = 13$
Answer: C) 13

3. $A = \{2,4,6,8\}$, $B = \{1,2,3,4,5\}$
$A \cup B = \{1,2,3,4,5,6,8\}$
Complement in $U = \{1,2,3,4,5,6,7,8,9\}$:
$(A \cup B)' = \{7,9\}$
Answer: B) $\{7, 9\}$
$A = \{3,6,9,12\}$(multiples of 3 less than 15)
$B = \{1,3,5,7,9,11,13\}$(odd numbers less than 15)
Intersection: $A \cap B = \{3,9\}$
Answer: A) $\{3, 9\}$

5. Use the formula:
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
=28+19−7=40
Answer: C) 40

6.
$A = \{45,54,63,72\}$
$B = \{54,81,90\}$
$C = \{45,54,60\}$
First find intersection:
$A \cap B = \{54\}$
Now union with C:
$(A \cap B) \cup C = \{45,54,60\}$
Answer: D. $\{45, 54, 60\}$

7.
$A = \{1,2,3,4,5,6\}$, $B = \{2,3,5,7\}$
$A - B = \{1,4,6\}$
$B - A = \{7\}$
$(A - B) \cup (B - A) = \{1,4,6,7\}$
Answer: C. $\{1, 4, 6, 7\}$

8. Use the formula:
$n(A - B) = n(A) - n(A \cap B)$
$= 32 - 12 = 20$
Answer: A. 20

9. Given:
$n(A) = 50$
$n(A - B) = 18$
Using:
$n(A \cap B) = n(A) - n(A - B)$
$= 50 - 18 = 32$
Answer: A. 32

10. Use the formula:
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Substitute values:
$90 = 52 + n(B) - 18$
$90 = 34 + n(B)$
$n(B) = 56$
Answer: C. 56

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11.
$A = \{2,4,6,8,10\}$, $B = \{4,8,10\}$
$B \subset A$
$A \cap B = \{4,8,10\}$
$A - B = \{2,6\}$
$A \cup B = \{2,4,6,8,10\}$
Correct Answers: A, B, C, D

12.

$A = \{1,3,5,7,9\}$, $B = \{2,4,6,8\}$

Since there are no common elements:

$A \cap B = \phi$

$A \cap B = \phi$

$A \cup B = \{1,2,3,4,5,6,7,8,9\}$

$A - B = A$

$B - A = B \neq \phi$

Correct Answers: A, B, C

13.

Factors of 18:

$A = \{1,2,3,6,9,18\}$

Even numbers less than 10:

$B = \{2,4,6,8\}$

$A = \{1,2,3,6,9,18\}$

$B = \{2,4,6,8\}$

$A \cap B = \{2,6\}$

$A \cup B = \{1,2,3,4,6,8,9,18\}$

Correct Answers: A, B, C, D

COMPREHENSION TYPE

PASSAGE

14. Students who use only YouTube:

$n(Y - B) = n(Y) - n(Y \cap B)$

$= 45 - 20 = 25$

Answer: B) 25

15. Students who use only Byju's:

$n(B - Y) = n(B) - n(Y \cap B)$

$= 38 - 20 = 18$

Answer: A) 18

16. First find students using at least one platform:

$n(Y \cup B) = n(Y) + n(B) - n(Y \cap B)$

=45+38−20=63

Students using neither platform:

$80 - 63 = 17$

Answer: A) 17

MATRIX MATCHING TYPE

17. P. If $n(A) = 28$, $n(B) = 20$, $n(A \cap B) = 4$. Find $n(A \cup B)$

Using formula:

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$n(A \cup B) = 28 + 20 - 4$

$= 48 - 4$

$= 44$

Matches with **A. 44**

Q. If $n(P) = 50$, $n(Q) = 35$, $n(P \cup Q) = 70$. Find $n(P \cap Q)$

Using formula:

$n(P \cap Q) = n(P) + n(Q) - n(P \cup Q)$

$n(P \cap Q) = 50 + 35 - 70$

$= 85 - 70$

$= 15$

Matches with **B. 15**

R. If $n(M) = 38$, $n(N) = 24$, $n(M \cap N) = 9$. Find $n(M - N)$

Using formula:

$n(M - N) = n(M) - n(M \cap N)$

$n(M - N) = 38 - 9$

$= 29$

Matches with **C. 29**

S. If $n(X) = 31$, $n(Y) = 21$, $n(X \cap Y) = 8$. Find $n(Y - X)$

Using formula:

$n(Y - X) = n(Y) - n(X \cap Y)$

$n(Y - X) = 21 - 8$

$= 13$

Matches with **D. 13**

INTEGER TYPE

18. During a food festival

50 students like Pizza

18 students like both Pizza and Burgers

Find students who like only Pizza.

Using formula:

$n(Pizza\ only) = n(Pizza) - n(Pizza \cap Burgers)$

$50 - 18 = 32$

Answer: 32 students

19. If $n(A) = 22$, $n(B) = 17$, $n(A \cap B) = 9$. Find $n(A \cup B)$

Using formula:

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

22+17−9

$39 - 9 = 30$

Answer: **30**

20. Two sets are disjoint.

If $n(A) = 14$, $n(B) = 19$. Find $n(A \cup B)$

For disjoint sets:

$n(A \cup B) = n(A) + n(B)$

$14 + 19 = 33$

Answer: **33**

STUDENT PRACTISE CORNER

1. The intersection of the sets {2, 4, 6} and {4, 6, 8} is the set.

   Intersection means common elements in both sets.

   {2, 4, 6} ∩ {4, 6, 8} = {4, 6}

   Answer: b) {4, 6}

2. The union of the sets {3, 5} and {5, 7, 9} is the set.

   Union means all distinct elements from both sets.

   {3, 5} ∪ {5, 7, 9} = {3, 5, 7, 9}

   Answer: a) {3, 5, 7, 9}

3. If A = {2, 5, 8}, B = {1, 5, 7}, C = {2, 5, 9} then A ∩ (B ∪ C) is

   Step 1: Find B ∪ C

   B ∪ C = {1, 5, 7} ∪ {2, 5, 9}

   = {1, 2, 5, 7, 9}

   Step 2: Find intersection with A

A ∩ (B ∪ C)
= {2, 5, 8} ∩ {1, 2, 5, 7, 9}
= {2, 5}
Answer: C. {2, 5}

4. If A = {x : 0 ≤ x ≤ 2} and B = {y : y is a prime number}, then what is A ∩ B equal to?
A = {0, 1, 2}
Prime numbers are numbers greater than 1 having exactly two factors.
Among 0, 1, 2 only 2 is prime.
So,
A ∩ B = {2}
Answer: C. {2}

5. Let A be the set of natural numbers less than 10 and B be the set of odd numbers less than 10. Find A − B.
A = {1, 2, 3, 4, 5, 6, 7, 8, 9}
B = {1, 3, 5, 7, 9}
A − B means elements in A but not in B.
A − B = {2, 4, 6, 8}
Answer: A. {2, 4, 6, 8}

6. If n(A) = 40, n(B) = 30 and n(A ∪ B) = 55, find n(A ∩ B).
Use the formula:
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Substitute the values:
55 = 40 + 30 − n(A ∩ B)
55 = 70 − n(A ∩ B)
n(A ∩ B) = 70 − 55
n(A ∩ B) = 15
Answer: A. 15

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. Let $A = \{2, 4, 6, 8, 10\}, B = \{1, 2, 3, 4, 5\}$. Choose all correct statements.
Step 1: Find common elements
A ∩ B = {2, 4}
So A is correct.
Step 2: Find A − B
A − B means elements in A but not in B.
A − B = {6, 8, 10}
So B is correct.
Step 3: Find B − A
B − A = {1, 3, 5}
So C is correct.
Step 4: Find n(A ∪ B)
A ∪ B = {1,2,3,4,5,6,8,10}
Number of elements = 8
So D is correct.
Answer: A, B, C and D

8. Let n(A) = 42, n(B) = 30, n(A ∩ B) = 14. Choose all correct options.
Use the formula:
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
n(A ∪ B) = 42 + 30 − 14
= 58

So A is correct and D is wrong.
Find n(A − B)
$n(A - B) = n(A) - n(A \cap B)$
n(A − B) = 42 − 14 = 28
So B is correct.
Find n(B − A)
n(B − A) = 30 − 14 = 16
So C is correct.
Answer: A, B and C

9. Let n(A) = 18, n(A ∩ B) = 5. Choose the correct options.
Use the formula:
$n(A - B) = n(A) - n(A \cap B)$
n(A − B) = 18 − 5 = 13
So A is correct.
Check C:
n(A − B) = n(A) − n(A ∩ B)
This formula is true.
So C is correct.
Answer: A and C

COMPREHENSION TYPE

PASSAGE

10. How many students use only Instagram?
Given:
n(I) = 40
n(I ∩ W) = 25
Students using only Instagram:
$n(I - W) = n(I) - n(I \cap W)$
n(I − W) = 40 − 25
= 15
Answer: B) 15

11. Which set operation represents students using only Instagram?
Only Instagram means students in I but not in W.
So the operation is:
I − W
Answer: C) I − W

12. Students using both apps belong to:
Students using both Instagram and WhatsApp are common in both sets.
Common elements belong to the intersection set.
Answer: B) Intersection set

13. MATRIX MATCHING TYPE
P. $n(A) = 20, n(B) = 15, n(A \cap B) = 5$
Using
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 20 + 15 - 5 = 30$
P → b
Q. $n(A) = 25, n(A \cap B) = 10$
Using
$n(A - B) = n(A) - n(A \cap B)$
$= 25 - 10 = 15$

Q → a

R. $n(B) = 24, n(A \cap B) = 8$

Using

$n(B - A) = n(B) - n(A \cap B)$

$= 24 - 8 = 16$

R → d

S. $n(A) = 12, n(B) = 14, n(A \cup B) = 20$

Using

$n(A \cap B) = n(A) + n(B) - n(A \cup B)$

$= 12 + 14 - 20 = 6$

S → c

INTEGER TYPE

14. If $n(A \cup B) = 52, n(A) = 30, n(B) = 28$

Using

$n(A \cap B) = n(A) + n(B) - n(A \cup B)$

$= 30 + 28 - 52 = 6$

Answer: 6

15. $A = \{1,2,3,4,5,6\}, B = \{4,5,6,7,8\}$

$A - B = \{1,2,3\}$

Number of elements in $A - B$

$= 3$

Answer: 3